

Biomarkers in SHARE: Documentation of implementation, collection, and analysis of dried blood spot (DBS) samples 2015 – 2023

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Author contribution:

MBS oversaw the biomarker project, LW and MBS designed, organized and managed the data collection processes, ABS coordinated SHARE, designed the data adjustment processes and was responsible for funding, KAR coordinated the biobank and punching in Denmark, NBL and KS managed the subdivision/punching of the samples and executed the analyses of the laboratory analyses in Copenhagen, JC, EK, TMK, AP, and MW executed the laboratory analyses in Seattle, RG developed the spot size correction, HMH, AS, and LW performed the regression analyses, and DS took care of ethics and data protection issues. All authors participated in drafting and critically reviewing the manuscript.

Abstract

SHARE, the “*Survey of Health, Ageing and Retirement in Europe*”, is a large population-based panel survey among people aged 50 and over with data from 28 European countries and Israel. It investigates individual, economic, health-related, and social life-course circumstances in order to shed light on the challenges of population aging for individuals and society as a whole. Understanding aging per se, and how we age differently over the life course given our individual backgrounds, current health, and socio-economic factors, are the aims of SHARE.

In order to maintain intertemporal, international and intercultural comparability, SHARE has adopted collection of objective data in the health domain. SHARE measures physical performance, such as grip strength, peak expiratory flow, walking speed, chair stand, word recall, and Euro-D depression among others in the physical, cognitive and mental health modules. In 2015, SHARE collected dried blood spot (DBS) samples as an additional objective measure of health. Eleven European countries and Israel participated in the DBS collection. The collection was harmonized in terms of designing documents, gaining consent, procuring blood-collection material, and training interviewers how to collect DBS samples while observing the national ethic and administrative regulations in all countries. Altogether, approximately 27,200 respondents consented, resulting in an overall participation rate of 67% with considerable differences between countries and interviewers.

This report describes the carefully monitored processes of gaining consent for DBS collection and for the implementation, collection and evaluation of the to-date largest DBS-based study

of a representative adult population in Europe. We also describe the choice of blood biomarkers, the assays employed to determine the blood biomarker concentrations in DBS collected in the home of survey respondents and the validation and adjustment of the laboratory results for the impact of sample collection in a non-medical environment. Finally, we present the data obtained for seven out of 17 blood biomarkers. The data for the remaining ten biomarkers analyzed at the Staten Serum Institut will follow in a separate release.

Keywords:

Dried blood spots (DBS), biomarkers, multinational population-based surveys, age-related health conditions, objective health measurement

Abbreviations:

HbA1c or A1c: glycated hemoglobin A

CAPI: Computer-assisted personal interviewing

BMI: body mass index

CHO or TC: total cholesterol

CRP: C-reactive protein

CVD: cardiovascular disease(s)

CYS or CysC: cystatin C

DBS: dried blood spot

DBSS: dried blood spot samples

EDTA: Ethylene Diamine Tetra Acetic acid

ELISA: enzyme-linked immunosorbent assay

HDL(-Cholesterol): high density lipoprotein(-Cholesterol)

HRS: The Health and Retirement Survey

HPLC: high-performance liquid chromatography

THB: total hemoglobin

TRG: triglycerides

TTT: Train-the-Trainer

SHARE: The Survey of Ageing, Health and Retirement in Europe

VBS: venous blood sample

Countries:

BE: Belgium

CH: Switzerland

DE: Germany

DK: Denmark

EE: Estonia

ES: Spain

FR: France

GR: Greece

IL: Israel

IT: Italy

SE: Sweden

SI: Slovenia

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We thank the country teams and survey agencies of the SHARE countries participating in the DBS collection for the implementation of the new module in SHARE Wave 6, for training their interviewers and supporting them to collect DBS. In addition, our appreciation goes to the interviewers, who convinced the respondents to participate in this project and collected the blood during the survey.

Special thanks go to the SDU staff and students at the Biobank in Odense, DK: Susanne Knudsen, Stine O. Høgh and, foremost, Nynne E. Ustrup for receiving, registering, safely storing, and later preparing thousands of DBS cards for shipment to the laboratories. Thanks to Mads Nybo (SDU and OUH, Odense, DK) for valuable discussions. A great thank goes to Nicklas Petersen at Statens Serum Institut, Copenhagen, DK, for sorting and punching all DBS samples.

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1 Introduction

1.1 SHARE, the Survey of Health, Ageing and Retirement in Europe

This report describes the blood-biomarker data collected in SHARE Wave 6 in 2015 in 12 countries, which was assayed from ca. 24,000 dried blood spot (DBS) samples in 2017/2018 and 2020/2021.

The Survey of Health, Ageing and Retirement in Europe (SHARE) is an infrastructure of micro data addressing the health, social and economic impacts on growth and prosperity of the aging populations in Europe (A. Börsch-Supan et al., 2013.). With the release of Wave 8 in 2022, the SHARE database contains ca. 530,000 interviews from approximately 140,000 respondents aged 50 and over from 28 European countries and Israel. (www.share-project.org; Bergmann, M., A. Börsch-Supan (Eds.) 2021; Bergmann, Scherpenzeel, & Börsch-Supan, 2019; A. Börsch-Supan, Brandt et al., 2011; A. Börsch-Supan, Brandt et al., 2013; A. Börsch-Supan, Brandt, & Schröder, 2013; A. Börsch-Supan, Bristle et al., 2019; A. Börsch-Supan, Brugiavini et al., 2008; A. Börsch-Supan, Brugiavini et al., 2005; A. Börsch-Supan & Jürges, 2005; A. Börsch-Supan, Kneip et al., 2015; Malter & Börsch-Supan, 2013; Malter & Börsch-Supan, 2015; Malter & Börsch-Supan, 2017).

1.2 Self-reported and objective health measures

Since its first wave in 2004, SHARE respondents are asked to self-report current general health, diseases and conditions, use of medication, as well as functional limitations and disability, and are asked for height and weight. They also self-reported health behaviors such as smoking, drinking, exercising, and physical fitness, and self-rate their health. Although individuals are likely to evaluate their own health status differently even when they have the same objective condition (Jürges, 2007), self-reported information is of great value in itself (Doiron, Fiebig et al., 2015; Jylhä, 2009). The accuracy of self-reported health data depends on individual health literacy and access to health care and treatment, which are correlated with socio-economic factors. This correlation may lead to spurious associations between health outcomes and socio-economic factors in the observed data (Bound, 1991; Sen, 2002). Therefore, SHARE had decided to spend major efforts to collect objective information on physical, cognitive and mental health, and, additional measures such as grip strength, peak expiratory flow, walking speed, and chair stand as physical measurements, and other measures, e.g., word recall, Euro-D depression battery, etc., in the physical, cognitive, and mental health modules.

In addition, SHARE early on developed a plan to obtain blood-based health measures. In Wave 4, a small feasibility study was performed in Germany, and it was decided to collect DBS samples in Wave 6 in more countries, since collecting venous blood was both too expensive and too complicated given different national restrictions in doing so.

While the collection of health data in a population-based survey is already a complex enterprise in a single country, the effort in a pan-European study like SHARE is even greater and requires extensive planning, as it is crucial to ensure the comparability of collection protocols across countries (Weiss, Sakshaug, & Börsch-Supan, 2018). This applies in particular to obtaining blood-based biological markers detectable in dried blood spot samples (DBSs) as was done in Wave 6 in 2015 (Börsch-Supan, M & Andersen-Ranberg, Chapter 6.1, in: Malter & Börsch-Supan, A (Eds.) 2017; Börsch-Supan, M et al., 2020).

1.3 Combining bio-medical with socio-economic research: Collection of DBS samples in a multinational population survey

Blood biomarkers shed light on the mechanisms that relate socio-economic factors influenced by lifestyle and environment to morbidity and mortality, which are not confounded by the shortcomings related to self-reported health information.

Collecting blood is the method of choice when combining bio-medical and socio-economic research (McDade et al. 2007; Williams & McDade, 2009; Kim et al. 2023). From a laboratory point of view, venous blood samples are best suited for analyzing blood biomarkers. However, collecting venous samples in very large community-based setting in so many countries as in SHARE entails major logistic challenges of collection, transportation and timely laboratory analyses as well as large expenses for venipuncture performed by certified health professionals. Moreover, a venipuncture is invasive and may potentially cause harm to the individual. It may also be regarded as intrusive by the respondent and thus yield higher proportions of non-participation in the SHARE survey itself.

Instead of drawing venous samples, SHARE decided to collect blood using a method more adaptable to the circumstances of a very large international population survey. The sampling of DBS offers numerous advantages over conventional blood collection (Mei et al. 2001; Li & Lee 2014; Li & Tse 2010). DBS samples are drops of capillary whole blood collected on filter paper from a simple finger prick. The method is regarded as less intrusive than venipuncture and easily applied. Blood collected on filter cards can be shipped by regular mail to a laboratory and then stored frozen until analyses. This makes DBS samples well suited for many screening programs or population-based studies in the area of public health research.

The blood-based biomarkers shall provide SHARE and its users with additional objective health measures to invest more closely the most common aging-related health conditions in the context of the socio-economic information gained from its survey. Those health conditions and aging-related diseases include diabetes (e.g., Kalyani, Golden, & Cefalu, 2017), cardiovascular disease (CVD) (e.g., North & Sinclair, 2012), decline of kidney function, and cognitive decline, but also loss of strength and muscles (frailty, sarcopenia). A detailed description of the chosen biomarkers is given in Section 4 below.

Surveys such as SHARE operate in the field, i.e., respondents are interviewed in their homes. Instead of certified health professionals, trained interviewers collect the blood samples following a fixed protocol but not under controlled and standardized conditions as in clinics or laboratories. The conditions during fieldwork and shipment may vary namely the environmental temperature and humidity; those conditions are difficult to control. In the case of SHARE, the geographical and climatic regions stretch from northern Sweden to Mediterranean countries like Spain and Israel. In addition, best care and attention exercised by the interviewer while collecting the blood and handling the sample is very important (Crimmins, Zhang et al., 2020). The interviewer may influence the drying time of the blood samples, humidity protection during shipment, or other quality measures, shortcomings which lead to avoidable damage (see 2.2.3 and 3.2).

Not all collected samples may contain sufficient material to permit analyzing all the biomarkers of interest as not all respondents bleed well; furthermore, sampling mistakes happened or the sample is of bad quality. This and certain medical conditions, e.g., bleeding disorders prohibiting blood-collection, and non-consenting led to biomarker values being obtained from only a subsample of the DBS-eligible respondents.

In awareness of the non-medical circumstances of blood collection during fieldwork, SHARE and the two laboratories in Seattle and Copenhagen (entrusted with the blood biomarker analyses; see Section 4) performed validation experiments in 2018 and 2019, respectively. The influence of single conditions, such as temperature, humidity, and drying and shipping times, was tested as well as were interactions of these conditions (Börsch-Supan, Weiss et al., 2019; Weiss & Börsch-Supan, Chapter 37, in: Börsch-Supan, A et al. (Eds.), 2019). In 2021, Börsch-Supan, A. et al., published a detailed description of the validation experiments.

In Section 2.1 below, we will describe the steps preceding the blood collection: the preparation of all legal, ethical, and administrative documents required, the compilation of the DBS collection kit, the implementation of a DBS module in the CAPI instrument, the training of the interviewers on how to collect DBSs on a filter card and send the sample to the biobank. Section 2.2 describes the collection of DBSs during the survey and the fieldwork monitoring. In Section 3, we describe how the final DBS sample emerged: not all respondents were eligibility due to health reasons, not all eligible respondents gave their consent to collect DBSs, and not all collected DBSs had sufficient quality to be analyzable. This section ends with the computation of DBS weights accounting for this selection process. Section 4 turns to the laboratory analysis of DBSs: In Section 4.1, we present the blood biomarkers that SHARE had chosen to analyze and the motivation for our decision, and, in Section 4.2, we describe the preparation of the DBS samples for laboratory work. Unfortunately, the DBS values obtained from the laboratory analysis cannot be used straightforwardly in substantive research, since biomarker values derived from field-collected DBSs have analysis levels and variances that differ from biomarker values derived from venous blood samples (“gold standard”). Section 5 describes the reasons for these differences and the necessary adjustments for field

conditions, for the size of a collected DBS, for the storage time of the blood samples before analysis, for lab effects particular to DBS analysis, as well as corrections for the type of blood sample, based on a set of validation experiments (Börsch-Supan, A. et al., 2021). Finally, in Section 6 we present descriptive statistics of the final data set with the SHARE DBS-derived biomarkers.

2 Methods

2.1 Implementation

2.1.1 Legal, ethical and administrative requirements

Collecting data from respondents in a survey, particularly when collecting blood samples, needed the approval of ethical committees in all participating countries and the Ethics Council of the Max Planck Society. DBS collection is considered an invasive, albeit only minimally invasive method. It therefore requires careful legal, ethical and administrative preparation.

For the DBS collection in SHARE Wave 6, the CAPI¹-assisted questionnaire addressing blood collection, consent documents, and other training and field material, as well as a survey protocol were prepared centrally in English. The documents were translated into the national languages of the SHARE countries and – depending on the requirements of each country – compiled, adapted, extended, and then presented to the responsible national ethics committees and/or other authorities to obtain ethics approval (Schmidutz, 2016; Schmidutz, Ryan et al., 2013; Schmidutz & Weiss, 2017).

Obtaining informed written consent was obligatory in all participating countries. The consent defined how the sample collection had to be conducted in the field and how the samples can be handled and analyzed in the future. However, the countries differed regarding the information that had to be provided to the respondents and the content and form of the consent documents. In some countries (Switzerland, Denmark, Italy, Sweden) the respondents had to be informed about the blood collection with an advance letter; in other countries it was sufficient to inform the respondents during the interview prior to the planned DBS collection.

SHARE decided not to report lab values to the respondents in case of a level of a blood biomarker value indicating a possible health concern. This was done for several reasons: the methods of blood collection used in SHARE are not suitable for individual disease diagnosis; instead, they will shed light on population health. Diagnosis of a serious health condition has to be performed under standard clinical conditions with specific testing requirements. In addition, the blood biomarker data obtained during the survey are only available a long time after blood was collected; at the time of back-reporting, the health conditions of the respondent may be different. In addition, experiences in Germany while piloting the DBS collection in Wave 4 showed that back-reporting is complicated since the German respondents had to consent to reporting blood biomarker values to a third institution/person, e.g., via his/her general practitioner. In many cases, the contact with the GP failed. All ethic committees agreed with our reasoning not to report blood biomarker values to the respondents, with a single exception: in Denmark, the respondents have to be notified in case unknown diabetes was detected by the SHARE DBS analyses; this has to be carried out via the Danish survey agency, as SHARE has no access to the addresses of respondents.

¹ CAPI: Computer-assisted personal interviewing

The interviewers play a crucial role in the process of gaining consent, because they are the only personnel in direct contact with the respondent and are those collecting the DBS sample during the interview. Hence, the aspect of successfully obtaining consent was an important part of the DBS collection training (Korbmacher, 2014; Schmidutz & Weiss, 2017), see Section **2.1.6**.

Most countries allowed the collection of blood by trained interviewers. In France, however, interviewers were not allowed to touch the respondents. French respondents therefore pricked their own finger with the lancet, guided by the interviewers or could decide to be pricked by a third person e.g., a confidant. In Austria, the Czech Republic, Poland, and Luxembourg even this process was not possible. Only medically trained personnel are allowed to take any kind of blood samples. Since this was neither practical nor financially feasible, no DBS samples could be collected in those countries. And, lastly, Croatia, Hungary, The Netherlands, and Portugal could not be included in the DBS collection of the main survey due to financial reasons. Twelve SHARE countries² (Belgium, Denmark, Estonia, France, Germany, Greece, Israel, Italy, Slovenia, Spain, Sweden, and Switzerland) remained and a total of ca. 27,200 DBS samples were collected. In addition to the main DBS collection, we conducted a small study in Poland in 2016 (under the supervision of Dr. Roman Topór-Mądry, Department of Epidemiology and Population Studies, Jagiellonian University Medical College, Krakow, Poland) with 37 participants, in which nurses collected venous and capillary (DBS) blood. The Polish study could therefore be used for validation purposes (see **5.4** and Weiss, Börsch-Supan, M et al., 2019).

We developed a detailed plan how to electronically follow processing and storage of the blood samples as well as the corresponding laboratory data. To ensure data privacy of the respondents, the signed consent forms, the blood material, and the CAPI interviews are each stored at separate locations: with the national survey agencies, the biobank, and on the SHARE server, respectively. All three data sources (the sample itself, the consent form and the interview data) are labelled with a barcoded identification numbers for each participating respondent. This number is used for the future linkage of the biomarker analyses results to the survey data.

For the secure storage of DBS samples from all participating SHARE countries, the SHARE biobank was established at the Department of Public Health at the University of Southern Denmark (SDU) in Odense, Denmark. It is affiliated with the biobank of the Danish Twin Registry, hosted by SDU. All blood samples collected from respondents during the field phase of Wave 6 have been sent there, where they are stored at -20°C, each together with desiccant for humidity protection in laboratory-style freezers owned by SHARE-ERIC until 2023. From 2024 on, storage at SDU will be in -20°C cold rooms. For surveillance, all freezers or cold rooms, respectively, are connected to an alarm system, which notifies a biobank staff member in the case of a temperature rise or another hazard.

² Country Code of all SHARE countries in wave 6: AT Austria, BE Belgium, CH Switzerland, CZ Czech Republic, DE Germany, DK Denmark, EE Estonia, ES-sp Spain, ES-cat Catalan province of Girona, FR France, GR Greece, HR Croatia, HU Hungary, IL Israel, IT Italy, LU Luxembourg, NL Netherland, PL Poland, PT Portugal, SE Sweden, SI Slovenia.

2.1.2 Required documents

The SHARE biomarker team centrally prepared and compiled the material needed by the SHARE interviewers to collect the DBS from the survey respondents during the interview. This material included the blood-sampling equipment, the DBS collection kit (see 2.1.3) but also documents to inform the respondent about the DBS collection, processing and subsequent analyses of the blood-samples and the consent forms, which had to be signed by consenting respondents before DBS collection.

- (1) Information leaflet
- (2) Consent form
- (3, 4, 5) Interviewer instructions: DBS manual, step-by-step instructions written and as illustrated flyer

The *information leaflet (1)* explained the purpose and DBS collection method, and provided respondents with contact details as well as information about their rights and the privacy practices within the project. The DBS collection *consent form (2)* recorded the information in an abbreviated version and had to be signed by both the respondent and the interviewer. The leaflet and one copy of the consent form remained with the respondent; the second copy was mailed to the national survey agency by the interviewer.

In addition, *information material (3)* with a *step-by-step guide (4)* of the blood collection procedure was prepared for the interviewers. As a quick reference, interviewers also received a brochure – an *illustrated flyer (5)* – illustrating the most important steps of the blood collection procedure. This material is provided in the Appendix.

2.1.3 The collection kit

We equipped the interviewers of each country with ready-to-use DBS collection kits (Figure 2-1). The coordinating SHARE biomarker team in Munich designed the kit as no suitable kit was commercially available. The kit contained all items needed for the blood collection and subsequent shipment of the samples to the SHARE biobank. To facilitate decisions, which specific items had to be included in the kit, we investigated various shipment formats – namely envelopes, bags and different desiccant materials – with regard to their temperature- and humidity-protection capacity (see 2.1.4). To obtain information about the temperature exposure of samples during shipment, we used temperature-tracking devices (Figure 2-3) in the field during the pretest of SHARE Wave 6, and, in Israel, we used long-run temperature tracking starting with the summer months during the SHARE Wave 6 main survey (see 2.1.4 and 2.2.3).

The blood-collection items were bought centrally and assembled into kits by a local Munich company. This ensured the use of identical material in all countries and by all interviewers. Altogether, 40,000

kits were delivered to the national survey agencies that were commissioned with the SHARE data collection in Wave 6 in the participating countries.



Figure 2-1: Items in a SHARE W6 DBS collection kit. A large Ziploc plastic bag containing all necessary material for DBS collection, which later served as a disposal bag for used items after the DBS collection; the SHARE DBS filter card with pre-printed 1-cm circles (see Figure 2-7) for collecting the blood drops; blue disposable protective cloth used as a clean base for material and as surface protection from any small blood stains; disposable protective gloves; disposable disinfecting wipes; two semi-automatic, one-time use lancets; sterile gauze pads; and plasters; 2g of molecular-sieve desiccant contained in a small Tyvek® bag. To prevent untimely humidity absorption and extend the shelf life, the desiccant bags were individually sealed in a larger aluminum pouch, from which the small – 25x55mm – Tyvek® bag was to be removed before use. For later shipment of the sample to the biobank, the kit also contained a small plastic bag for the DBS card and desiccant and a special tear-proof and water-resistant Tyvek® mailing envelope (Figure taken from Börsch-Supan, M et al., 2020)

Not shown in Figure 2-1, but also part of the interviewers' collection material: reusable hand warmers to be used to warm the respondents' hands before the DBS collection, since warm hands exhibit better blood flow. The hand warmers stayed with the respondents after the interview as a small token of appreciation.

In addition, the interviewers were provided with a sheet with barcode stickers (all with the same barcode number) unique to each respondent; pre-printed address stickers for mailing the sample to the SHARE biobank; and stamps, which were supplied by the survey agencies of each country for international postage to Denmark. Temperature strips used in Israel were bought while the survey was in the field and were added to the Israeli test kits.

2.1.4 Investigating envelopes, temperature-control devices and desiccant materials

Envelopes and bags: Different shipment-material formats were tested before preparing the collection kits in order to investigate temperature and humidity protection for the DBS samples. We

were interested to learn how well envelopes made of different material and different kinds of plastic bags would protect the sample against temperature change. We acquired various kinds of envelopes (paper padded, bubble-foil padded, unpadded Tyvek®) and bags (polyethylene PE, Mylar, insulating material). We created various combinations of envelope/bag each with an “empty” DBS card and desiccant inside the bags – as if ready for shipment. A temperature/humidity logger was added to the envelopes, which were then exposed to various controlled temperatures between room temperature and up to 35°C for various time periods in a laboratory oven³. Different material combinations provided temperature protection within a very narrow range (Figure 2-2). The temperature inside an envelope/bag combination exposed to 35°C reached a threshold of 33°C in a time span between 21 and 44 minutes. In summary, compared to the expected shipping times of international mailing the environmental temperature was reached inside the envelopes within a very brief time. Therefore, we chose for the study envelope material Tyvek®, being tear- and water-resistant, light, and easy to close in combination with a simple PE bag for the DBS card.

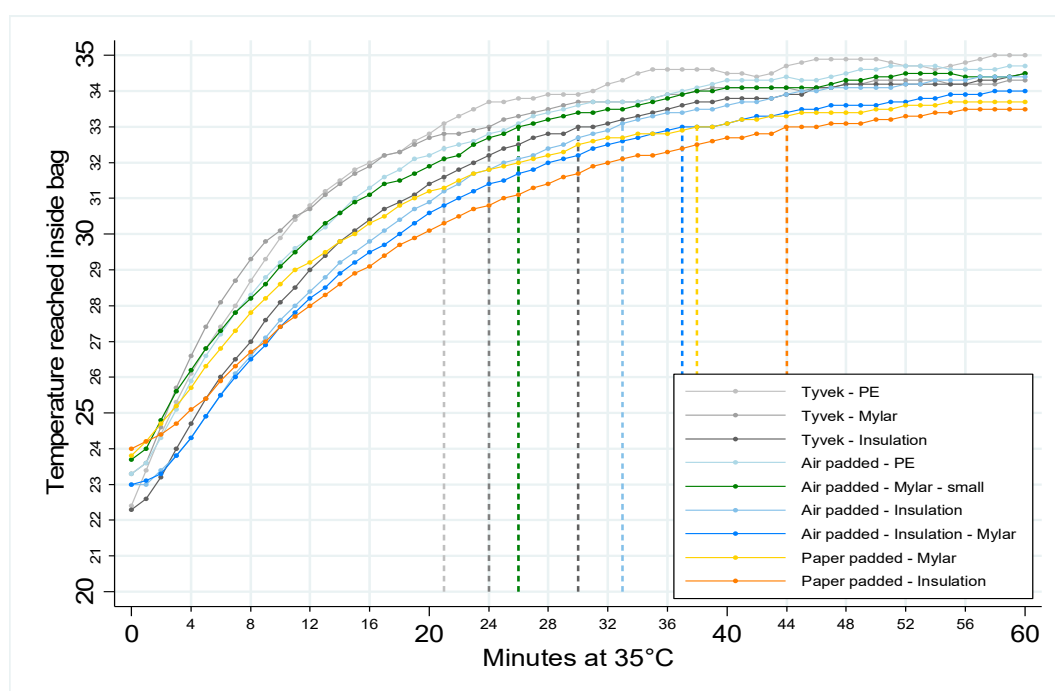
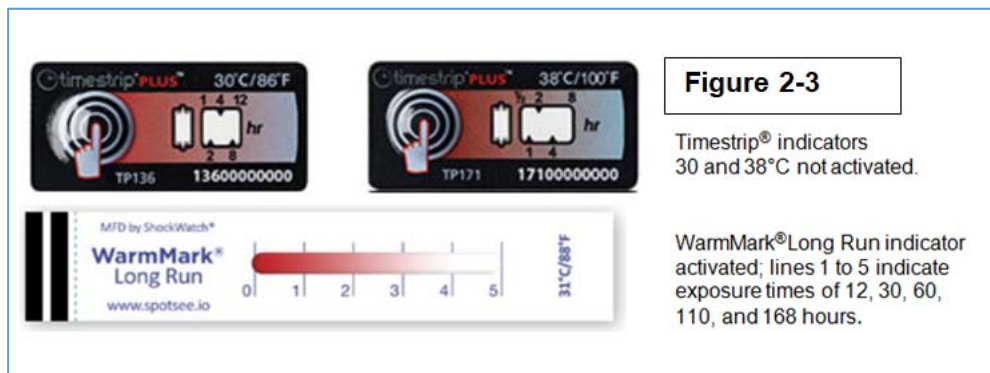


Figure 2-2: Envelope testing for temperature protection. Various envelope/bag combinations were exposed to 35°C. Temperature inside the bags reached a threshold of 33°C after 21 to 44 minutes (dashed vertical lines) (Figure taken from Börsch-Supan, M et al., 2020)

Long-term temperature control during shipment: During shipment to the biobank, which, dependent on the country of origin, could be by plane, train or truck, the DBS samples are exposed to varying – unknown – temperatures. However, the exposure to high temperature has been shown to influence the subsequently measured levels of thermally labile biomarkers in DBSs (Börsch-Supan, Weiss et al., 2021; Crimmins, Zhang et al., 2020). We included long-term temperature

³ The oven was kindly loaned to us by the Max Planck Institute of Neurobiology in Munich.

trackers with the DBS samples during shipments to learn more about the temperature levels the samples are actually exposed to and the association of those temperatures with the ambient temperature at the time of collection of the blood sample. The trackers are small plastic strips (Figure 2-3; Timestrip Label 30 and 38°C; Timestrip UK Ltd, Cambridge, UK, and Shockwatch WarmMark® Long-Run indicator, Hanwell Solutions Ltd. UK, for 31°C) with an activating mechanism to be initiated by the interviewer when preparing the blood sample for shipment.



(Figure taken from Börsch-Supan, M et al., 2020)

The tracking strips indicated by the spread of colored indicator if a certain temperature threshold was surpassed and its duration. This information was recorded upon receipt of the sample by the biobank staff. The local outside temperature during the interview and blood collection was estimated by the interviewer and entered into the CAPI.

During the pretest of Wave 6, we used two different temperature strips to indicate exposure to 30°C, or 38°C or higher temperatures, respectively. The duration of the exposure was measured in steps of one hour for up to 12 (or 8) hours overall.

During the main data collection, we tracked the temperature during shipment in some Israeli samples. For those, we used long-run trackers indicating the exposure to 31°C or higher for 0, 12, 30, 60, 110 (2.5 days), or 168 (7 days) hours overall.

Only a few samples reached the threshold of 38°C during the pretest (11 out of 223 samples with linkable interview data). For 30°C and 31°C, we created dummy variables to indicate whether the respective threshold was reached at all and calculated a logistic regression of the dummy variable on the estimated outside temperature, controlling for shipment time. The correlations between the outside temperature and the dummy variables were both significant. The predictive margins of the 30°C and 31°C dummy variables over the estimated outside temperature (controlled for shipment time) are shown in Figure 2-4.

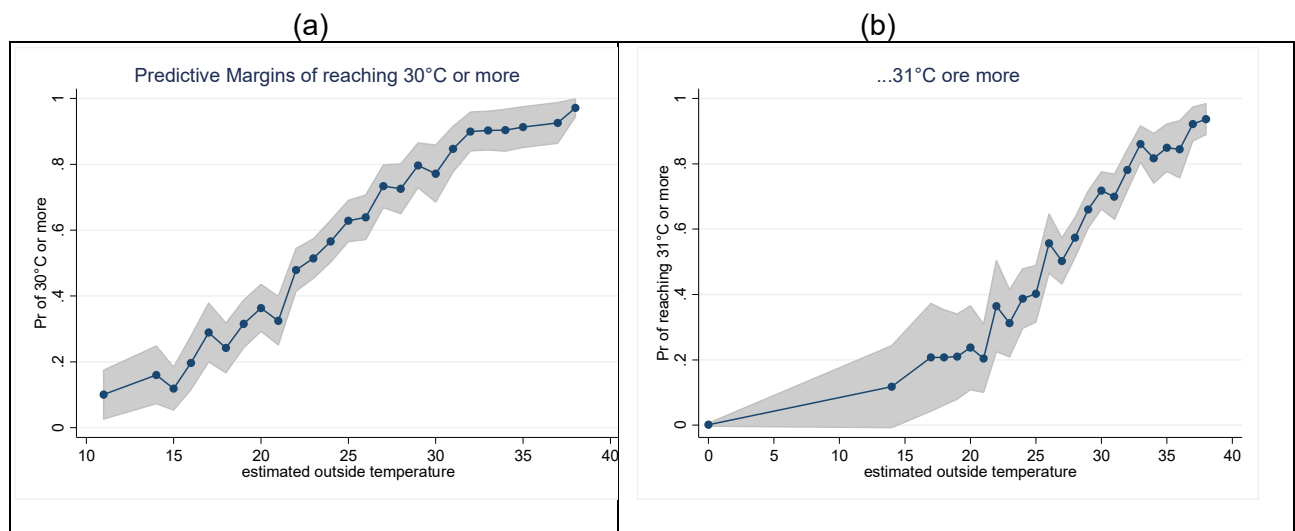


Figure 2-4: Predictive margins of reaching a temperature threshold after logistic regression of the according dummy variable on the estimated outside temperature and controlling for shipment time. (a) threshold at 30°C, Wave 6 pretest samples, N=288; (b) threshold at 31°C, Wave-6 Israeli main survey samples, N=299 (Figure taken from Börsch-Supan, M et al., 2020)

We conclude that the outside-temperature estimation by the interviewer during blood collection provides valuable information on the temperature pattern the sample is exposed to during the entire shipment time.

Desiccant: We studied product comparisons of silica gel vs. molecular sieve (see below). Molecular sieve adsorbs water from the surrounding faster and more aggressively than silica gel, though its equilibrium capacity for water is slightly lower than silica gel. On the other hand, molecular sieve does not release water at higher temperature as easily as does silica gel. We calculated that 2 g of molecular sieve in a sachet per DBS card will be sufficient for a dried DBS sample containing up to five blood spots. (e.g., https://www.sorbentsystems.com/desiccants_charts.html#table1; <https://www.multisorb.com/blog/pharmaceuticals/molecular-sieve-vs-silica-gel-whats-the-difference/>).

2.1.5 The DBS module in the CAPI instrument

To conduct the blood sampling, SHARE set up a short questionnaire module within the CAPI instrument, the BS module, closely following the DBS collection module used in the Health and Retirement Study (HRS⁴, Crimmins, Guyer et al., 2008). For the SHARE English version, see: http://www.share-project.org/fileadmin/pdf_questionnaire_wave_6/Generic_main_qnn_6_3_13.pdf. The module instructions guided the interviewers through the DBS collection: from introducing the purpose of the DBS collection to gaining consent, and included a section for the interviewer to report possible problems that may have occurred during the blood collection.

The BS module was routed to interviewers via a preloaded variable; it only appeared for DBS-eligible respondents (see 2.2.1). As a reminder to the interviewer, instruction at the beginning of the BS

⁴ HRS is the SHARE “sister” study in the United States of America

module denoted it as “non-proxy module”: In case of the respective interview being a proxy⁵ interview, no blood was collected, and the interviewer skipped the entire BS module.

The first item to be read to the respondent was a short introduction with two versions: (1) for countries that announced in their advance letters that DBS were to be collected (see 2.1.1) and (2) for all other countries. For the actual blood collection, the CAPI referred the interviewer to the printed step-by-step instructions, eliminating the need to handle the laptop while wearing the protective gloves worn during the blood collection. After the blood collection, the printed instruction leads the interviewer back to the CAPI to twice enter the barcode number of the blood sample. A programmed soft-check verified that the barcode numbers entered were identical or indicated to the interviewer that a barcode number entry correction was necessary. The regular interview then proceeded to the following modules. Two items at the end of a regular SHARE interview of DBS-eligible respondents were not part of the actual BS module but was routed in the same fashion. Item one asked the interviewer to estimate the current outside temperature. This information was used to obtain information on the temperature the sample was exposed to during local handling and at the start of shipment (Weiss & Börsch-Supan, Chapter 37, in: Börsch-Supan, A et al. (Eds.), 2019). Item two referred the interviewers back to the printed step-by-step instruction, where they were guided through the preparation of the sample for shipment. Having these items not appear directly at the end of the BS module but instead at the end of the regular SHARE interview (with other modules in between) helped to ensure that the blood sample was dried for an adequate length of time.

2.1.6 Training the SHARE interviewers to collect DBS

Although survey agencies and their interviewers had experience from previous SHARE waves in collecting physical performance measurements, the collection of DBSs was novel. With the exception of six countries (BE, CH, DE, DK, IT, PT), which had participated in a logistics test run in the pretest of SHARE Wave 5, most survey agencies in SHARE Wave 6 had not previously collected DBS.

All interviewers of the countries participating in the SHARE Wave 6 main survey had to be trained extensively and in their national languages. SHARE works with a hierarchical “train-the-trainer” (TTT) scheme: First, members from the SHARE country teams and their survey agencies were trained and prepared in a central TTT workshop in Munich using the English language. The TTT agenda served as a template for the subsequent national training sessions that were conducted by the respective national survey agencies and in the languages of the interviewers.

The training session for the DBS collection was part of the general interviewer training for SHARE Wave 6. It provided the interviewers with background knowledge about the DBS collection, the biomarkers to be analyzed as well as biosafety concerns and data privacy rules. They were trained

⁵ In an interview a proxy respondent answers in lieu of the person, from whom information is being sought; this could be e.g., a partner, spouse, confidant.

to answer DBS-specific questions, dispel doubts and explain procedural and medical facts. The process of consent gaining was carefully scripted and required more skills on part of the interviewer than did the other survey items (Korbmacher, 2014). Being well prepared and well trained played an important role for successfully gaining consent from the respondents.

A central part of the DBS training consisted of a thorough hands-on training. The collection kit was introduced followed by a mock interview and blood collection. Every interviewer had to perform DBS collection under supervision. Interviewers making mistakes or missing steps were retrained. After the interviewers could successfully perform the blood-collection procedure, they were certified. Only certified interviewers were allowed to collect DBS samples for SHARE. Moreover, interviewers were trained for particular situations, like advising participants taking blood thinners or helping participants who do not bleed easily, and were prepared to physically support respondents who might feel dizzy following the finger pricking. All interviewers were provided with printed DBS interviewer instructions as a reference for self-training or as preparation for questions that could arise during the fieldwork. Attention was given to the interviewers' attitude towards the blood collection and feeling comfortable in performing it. To address those items, we worked closely together with the German survey agency. German interviewers were questioned after the pretest DBS training to suggest improvements. We considered their suggestions for the main survey training. Most of those interviewers, who had stated before the training session to feel somewhat worried with the idea of collecting blood samples, told us afterwards that they now felt confident about it when thinking of the upcoming DBS collection. Interviewers who were still worried were not assigned to interview DBS-eligible respondents.

2.2 Collecting DBSs

2.2.1. Eligibility for DBS sampling

In the countries participating in the DBS collection, all respondents in panel households were asked to consent to blood collection, i.e., all respondents who had previously participated in any SHARE wave and their partners (for the latter, independent of previous participation in SHARE). No blood was collected from respondents who refused to participate or were not able to give consent themselves. Respondents who directly or via proxy mentioned a medical reason to avoid blood collection (such as bleeding disorder) were excluded from the DBS collection. In France, DBSs were collected in only four districts: Aquitaine, Bourgogne, Bretagne, and Ile de France. From the 884 DBS-eligible respondents in those districts, 569 respondents consented to blood collection. Some respondents from other districts, thus actually not DBS-eligible, had been inadvertently asked to consent to blood collection by interviewers working in DBS-eligible and non-eligible districts and nine of those respondents consented. Those samples were included, making a total of 578 French DBS.

2.2.2 Blood collection from respondents and dispatch of DBS cards to the biobank

Starting the DBS collection, the interviewer provided a printed information leaflet, explained the procedure, asked for consent, but expressed that participation was voluntary. Upon agreement of the respondent, written consent was collected by having the respondent sign the consent form.

The interviewer then prepared the blood collection: an activated warming pad was handed to the respondent as warming the hands increases the blood flow into the fingers, thereby increasing the likelihood that a sufficient amount of blood would be collected. The date of the blood collection was noted on the DBS card, which was labelled with a respondent-specific barcode sticker.

While wearing protective gloves, the interviewer sterilized the puncture site on a finger - preferably the ring finger on the non-dominant hand - and used a semi-automatic lancet for pricking. Self-pricking by the respondent or pricking by another person was an option. The first drop of blood was wiped off to eliminate remaining alcohol from the sterilization. The next blood drops were collected by permitting them to fall onto the absorbent filter paper of the SHARE DBS card (one drop per pre-printed circle). The filter card was then left open to permit the blood to dry until the end of the regular SHARE interview, but for at least 15 minutes. At the end of the interview, a CAPI item reminded the interviewer to pack the DBS card by slipping it into a small polyethylene (PE) bag together with desiccant; the bag was then zip-closed and placed into the Tyvek® envelope.

Another sticker with the same barcode number was attached to the consent form and the barcode number was entered into the CAPI instrument. The signed consent form was sent to the national survey agency and the DBS card to the SHARE biobank in Odense, DK. The barcode number ensures that the CAPI data, the consent form and the collected DBSs can be matched, while guaranteeing respondents' anonymity. All three data sources (DBS card, consent form, CAPI interview) have to be available for the sample to be considered "complete" and to be considered for laboratory analyses.

2.2.3 Monitoring fieldwork in SHARE Wave 6 - in particular DBS collection

The SHARE Wave 6 all-data collection lasted from January until November 2015, and fieldwork was actively managed by SHARE Central (Munich, Germany) through monitoring interviewer activity, e.g., contact and cooperation rate, response/retention rate, as well as consent for blood collection and quality of DBS samples. Reports were mailed bi-weekly to all survey agencies and country teams and served as feedback of interviewer performance (Malter & Sand, 2017). Figure 2-5a shows the different DBS-collection periods of the participating countries during SHARE Wave 6 fieldwork in 2015. While several countries (SE, FR, BE, CH, SI) had almost completed DBS collection in the first part of the year, GR and EE started collecting blood samples with the beginning of summer 2015 until end of November, and DE, ES, IT, DK, and IL collected during the entire field phase of Wave 6. Figure 2-5b shows the total number of DBS cards received in the Biobank in Odense, DK, by calendar week in 2015. The collection month and date as well as the climate region of the

participating countries have to be considered when analyzing blood biomarker data. Various blood biomarker levels show seasonal variation, which depend on the regional temperature (see 2.2.5).

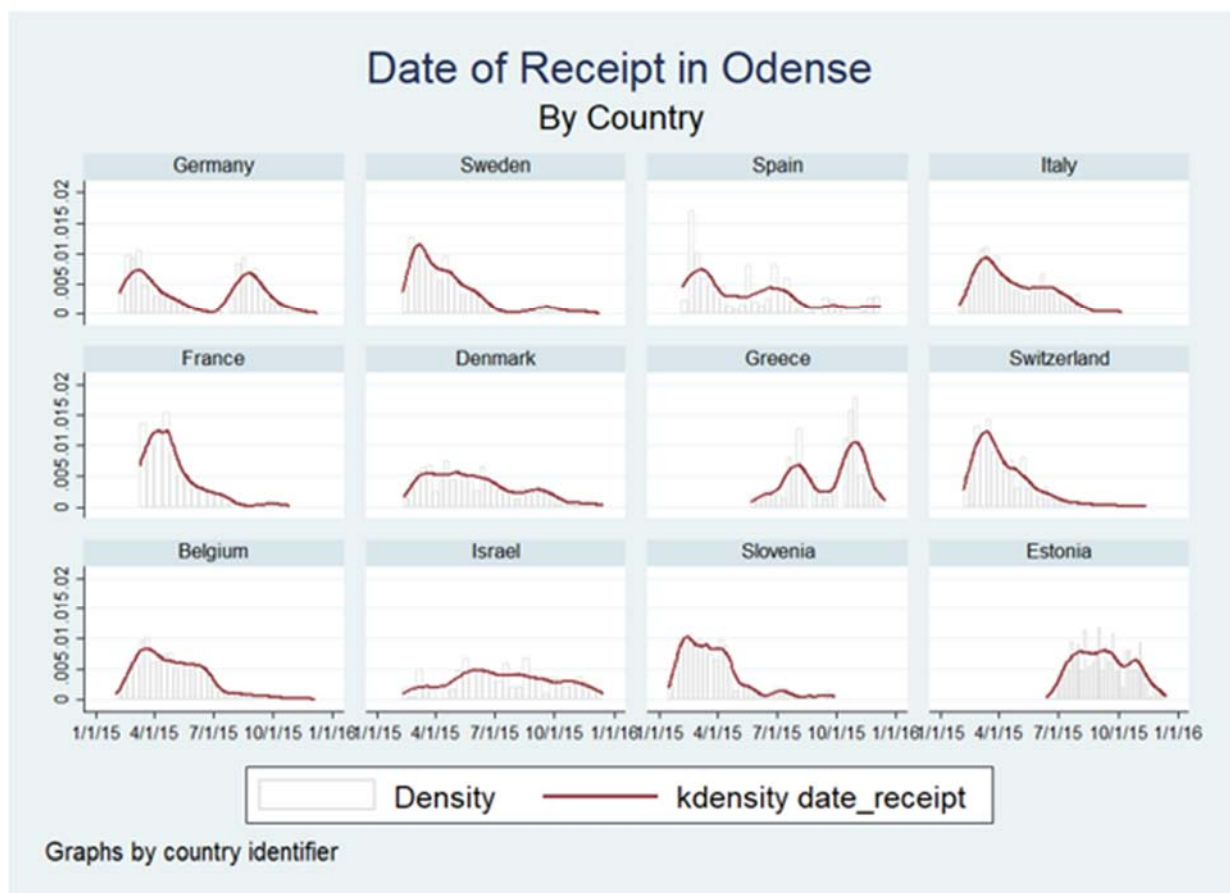


Figure 2-5a: DBS collection activity by country. Cards received in the biobank in Odense during 2015.

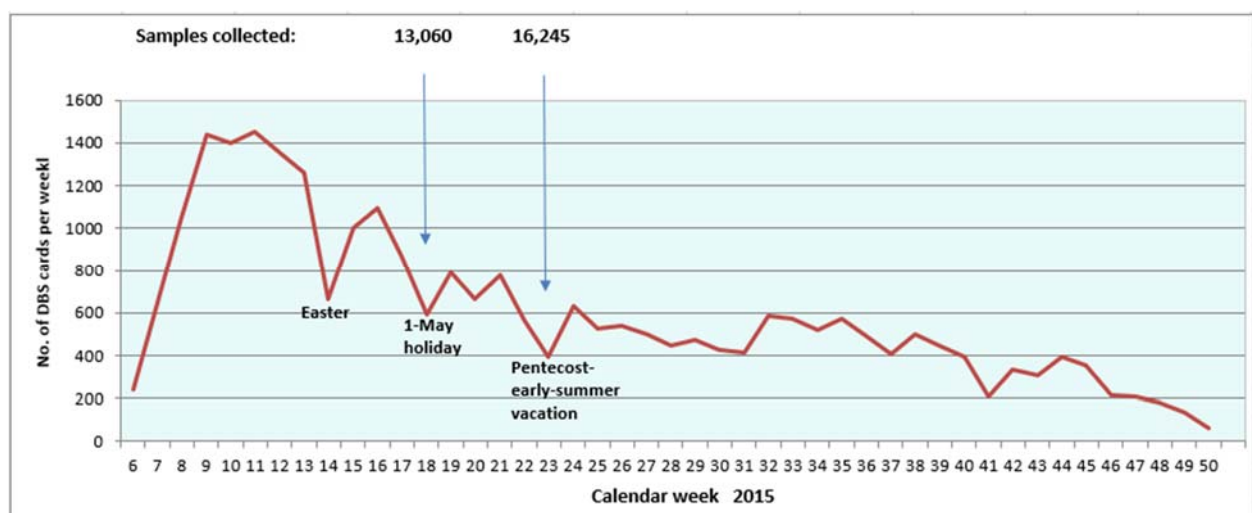


Figure 2-5b: Total number of DBS cards received in the SHARE Biobank in Odense, DK, by calendar week (in 2015). The reduced number of received DBS cards during the first half of 2015 (around weeks 14, 18, and 23) coincide with vacation periods or public holidays in Europe. Almost 50% of all DBS blood samples were collected until beginning of May (week 18) and almost 60% before the onset of summer (week 23). A few more cards were received after November 2015, the end of the collection period, during December 2015 and even in January 2016.

Within this frame of the general participation monitoring, we reported the number of collected DBS samples in each country as part of the bi-weekly reports together with the country-specific consent rate, which was calculated by the DBS team in Munich (see Section 3). The survey agencies were explicitly requested to get in touch with under-performing interviewers as SHARE was not only striving for a high participation rate but also wanted to ensure the collection of high-quality DBS samples and enough blood spots for later biomarker analyses. Our goal was to detect sampling mistakes in a timely manner and correct them by reminding or retraining under-performing interviewers. The monitoring of sample quality and quantity involved the staff at the SHARE biobank, who visibly checked all DBS samples arriving in Odense for shortcomings (Friedel & Weiss, Chapter 6.3.2, in: Malter & Börsch-Supan, A. (Eds.) 2017).

Biweekly, the biobank staff submitted a list reporting received DBS samples (identified by the barcode number attached to each DBS) together with quality information collected for each incoming sample to the biomarker team in Munich.

Upon arrival at the biobank, the following information was recorded for each incoming sample:

- Any damage to the Tyvek® mailing envelope containing the PE bag with the DBS filter card.
- Information on proper humidity protection during shipment:
 - o The status of the small PE bag, i.e., if it contained the DBS card and desiccant and had been properly zip-closed
 - o The presence of a desiccant and whether it had been taken out of its own humidity protection, the aluminum pouch
- In case a temperature indicator had been added (only during pretest and in Israel during the main survey), the reported duration of any temperature-threshold exceedance was noted. This provided some insight into temperature exposure of the DBS cards during shipment (see 2.1.4).
- Information on DBS quality, quantity and any visible damage to the card or blood spot(s), namely
 - o Number and quality of blood spots (size, smeared blood or overlapping spots) on the DBS filter card (Figure 2-7), which provided information on the amount of blood available for analyses for each barcoded sample. In some cases, this information was used to disentangle erroneously doubled barcode numbers.
 - o Imprints of the blood spots on the filter-paper flap covering the blood-spot circles, a sign that the card was closed while the blood was still wet. This indicated a too brief drying time before the DBS card was packed for shipment.
 - o The dates of blood collection (written on the DBS card) and of final freezer storage; these dates were used to calculate shipment times by sample and country-specific averages (Figure 2-6).

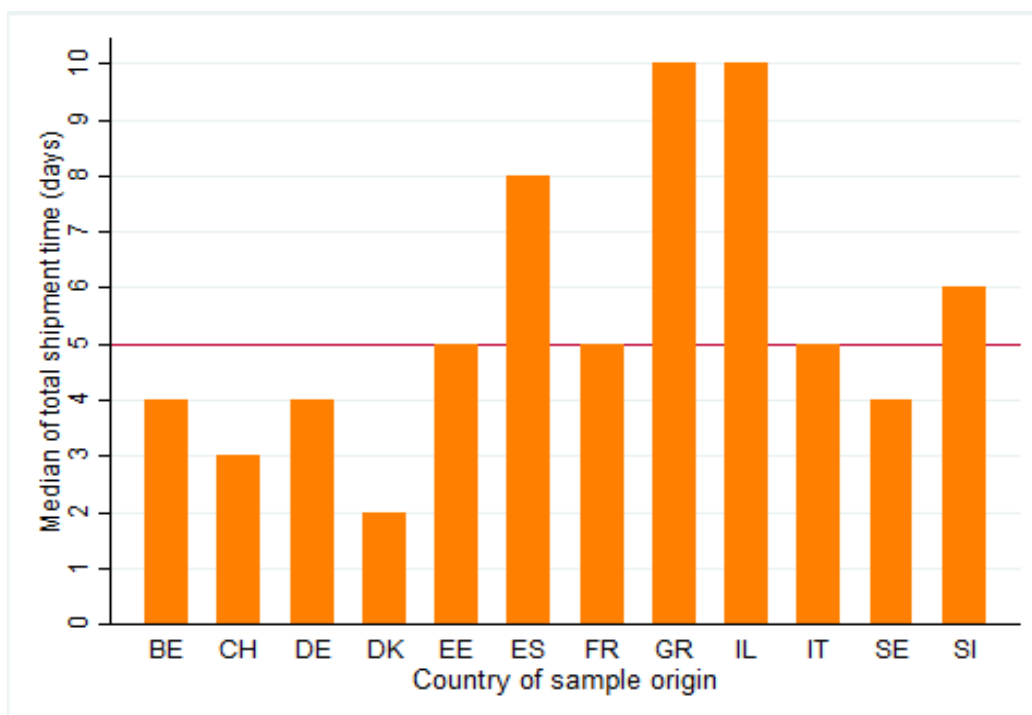


Figure 2-6: Median shipment time of DBS samples. The Y-axis shows the time (in days), which elapsed between the date the blood was taken (date on the DBS cards) and the date the card had arrived at the biobank in Odense. The horizontal line shows the median shipment time of five days for all countries (Figure taken from Börsch-Supan, M et al., 2020).

Figure 2-6 shows the time (in days) which had elapsed between blood sampling (date written on the DBS cards) and arrival of the sample at the biobank in Odense. Shipping time reflects interviewer performance (how quickly did the interviewer deliver the DBS card to the postal office) and the functioning of the national postal systems (opening hours of postal offices, availability of post-boxes and frequency of being emptied, postal strikes, international delivery). For the summer months, interviewers were instructed to deliver the envelopes with the blood samples directly to a post office instead of dropping them into heat- or sun-exposed mailboxes.

As mentioned in Sections 2.1.4 and 2.1.5, the interviewers were asked to enter the current local outside temperature at the time of the blood-collection into the CAPI. To check the usability of this information and to gain some insight into the temperature exposure of the DBS cards during shipment, we had equipped the SHARE Wave 6 pretest DBS kits with temperature indicators (Timestrip Label 30 and 38°C; Timestrip UK Ltd, Cambridge, UK) to be activated by the interviewers and then added to the PE-bag with the DBS card before shipment to the biobank; the Israeli country team acquired temperature indicators during the SHARE Wave 6 main survey DBS collection for temperature monitoring of the DBS samples starting in June 2015 (Shockwatch WarmMark® Long-Run indicator showing temperature exposure of $\geq 31^{\circ}\text{C}$ for up to 168 hrs.) Both kinds of strips had to be activated by the interviewers. The temperature-checking experiments are described and evaluated in 2.1.4 above.

The described sample-quality checks provided us with information about the total number of blood spots per card, the number of blood spots that filled the pre-printed circles (full-size blood spots), whether any blood spots were smeared (by touching the card), and the presence of overlapping DBS (multiple drops on the same spot). Figure 2-7 shows examples of DBS samples of various qualities.

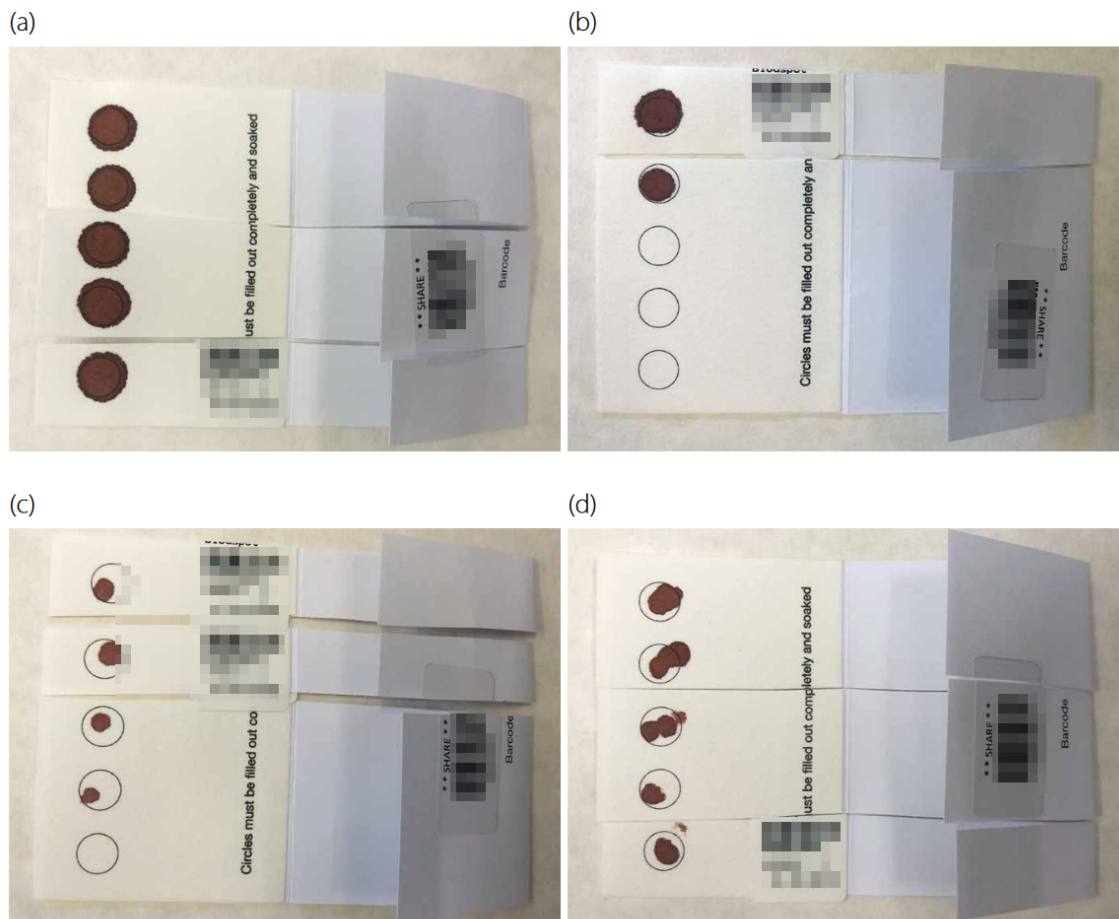


Figure 2-7: Quality differences of DBS filter cards received at the Biobank. Filter cards arrived with different amounts of blood/number of blood-filled pre-printed circles and in varying quality. (a) This card shows spots of best quality; the amount will allow a large number of analyses. (b) Fine quality but only two spots. (c) Minimal amounts of blood (all spots not even roughly fill the circle). (d) More blood than on the card (c) but the card was touched in circles 1, 2, 3, and drops are overlapping in circles 3, 4, 5; this card is unsuitable for biomarker analyses except for HbA1c (see 3.2). After arrival and ahead of freezer storage, cards were cut into fractions. Different fractions were labelled with the same barcode and stored for security reasons in separate freezers. In addition, for punching and analyses the necessary amount of blood (one or two fractions) were used without thawing the entire DBS sample (Figure taken from Börsch-Supan, M et al., 2020)

2.2.4 Fieldwork conditions

The monitoring activities described in the previous subsection yielded a set of fieldwork variables that are crucial to correct for adverse fieldwork conditions, see Section 5, Table 5.4. Implausible values of the fieldwork variables given the fieldwork circumstances were truncated at the lower and upper end.

Drying time (minutes): Interviewers were instructed to let the blood sample dry until the end of the interview but at least 15 minutes. This followed the protocol that was used at that time by the HRS (Crimmins et al. 2015). As an estimation of the drying time, we used the interview length between the blood spot module, programmed in CAPI (Computer-assisted personal interviewing), and the end of the interview (or the time until the end of the interview session if the interview was conducted in more than one session on several days).

Mean=20.7 minutes, Std.Dev=9.7, Range=5 - 40

Shipment time (days) was measured as the time between sample collection (date noted on card) and arrival at the biobank, where the card with the DBS sample was stored frozen until laboratory analysis.

Mean=5.5 days, Std.Dev=3.8, Range=1 - 20

Outside temperature (°C) was estimated by the interviewer at the time of blood collection. The estimate was entered into the CAPI. During Wave 6 pretest, we double-checked these estimates by using long-term temperature trackers and found no substantive deviations (Börsch-Supan, M. et al. 2020).

Mean=14.0 °C, Std.Dev=8.2, Range=0 - 35

Humidity protection during shipment: A small pouch with desiccant was added to the plastic bag containing the DBS filter card. We had chosen molecular sieve, which absorbs humidity faster than silica gel. The indicator for humidity protection consisted of two dummy variables: one for a not tightly closed plastic bag containing the DBS sample and one for a missing desiccant pouch inside the bag. Information on complete or missing components was registered for each sample on arrival at the biobank.

Open bag=0.11, Std.Dev=0.31, Range=0, 1

Desiccant missing=0.02, Std.Dev=0.13, Range=0, 1

Spot size (cm²) could not be measured during fieldwork. For the validation exercises described in Section 5, we created three precisely defined spot sizes in the validation experiment by pipetting 10, 30 and 60µl of blood. This corresponds to diameters of 6, 10 and 13.5 mm or areas of 30, 83 and 143 mm², respectively. Spot size of the field samples was determined by photometric methods as described in Section 5.3.

2.2.5 Daily and seasonal variation of blood biomarkers; the climate region; the impact of medication and food consumption

Apart from environmental impact, shipment or influence on the blood biomarkers caused by the interviewer (factors, which may be observed by monitoring) daily or seasonal variations of blood biomarker levels have to be considered. Seasonal variations in ambient light and/or temperature impact many blood biomarkers. Sunlight plays an important role for the body to generate sufficiently high levels

of Vitamin D, for example; values increase with the elevated exposure to the sun in summer. Brain-derived neurotrophic factor (BDNF; see 4.1.2) also strongly correlates with the increase of ambient sunlight. Levels for BDNF are lower between January and May and higher from June to December (Molendijk et al. 2012). A1c and plasma lipid values for example vary with the seasons as the body reacts to intake of seasonal food and adjusts to warm and cold temperatures. A large part of the latter adjustment can be explained by hemodilution in summer (more plasma, resulting in lower concentration of red cells and plasma lipids) and hemoconcentration in winter (accordingly, less plasma by losing fluid to the tissues and increases in cells and solids), with resulting in lower or higher biomarker levels, respectively. This may have consequences for, e.g., rated as being anemic, jumping a risk category for cardiovascular events or treatment implications. Therefore, the consideration of the climate region is important as monthly temperatures differed greatly during the DBS collection period of SHARE between northern Europe and the Mediterranean countries. (Donahoo et al. 2000; Ockene et al. 2004; Nayha 2002; Letellier and Desjarlais 1982; Maguire and Edwards 2001; Tseng et. 2005; Rocha e Pinho Pereira et al. 2015). For the analysis of total hemoglobin values (THB) the altitude at which respondents live (high elevations) may have to be considered.

Blood pressure varies with the time of the day or just concluded activities of the respondent. Food intake plays an important role for blood pressure, but mainly affects the level of blood sugar and lipids. Glucose blood levels increase soon after sugar consumption, and cholesterol and triglyceride levels increase after fat intake. The time of the blood collection during a survey should be known and if the blood sample was taken from a fasting (no food intake for 8-12 hrs.) or non-fasting participant. Fasting also applies to smoking and alcohol consumption; the latter has short and long-term impacts on many blood biomarkers. Medication, especially regularly taken drugs, may impact the biomarker of interest. Statins for example, are prescribed to bring down high total cholesterol levels, but they also lower the blood level of CRP. Other medication may raise the levels of those and other biomarkers (e.g., Herink & Ito, 2018). The hydration state of a person will influence the concentration of markers in the blood; this may be an important consideration, when looking at blood levels of very old people, who may have impaired thirst sensation, neglect drinking enough fluid or lower the fluid intake due to urinary incontinence problems.

Not only daily but also seasonal food consumption accounts for further fluctuation of the blood biomarker levels – which is important for a survey which collects blood material throughout a year. Festive and fasting seasons in the general population and among ethnic and religious groups or for individuals can increase or decrease blood biomarker values.

2.2.6 Lab-related impact

Last not least, all chemical assays show inter- and intraassay variation, and any laboratory produces its own lab-specific values: it uses particular analyzers, chemical brands and batches, and lab protocols. Values of biomarker differ between laboratories and the lab-

specific reference ranges should be used. Even different instruments in the same lab will produce different results; see Section 5.2, Crimmins et al., 2014 and Kim et al., 2023.

3 The DBS sample: Eligibility, consent, analyzability, and DBS-specific weights

3.1 Consent of eligible respondents

Among all individuals participating in the SHARE W6 interview, 40,049 of 52,374 were eligible for DBS collection. The target participants for collecting blood samples were the entire group of eligible individuals of the SHARE-W6 survey. Among the eligible individuals in each country, we issued DBS consent forms and asked if they would like to participate in DBS collection. We collected blood samples only from those who had signed a consent form. In France only four districts - not the entire country - were eligible for DBS collection. Therefore, we defined eligibility in France based on the actual eligible subsample. Participation in the DBS collection among eligible respondents was voluntary. We distinguished non-participation for medical reasons and lack of consent for any other reason.

Therefore, we calculated two different consent rates (per country and per interviewer):

$$\text{consent rate}_1 = \frac{N_{\text{consent given}}}{N_{\text{DBS eligible}} - N_{\text{prevented by medical reasons}}} \quad (1)$$

$$\text{consent rate}_2 = \frac{N_{\text{consent given}}}{N_{\text{DBS eligible}}} \quad (2)$$

Consent rate₁ is based on eligible DBS respondents who were able to give consent on their own but *excludes* those who mentioned medical reasons that prevented them from participating in blood collection. Consent rate₂ *includes* those respondents in the denominator (i.e., respondents who abstained from participation due to medical reasons). Consent rate₁ is therefore always higher than consent rate₂ (see orange vs. dark red bars in Figure 3-1). We instructed interviewers not to cajole respondents who had declined due to medical reasons to provide DBS; therefore, consent rate₁ may more precisely reflect the success of interviewers in gaining consent. However, consent rate₂ is the effective rate of consent among eligible interviews and, hence, the outcome of interest for the project. Not gaining consent from all respondents always bears the risk of biases caused by systematic differences between consenting and non-consenting respondents (Sakshaug, Couper & Ofstedal, 2010).

Table 3-1 shows the number of eligible and consenting individuals (according to consent rate₂, called “responses”).

Table 3-1: Number of eligible individuals, consenters and finally assayed blood samples in each participating country

	SHARE W6 completed interviews	Eligible for DBS collection	Consented to DBS collection	Consent rate 2 (consented/eligible)	Number of DBSS assayed	Assay rate of DBS consented	Assay rate of W6 interviewed
Belgium	5,815	4,743	3,729	78.62 %	3,118	83.61%	53.61%
Switzerland	2,803	2,797	2,130	76.15 %	1,905	89.44%	67.96%
Germany	4,412	4,408	3,163	71.76 %	2,867	90.64%	64.98%
Denmark	3,733	3,483	2,864	82.23 %	2,705	94.45%	72.48%
Estonia	5,638	4,991	3,684	73.79 %	3,450	93.67%	61.19%
Spain	5,623	3,041	1,871	61.53 %	1,379	73.70%	24.52%
France	3,947	884	569	64.37 %	410	72.06%	10.38%
Greece	4,928	2,259	758	33.47 %	650	85.75%	13.18%
Israel	2,035	2,031	1,072	52.78 %	887	82.74%	43.58%
Italy	5,311	4,054	2,223	54.83 %	1,733	77.96%	32.63%
Sweden	3,906	3,895	3,054	78.41 %	2,543	83.27%	65.10%
Slovenia	4,223	2,897	2,256	77.87 %	2,062	91.40%	48.82%
Total	52,374	39,483	27,373	67.18 %	23,709	84.89%	46.53%

DBSS: dried blood spot sample

The cooperation of all countries participating in the DBS collection is illustrated in Figure 3-1. For each country, consent rate₁ (orange bars) as well as consent rate₂ (dark red bars) are shown. The horizontal line is the mean of the dark bars over all countries with a consent rate₂ of 67.18%. This overall consent rate was very high, demonstrating that respondents' and interviewers' acceptance of this new measurement was very good. More than 80% consent was reached by BE, CH, DK, EE, SE, and SI when adjusting for medical reasons which prevented participation. The calculation of the French consent rates is based on the small, selected DBS-eligible sample of French respondents⁶.

The final consent rates in Figure 3-1, however, do not describe the large variation between interviewers. This is depicted in the box-plot diagrams of Figure 3-2. Some countries have little variation while others, predominantly in southern Europe, have very large variation in interviewer-specific consent rates. The small variation among the Danish interviewers for example may come from their experience as most had collected DBS previous to the SHARE blood collection. Their confidence with the procedure may have helped to confer a positive attitude to the respondents.

⁶ As described in Section 2.2.1, French panel households in only four districts were eligible for DBS collection. Nevertheless, a certain number of non-eligible respondents were asked and gave consent. The calculation of consent rates in France is based on the actually eligible subsample, while e.g., French numbers of available DBS samples refer to all received DBS cards.

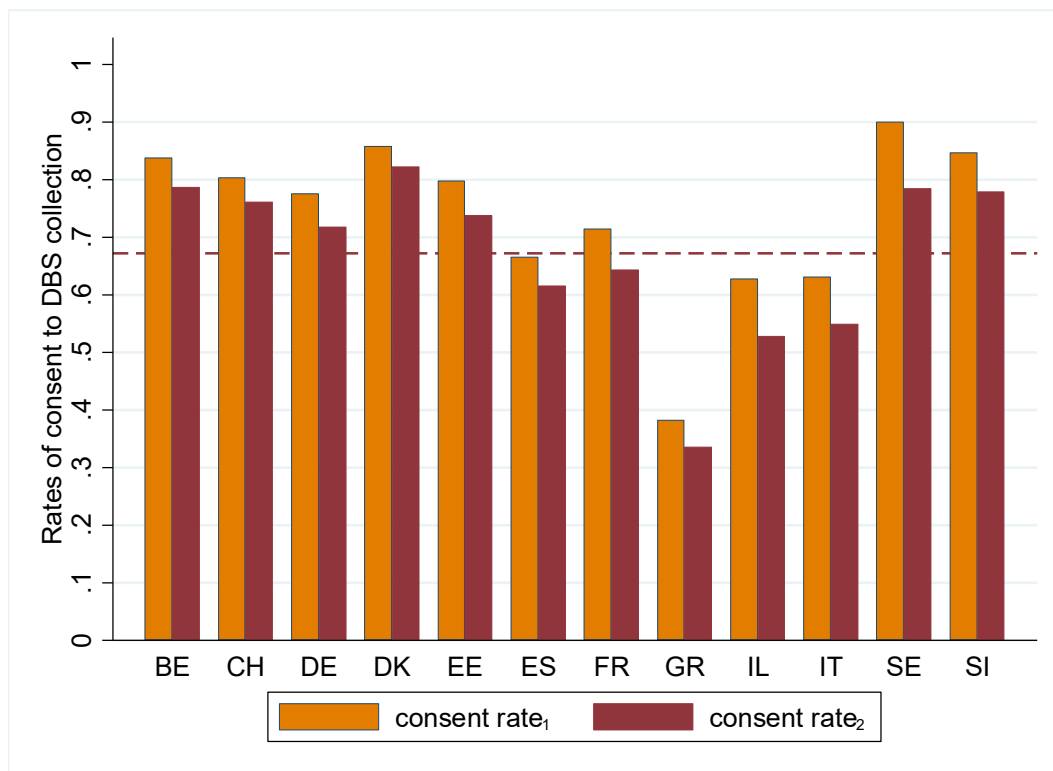


Figure 3-1: Final rates of consent to DBS collection in the participating countries. Orange bars: consent rate₁, see equation (1); dark red bars: consent rate₂, see equation (2); dashed line: overall consent rate of 67.18%. (Figure recalculated according to Börsch-Supan, M et al., 2020)

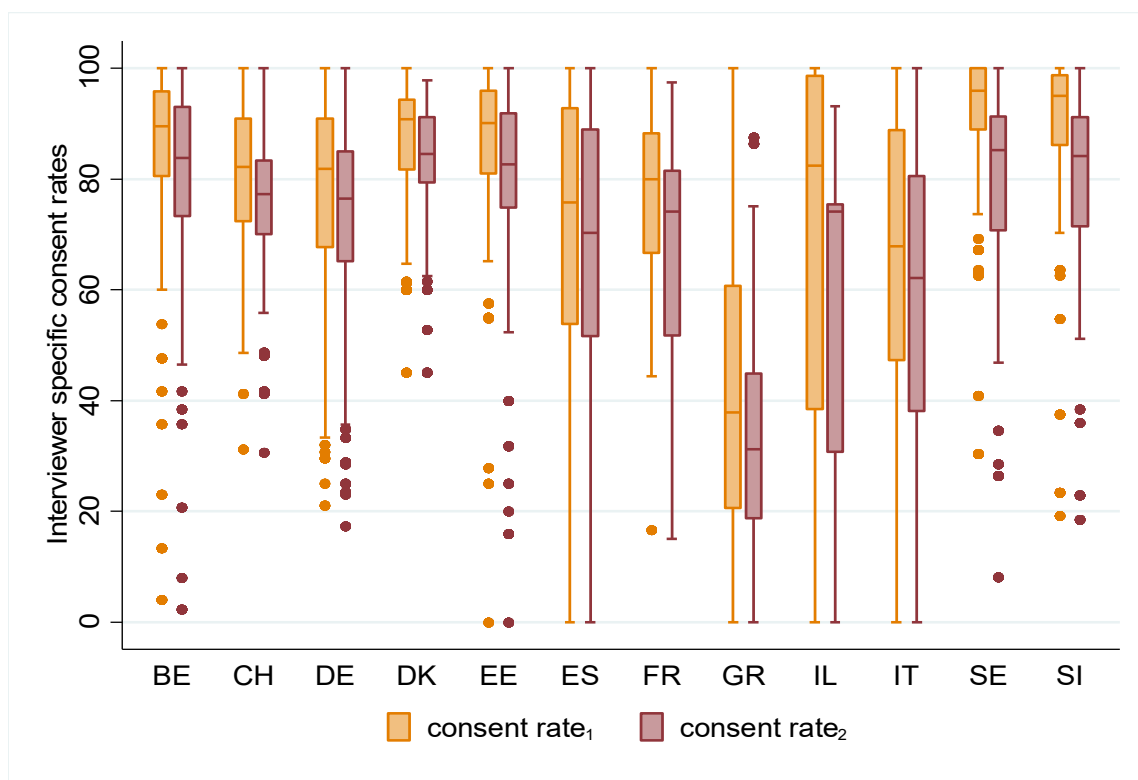


Figure 3-2: Box-plot diagram of consent to DBS collection in the participating countries. The diagram illustrates the strong involvement of interviewer performance on consent giving. Variation of consent per interviewer (interviewer performance) in southern Europe is much larger than in the North (Figure taken from Börsch-Supan, M et al., 2020)

3.2 Quality of the collected blood samples – recordable shortcomings and invisible damage

While almost 27,200 DBS cards were collected during the main survey of SHARE Wave 6, not all of these samples were complete or could be used for final laboratory assays. Three major criteria had to be fulfilled for a SHARE DBS sample to be counted as complete: 1) the consent for blood collection had to be given in writing and the signed consent form received at the national survey agency; 2) the DBS card was received at the biobank; 3) a CAPI interview was available. In addition, all three data sources had to be linkable by the same barcode. After fieldwork had concluded in all countries, the DBS samples were scrutinized for completeness according to the three criteria above. Incomplete samples were excluded from laboratory analyses.

Table 3-2 shows the numbers of received, complete and usable DBS samples per country.

In addition to being complete, a SHARE DBS sample had to meet certain quality requirements for laboratory analysis. Earlier, in 2.2.3, we mentioned the quality features and the amount of blood (number and size of spots) that the biobank team recorded for each DBS card arriving in Odense. The vast majority of the 26,705 samples were complete according to the three criteria but 354 of those had no blood⁷, leaving 26,351 complete samples with blood (columns 2 and 3, Table 3-2).

Further, we excluded cards of minor quality with too small or overlapping or smeared blood spots. This explains the 2528-samples difference between DBS counts (column 3 and 4, Table 3-2); thus, 23,823 DBS samples (column 4) had at least one good blood spot and were usable for analyses.

The card of a single SHARE respondent could contain a varying number of blood spots of the same or different sizes, optimally five completely filled pre-printed circles. On many cards, less blood was collected and a substantial number of the usable DBS cards had limited amounts of blood (i.e., $\leq$ one full circle, see Figure 2-6). 452 of those 2528 samples could be used, nevertheless. Overlapping or smaller spots may be analyzed for HbA1c as that measure is the ratio of glycosylated versus total hemoglobin and is thereby independent of the blood volume contained in a DBS (which is undefined in smeared or overlapping spots). Therefore, a total of 24,275 samples were prepared for laboratory analyses and data was obtained for at least one biomarker from 23,709.

Several of the visible shortcomings of the DBS cards, which were recorded on arrival at the biobank, included suboptimal amounts of blood on some DBS cards that, in some cases, limited the analyses of biomarkers. In worst cases, entire DBS samples were excluded from analyses due to smearing, overlapping drops, blood touching the DBS card cover, overly small spots, too brief drying time, or missing desiccant (see 2.2.3). These issues could have been avoided had better care been taken

⁷ Complete samples with no blood resulted, when the interviewer attempted to take blood by pricking the finger but the respondent did not bleed. This happened significantly more often with increasing age of the respondent. This is a no-blood result in contrast to a refusal, since consent was given.

by the interviewer, although the field situation and the interplay with the respondent may have influenced the outcome.

Much harder to control, and not readily identified, are “invisible” impacts on DBS samples, impacts that are seen later in the measured biomarker values. The biobank staff recorded possible sources of those impacts: missing desiccant, which means missing humidity protection; long shipment times indicating extended exposure to, e.g., high temperatures in summer. Environmental conditions could also impact DBS samples, such as high ambient temperatures and humidity during sample collection and/or during shipment, as well as postal service unreliability leading to long shipment times.

Table 3-2: Overview of DBS samples collected and finally assayed by country

	Number of DBS cards received at Biobank	Number* of complete** DBS samples	Number of complete DBS samples with blood	Complete DBS samples with at least one good-quality blood spot	DBS Assayed for at least one blood biomarker
Belgium	3690	3604	3484	3166	3,118
Switzerland	2132	2095	2051	1953	1,905
Germany	3147	3124	3095	2872	2,867
Denmark	2861	2842	2833	2715	2,705
Estonia	3683	3670	3652	3285	3,450
Spain	1827	1771	1754	1462	1,379
France	578	552	507	429	410
Greece	758	736	735	580	650
Israel	1072	1030	1030	821	887
Italy	2175	2153	2121	1839	1,733
Sweden	3013	2907	2893	2600	2,543
Slovenia	2242	2221	2196	2101	2,062
Total	27 178	26 705	26 351	23 823	23,709

Column 1: total number of DBS samples received at the biobank

Column 2: *Including 354 cards with no blood; respondent had consented to blood collection, but interviewer did not manage to collect blood; ** number of samples fulfilling the three criteria for completeness

Column 3: number of cards received in Odense with any amount of blood

Column 4: number of samples with at least one good-quality blood spot for analysis (as reported by the biobank staff upon receipt of the sample).

Column 5: number of samples with at least one available biomarker in SHARE data set.

Source: SHARE data release 7.0.0 and internal data on DBS samples

3.3 Constructing DBS-specific weights

The lack of eligibility and consent, plus the failure to produce a DBS sample that could be analyzed for at least one biomarker (see Figure 3-3) may lead to sample selectivity, i.e., the risk of bias due to systematic differences between eligible and non-eligible respondents, consenters and non-consenters, and analyzed and non-analyzed DBS samples. To account for this selection, we computed DBS weights, which analysts should use when they analyze the DBS data. The DBS weights are the product of calibrated cross-sectional individual weights (*cciw_w6*) from the main SHARE survey and a correction factor for the lack of eligibility, consent or analyzability.

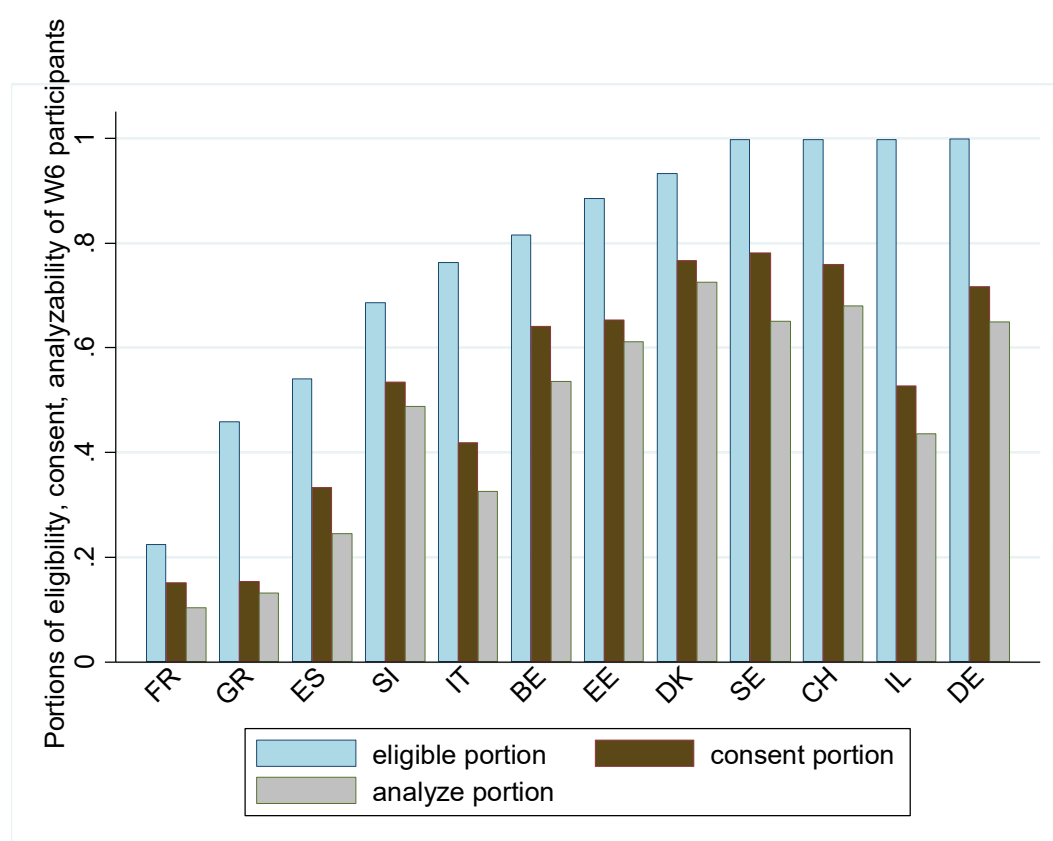


Figure 3-3: The proportion of DBS eligibility, consent and analyzability among SHARE W6 completed interviews in the participating countries.

For this correction factor, we use an inverse probability weighting method. We estimate a logistic model to predict the probability of having at least one biomarker value among all SHARE W6 participants in these 12 countries who completed interviews. The predictor variables for this model are demographic characteristics (gender, age, education) and country fixed effects, specified as dummy variables with K categories. Moreover, we included the interaction between education and gender. The dependent variable DBS_{ij} indicates that individual i in country j has at least one biomarker value.

$$\Pr(DBS_{ij}) = \text{logit} \left(\beta_0 + \beta_1 \text{Gender}_{ij} + \sum_{k=1..3} \beta_{2k} \text{Age}_{ijk} + \sum_{k=1..2} (\beta_{3k} \text{Edu}_{ijk} + \beta_{4k} \text{Gender}_{ij} \times \text{Edu}_{ijk}) + \sum_{k=1..11} \beta_{5k} \text{Country}_{jk} \right)$$

The regression was weighted with the baseline calibrated cross-sectional individual weights *cciw_w6* to match the population distribution of the predictors. Table 3-3 shows, that gender and education are not significantly affecting the DBS participation. Compared with respondents aged 50-59, respondents aged 60-79 are more likely to provide a DBS sample. Compared with Germany, people in Denmark and Switzerland are more willing to provide blood samples and also for samples to meet the analysis criteria. All other countries have a lower probability to provide analyzable DBS when compared to Germany.

The logit regression in Table 3-3 generates the predicted probability that an individual has at least one biomarker value. We then multiplied the calibrated cross-sectional individual weights *cciw_w6* from SHARE Wave 6 with the inverse predicted probability:

$$dbs_weight_{ij} = cciw_w6_{ij} \times \frac{1}{\Pr(\overline{DBS}_{ij})}$$

The actual *cciw_dbs_{ij}* were calculated by dividing this procedure into three steps: probability of respondent eligibility, probability of respondent giving consent given eligibility, and probability of having at least one biomarker value given that the respondent had given consent (see Section 3.2).

Table 3-3: Logit regression for inverse probability weights

DBS analyzed	Coefficient.	Std.Err.	[95% Confidence	Interval]
Gender				
Female (baseline)				
Male	-0.08	0.06	-0.197	0.037
Age				
50 – 59 (baseline)				
60 – 69	0.295***	0.049	0.199	0.391
70 - 79	0.439***	0.053	0.336	0.542
>80	0.003	0.064	-0.123	0.129
Education				
Low (baseline)				
Medium	0.113	0.086	-0.056	0.283
High	-0.158	0.101	-0.356	0.040
gender* education				
male*medium_edu	-0.111*	0.067	-0.241	0.020
male*high_edu	-0.002	0.079	-0.158	0.153
Country				
Germany (baseline)				
Sweden	-0.068	0.058	-0.181	0.045
Spain	-0.984***	0.077	-1.135	-0.832
Italy	-1.522***	0.058	-1.636	-1.409
France	-2.872***	0.068	-3.005	-2.740
Denmark	0.253***	0.054	0.148	0.359
Greece	-2.638***	0.06	-2.756	-2.520
Switzerland	0.098*	0.059	-0.017	0.214
Belgium	-0.509***	0.052	-0.612	-0.406
Israel	-1.12***	0.085	-1.287	-0.954
Slovenia	-0.621***	0.053	-0.725	-0.516
Estonia	-0.285***	0.047	-0.377	-0.192
Constant	0.552***	0.07	0.415	0.689
Observations	51,396			
R squared	0.1618			

Notes: Among all individuals participating in the SHARE W6 interview, 51,396 of 52,374 had information on calibrated cross-sectional individual weights (cciw_w6). The DBS-specific weights were calculated on this basis. Overall, 23,709 of the 51,396 samples have at least one assayed blood biomarker value. *** $p < .01$, ** $p < .05$, * $p < .1$

4 Assaying DBS samples for blood biomarkers of age-related conditions

SHARE chose to assay the DBS samples for 17 blood biomarkers (Börsch-Supan, M & Andersen-Ranberg, Chapter 6.1, in: Malter & Börsch-Supan, A. (Eds.), 2017). Priority in the selection process was given to relevance in evidence of diseases and health conditions common at older age, such as cardiovascular disease (CVD), diabetes and cognitive decline. Additional criteria were the existence of suitable assays of DBS (i.e., reagents and procedures to detect the biomarker) and the availability of comparative population values from (inter)national health registers (e.g., Eurohealth, WHO, the Robert-Koch-Institute in Germany, UK Biobank) and other population surveys, including HRS and its sister studies. Biomarker selection was also guided by the ability to share the same DBS eluent across multiple analyses, thereby limiting the amount of blood required.

Two laboratories, the Staten Serum Institut (SSI) in Copenhagen, Denmark, and the Department of Laboratory Medicine and Pathology, University of Washington (UW) in Seattle, USA, were entrusted with the DBS analyses. Both laboratories have the expertise to analyze DBS samples for various blood biomarkers and are able to handle such a huge number of samples. The SHARE DBS samples were apportioned among the two laboratories into two sets of biomarkers according to the laboratory's expertise in analyzing certain biomarkers (see 4.2.1). Admittedly, splitting DBS for analyses of different blood biomarkers in two different labs on two continents posed a challenge.

4.1 The biomarkers and aging

4.1.1 Routine blood biomarkers - assayed at the Department of Laboratory Medicine and Pathology in Seattle

The first dataset contains seven biomarkers measured in routine blood analyses. For these biomarkers, reference values from venous blood samples are well established.

Glycosylated hemoglobin (HbA1c or A1c, in %) is a biomarker for diabetes. An excess of sugar molecules in the blood irreversibly bind to hemoglobin. The so-formed A1c signals longstanding (the last 2-3 months) and chronically high levels of blood sugar (e.g., Florkowski, 2013).

High density lipoprotein cholesterol, (total) cholesterol, and triglycerides (HDL, CHO, TRG each in mg/dL) are molecules of the lipid panel and essential for participants in lipid metabolism, such as serving as building blocks for cells and transport molecules for lipids. Imbalances between the components of the lipid panel can lead to various diseases of the cardiovascular system (e.g., North & Sinclair, 2012; Upadhyay, 2015).

Cystatin C (CysC) is a biomarker of kidney function and is a risk factor for elevated morbidity and mortality. CysC, although a measure of the clearance of waste products from blood, is also a CVD

risk factor. Those with elevated CysC levels have been shown to be at highest risk for CVD, even when kidney dysfunction is mild; those with the highest levels of CysC are older and have hypertension, dyslipidemia, high BMI, and higher levels of C-reactive protein (e.g., Levin & Lan, 2016; Sarnak, Katz et al., 2008; Yaffe, Lindquist et al., 2008).

C-reactive protein (CRP) marks the general level of inflammation in the body caused by acute infections or chronic diseases. Inflammatory processes are involved in CVD, diabetes, chronic kidney disease, obesity, and cognitive decline (e.g., Ridker 2007; Lassale, Batty et al., 2019; Tang, Fung et al., 2017).

Total hemoglobin (THB) is a biomarker associated with diseases of the hematopoietic system (e.g., anemia, a decrease in red blood cells or hemoglobin, thereby lowering the ability to carry oxygen in the blood. Anemia may arise from loss of blood, pathological removal of blood cells, chronic inflammatory diseases, kidney disease, wasting diseases (e.g. cancers) and more (e.g., Goodnough & Schrier, 2014).

4.1.2 Biomarkers indicating risk of cardiovascular disease and cognitive decline – analyses at Staten Serum Institute (SSI) in Copenhagen

The second set of biomarkers is more innovative and the introduction of cytokine analyses from DBS to assess chronic inflammatory status, CVD, and cognitive status in a survey of aging is novel (Koelman, Pivovarova-Ramich et al., 2019; Rea, Gibson et al., 2018). Five biomarkers in this set are cytokines, small blood-borne proteins prominently involved in inflammatory processes; in addition, we analyzed the blood levels of three growth factors and two apolipoproteins. Cytokines are not routinely analyzed in clinical blood samples, but in research such measurements offer the opportunity to more closely investigate low-level chronic inflammation, such as in atherosclerosis, to fine-tune the search for the risk of CVD among the survey respondents beyond blood lipid values (such as cholesterol), and to biologically support the results of objective cognitive testing in the survey questionnaire. The selection and importance of the 10 biomarkers are described by Borbye-Lorenzen and Börsch-Supan (2019).

The five proteins *IL-8*, *IL-16*, *IL-18*, *IL-12/23p40*, and *MCP-1* indicate inflammation occurring in the body; all molecules are pro-inflammatory biomarkers but act in different tissues. Elevated blood levels of either one or several of these cytokines could better inform us about the type of inflammation and its degree; they may confirm the report of, for example, CVD or atherosclerosis by a respondent or point to yet undetected inflammation, cancer and/or (onset of) cognitive impairment (e.g., Brosseron, Krauthausen et al., 2014; Saleem, Herrmann et al., 2015; Scarabino, Peconi et al., 2020).

Vascular epithelial growth factor (VEGF) and *epidermal growth factor (EGF)* are proteins involved in angiogenesis and normal as well as pathologic cell and tissue growth. They participate in healing (e.g., in blood vessels) but may also indicate proliferating tissue in the case of cancer. In addition,

studies suggest that angiogenesis (VEGF) may be involved in the onset process of cognitive impairment (Huang, Jia, & Liu, 2013; Lange, Storkebaum et al., 2016).

Brain-derived neurotrophic factor (BDNF), the third growth factor, acts on neurons of the central and peripheral nervous system, supports the survival of neurons and encourages growth of neuronal tissue. Reports in the literature describe BDNF involvement in learning and memory. Its level increases with exercise and social embeddedness. In turn, the BDNF blood level is lowered in cognitive decline (e.g., Nascimento, Pereira et al., 2015; Salinas, 2016). Others relate BDNF to the progression of CVD (e.g., Bahls, Könnemann et al., 2019). As BDNF does not cross the blood-brain barrier, we need to stress that we are measuring the blood level of peripherally expressed BDNF and higher, lower, or unchanged circulating BDNF levels have all been described in AD patients compared to healthy controls (Baliatti, Giuli & Conti, 2018). In addition, BDNF shows a strong seasonal variation and correlates with the amount of ambient sunlight (Molendijk et al., 2012).

Apolipoprotein E type 4 (ApoE4) belongs to a class of proteins involved in the metabolism of lipids in the body and the brain, and is a component of the lipid panel. ApoE mediates the cholesterol metabolism, transports cholesterol to neurons and is the principal cholesterol carrier in the brain. ApoE4 is important in cardiovascular diseases. Carrying the *ApoE-e4* (allele) influences the lipid/cholesterol metabolism, increases the risk for cognitive decline and dementia, and lowers the age of AD onset (e.g., Corder et al., 1993; Strittmatter et al., 1996; Mooijaart, Berbee et al., 2006; Rasmussen, Tybjaerg-Hansen et al., 2015; Safieh, Korczyn, & Michaelson, 2019).

Clusterin (APOJ) has the general vital function in the organism to clear misfolded proteins and cell debris; through this function, clusterin is involved in aging and many diseases related to oxidative stress, including neurodegenerative and inflammatory diseases and cancers (e.g., Baralla, Sotgiu et al., 2015; Hsu, Lee et al., 2017; Jongbloed, van Dijk et al., 2015).

It is needless to say that we are investigating the blood levels of the peripherally expressed Apolipoproteins E and J. The analyses result of the above described biomarkers assayed at SSI will be released separately from the routine biomarkers (see 4.1.1) and together with their own assay description and validation

4.2 Analyses of the SHARE DBS samples

4.2.1 Planning blood-biomarker analyses and preparing the samples

For the analyses of a single biomarker only a small amount of a blood is needed, much less than one full-size DBS filling the pre-printed circle. This requires that the original DBSs collected during the survey has to be further divided, which is done by punching small discs (the punches) with a diameter of 3.2 mm (1/8 inch) from the blood spots (see Figure 4-1, and Börsch-Supan, M, Friedel, & Weiss, Chapter 6.4, in: Malter & Börsch-Supan, A (Eds.), 2017). The punching is followed by elution (extraction) of the dried blood in the disk with a buffer to yield a liquid sample.

Given the differences in quantity and quality between the individual DBS samples collected from the respondents during fieldwork (see Figure 2-7), we were aware that we would not be able to perform analyses for all 17 biomarkers from all DBS samples as good-quality DBS of sufficient size (not too small a diameter) is required. The Department of Laboratory Medicine and Pathology at UW developed SHARE-specific assays (for the first seven routine blood biomarkers described in 4.1.1), to allow the analyses of multiple biomarkers from one, resp. two punches; usually one punch is eluted for the analysis of one biomarker. Thanks to highly sensitive methods like DBS-LC-MS/MS (liquid chromatography–mass spectrometry), such small analyte amounts can be measured, but the methods also have their limitations (Li & Tse, 2010). A multiplex immunoassay, which was used to analyze the cytokines, is another very sensitive and specific method.

We spent considerable time carefully planning how to gain as much blood biomarker information as possible from all cards, and at the same time to cover the list of biomarkers which we intended to analyze from as many samples as possible. For the analyses of all 17 biomarkers described above, five 3.2-mm punches from each DBS sample, not necessarily from one spot, are required (Tables 4-1 and 4-2; Figure 4-1). The 17 biomarkers were divided into three biomarker groups A, B; C. Biomarkers in a group can be eluted together from the punched blood disc(s) as listed in Table 4-1; then, the analyses for the individual biomarker follow from aliquots of the eluent.

Table 4-1: Biomarkers/biomarker groups analyzed from SHARE DBS

Marker Group	Individual Biomarkers	Number of 3.2 mm discs needed for analyses
A-markers - or -	- HbA1c - total Hemoglobin (THB) - HDL-Cholesterol	2
HbA1c-only (in case of limited material)	- HbA1c	1 (can be taken from an overlapping or smeared blood spot)
B-markers	- total Cholesterol (CHO) - Triglycerides (TRG) - C-reactive protein (CRP) - Cystatin C (CysC)	1
C-markers	IL-16, IL-8, IL-12/23, IL-18, MCP1; VEGF, EGF, BDNF, Clusterin, ApoE4-protein.	2

A- and B-marker were analyzed at UW in Seattle; C-marker analysis was assigned to SSI in Copenhagen (Table taken from Börsch-Supan et al., 2020).

The assays of the seven “routine blood biomarkers” at UW required three punches: two for the analyses of the biomarkers HbA1c, HDL, and THB (A-marker group) and only one for TC, TRG, CRP, and CysC (B-marker group). Analyses were conducted according to published techniques, but adapted by UW for the SHARE-specific analyses of multiple biomarkers from the eluent from only one or two punches: HbA1c (Egier, Keys et al., 2011), THB (Frenchik, Tsonev, & McFaul, 2004), TC (Lakshmy, Gupta et al., 2010; Lakshmy, Mathur et al., 2012), HDL (Arranz-Peña, Tasende-Mata, & Martin-Gil, 1998; Huang, Kao, & Tsai, 1997), TRG / TG (Lakshmy, Gupta et al., 2010; Quraishi, Lakshmy et al., 2006), high-sensitivity CRP (Brindle, Fujita et al., 2010; McDade, Burhop, & Dohnal, 2004), and CysC (Vogl, 2014). The adapted assay protocols can be found in the Appendix

Table 4-2: Blood amount in a DBS sample and assignment for biomarker analyses

Possible number of punches per DBS sample	Punch 1	Punch 2	Punch 3	Punch 4	Punch 5
5 and more	C	C	A	A	B
4	C	C	A*	B	
3	A	A	B		
2	A*	B			
1	B or A* (depending on spot quality)				

With large enough samples (first row), the five punches necessary for the analyses of 17 biomarkers could be made. Limited amount of blood i.e., fewer or one DBS on a card and/or blood spots of small diameter also limit the number of punches. A choice for the biomarkers to be analyzed has to be made. Punching was conducted according to the scheme presented in this table. Therefore, not every DBS sample could be analyzed for all biomarkers (Table taken from Börsch-Supan, M et al., 2020).

The analyses of the second biomarker group at SSI required further two punches for the simultaneous analysis of 10 biomarkers. As cytokines and growth factors usually circulate in the body at very low concentrations, detecting them requires highly sensitive methods, such as immunoassays. Modern multiplex approaches analyze multiple biomarkers from very small amounts of blood, such as DBS (Borbye-Lorenzen and Börsch-Supan, 2019; Skogstrand, 2012; Skogstrand, Thorsen et al., 2005).

As mentioned above, the protocol for the SHARE multiplex immunoassays and the validation of these biomarkers for field conditions will be published later together with the release of the cytokine data. Here, we concentrate on the release of routine blood biomarkers analyzed at UW.

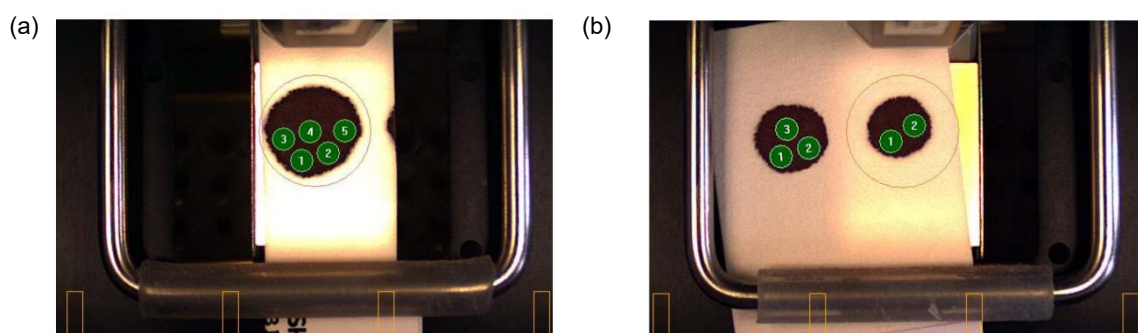


Figure 4-1: DBS cards in the puncher. Cards are turned over for aiming at the punch spot to place the punch at a fully soaked section of the DBS. While punching the DBS samples, the semi-automated robot punching system (Panthera Puncher 9, Perkin Elmer, Waltham, Ma, USA) used at SSI assigned the punch location and took a photo of the blood spot. This provided us with precise information of size and quality of all punched DBS samples. (a): Some, yet few, spots were large enough to gain all five punches needed for the SHARE assays from a single DBS. More frequently the punches for a DBS sample had to be taken from two or more (smaller) blood spots. (Figure taken from Börsch-Supan, M et al., 2020)

The subdivision – the punching – of the DBS samples (blood amount on an individual DBS filter card) was carried out at SSI in Copenhagen. In general: before the punching starts with a semi-automated robot punching system (Panthera Puncher 9, Perkin Elmer), the DBS cards have to be hand-sorted for the punching of the respective biomarker groups (see Table 4-2). Then, the sample is registered in the system by scanning the respondent-specific barcode. The card is manually positioned in the puncher; the 3.2mm disc is cut by the robot from the flipside⁸, and the punch drops into the well of a standard 96-deep-well microtiter plate (Figure 4-2). Each well of a microtiter plate contains one (for B-marker analysis), respectively two punches (for a- or C-marker analysis) out of the sample of a different respondent; the barcode scanning of the robot creates an electronic plate map and facilitates the later linkage of the analyses result to the interview data of a respondent. For the analyses of the A-markers, for example, two discs are cut and the wells contain two punches. Then the B- or C-marker punch(es) from the same DBS sample are cut into other plates.

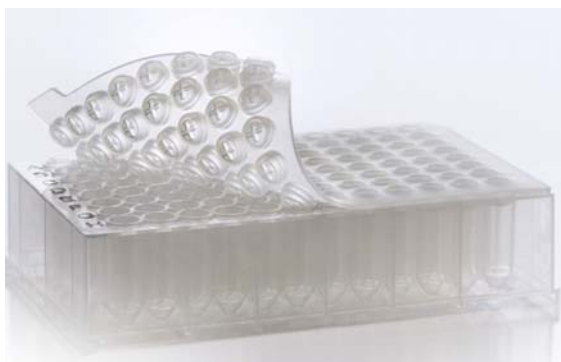


Figure 4-2: 96-well microtiter plate with cover.
(Figure taken from Börsch-Supan, M et al., 2020)

The procedure is repeated until 72 or 76 wells of a plate are filled (UW uses 24 and SSI 20 wells out of 96 for controls with known biomarker levels). Now the plate receives an identifying number and is sealed with a cover to prevent the discs from falling out or getting mixed during handling and future transport of the sample-punches to the laboratories.

The plates were frozen at -20°C until shipment of the A- and B-marker plates from Copenhagen to Seattle or in-house analyses of the C-markers at SSI.

The punching of all ca. 24,000 SHARE Wave 6 main survey DBS samples lasted from November 2016 till May 2018.

4.2.2 Laboratory analyses of the SHARE DBS samples

⁸ After falling onto the DBS card, the blood drop spreads laterally and vertically (soaking) into the paper. At the fringes of blood spot the vertical spread, the soaking, may not be as complete as in the middle, i.e., the diameter on the flipside may be slightly smaller than on top. Therefore, the filter card is turned and the punch is cut from the flipside to create a homogeneously soaked punch.

A random subsample of ca. 8000 samples was selected and assayed for A- and B-markers (listed in Table 4-1) in Seattle between October 2017 and May 2018, and the remaining ca. 16.000 samples from May 2020 till August 2021 (for assay descriptions see Appendix).

Analyses for all SHARE samples, which had been punched for C-markers (ca. 16,000) at SSI, were completed within three months (June to August 2018).

Table 4-3: Samples analyzed per biomarker group and analyses period

Marker groups	Batches	Period of Analyses
A-markers:	First: ca. 4200 samples Second: ca.10.600 samples	08/2017 – 02/2018 05/2020 – 07/2021
B-markers	First: ca. 7000 samples Second: ca.16.200 samples	11/2017 – 04/2018 09/2020 – 10/2020
C-markers	Ca. 16.000	02/2018 – 06/2018

5 Adjusting for field conditions, spot size, blood-specimen type, extended storage time, and change of analyser

Despite the advantages of collecting DBS for very large population surveys, especially in a cross-national context, DBS have unique limitations. The small quantities of blood collected per sample require the use of highly sensitive analytical instrumentation as described in the previous section. Different instruments may produce different results and hence data may need to be harmonized across platforms to permit comparisons between studies (Hu, Crimmins et al. 2023). Moreover, during sample collection in a non-clinical setting like during a population survey, DBS are inevitably exposed to varying fieldwork conditions such as differences in outside temperature, humidity, and shipping time (Hannon et al. 2013; Adam et al. 2011; Parsons et al. 2020). In addition, sample quality may be affected by abbreviated DBS drying times, missing humidity protection, or collection of smaller than optimal blood volumes. In fact, of the greater than 60,000 blood spots collected in SHARE, more than 57% had a diameter of less than 0.7 cm, which is considered suboptimal for biochemical analysis. Finally, it is necessary to convert the data obtained from the assays of the capillary blood eluted from the DBS into plasma, serum or blood values (standard equivalents) to allow results to be evaluated using metrics established for venous blood.

Börsch-Supan, A. et al. (2021) showed that environmental factors, as well as differences how DBS are collected by the interviewers, influence the quality of the samples and the therein-measured biomarker levels, although not all analytes are affected in the same way by these conditions. Moreover, they found that donor characteristics had no discernable influence on the relationship between environmental factors and biomarker values. These results are in accordance with the findings of the Health and Retirement Study (HRS, e.g., Crimmins et al. 2020) and other studies (Edelbroek et al. 2009; Freeman et al. 2018; Lim 2018). A major finding of those experiments was the importance of spot size.

There are many reasons why interviewers in SHARE and other surveys (e.g., Peck et al. 2009; Crimmins et al. 2020) experienced difficulties in collecting sufficiently large blood spots. A leading explanation is that interviewers did not wait for a sufficiently large blood drop to form and freely fall onto the filter card but instead applied a smaller volume of blood collected on a fingertip to the card. This is particularly relevant for large population surveys where data collection is outsourced to national survey agencies, which employ lay interviewers who are trained to collect DBS but cannot be monitored perfectly in the field.

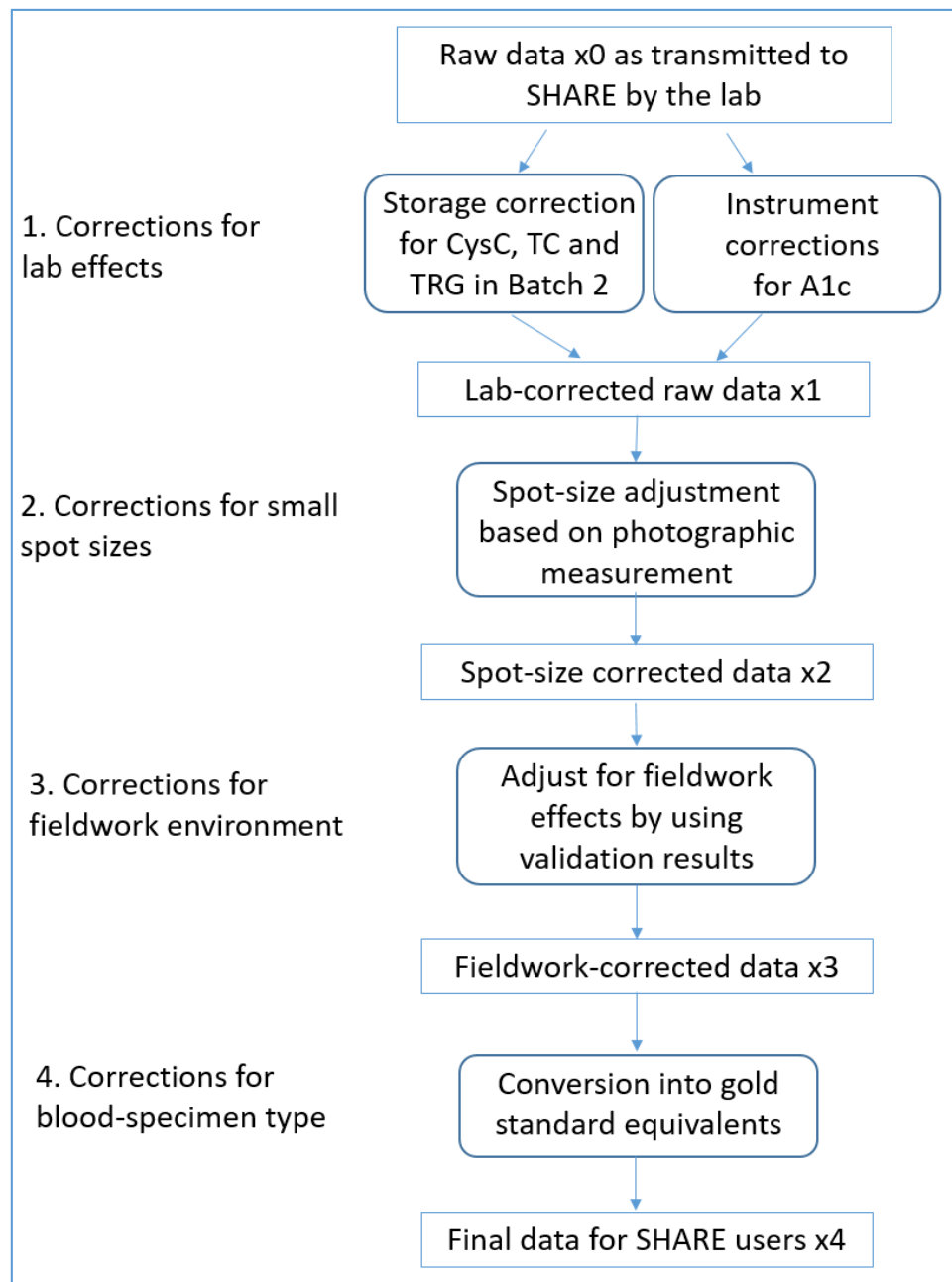
SHARE also faced other challenges. Not all DBS samples could be assayed immediately. For funding limitations, approximately one-third of the samples (Batch 1) were analyzed relatively quickly (in 2018), while the remaining two-thirds of the samples (Batch 2) was analyzed about two years later (2020/21) when sufficient funding was obtained. The degradation of biomarkers in the samples

during storage as well as changes of the analysing instruments (Kim et al. 2023) in the interim between analyses of the Batches required adjustment of the biomarker values. .

This section describes the necessary adjustments, which were based on a large set of validation experiments. Some validations are straightforward such as the standardization of analyzing instruments. Others are much more complicated, especially the correction for fieldwork effects. Börsch-Supan, A. et al. (2021) set up a large number of laboratory experiments to generate formulae that were employed to convert data obtained from DBS exposed to a given set of fieldwork conditions (e.g. temperature, humidity protection, spot size) into venous blood values. For these formulae to be applied to the survey-collected DBS, information on the fieldwork conditions must be collected. SHARE installed various monitoring mechanisms to that effect (s. 2.2.3 and Börsch-Supan, M. et al. 2020). The sequence of conversion steps based on these validations is summarized in Figure 5-1.

The conversion of the DBS raw data obtained by the laboratory (denoted by x_0 in Figure 5-1) into the final data provided to SHARE users (denoted by x_4) has four steps. First, we correct for the extended storage time and the change of one of the instruments in the lab (Sections 5.1 and 5.2). This produces the lab-corrected values x_1 . Second, we correct for small spot sizes using the photometric measurement of our DBS samples (Section 5.3). This produces the size-corrected values x_2 . The third step is the most involved correction. It adjusts for the conditions during and after the fieldwork, such as temperature, humidity, and transport conditions and duration. This adjustment is based on a large set of validation experiments in which field conditions were simulated in a lab (Section 5.4). This produces the fieldwork-corrected values x_3 . Finally, since the capillary blood eluted from DBS is different from venous blood, based on laboratory comparisons between DBS and venous blood samples (Section 5.5), we convert the units of the x_3 data into serum or whole blood values (standard equivalents). This produces the blood-specimen type-corrected final values x_4 .

Figure 5-1: Conversion of DBS raw data into final data for SHARE users



5.1 Storage correction for CHO, TRG and CYC

Given that the data was analyzed in two batches, some in 2018 and the rest in 2020/21, we may expect the length of time in storage to influence the analyte value. Upon further investigation, we found that three biomarkers were particularly negatively influenced by storage time: total cholesterol, triglycerides, and cystatin C. For each of these biomarkers, analyte values were significantly lower in Batch 2.

In order to control for differences in storage time, we used two approaches. First, we pulled DBS collected from 96 respondents that had been assayed in Batch 1 and had sufficient blood for re-

analysis and re-analyzed CHO, TRG and CYC in 2020. We then generated a time-in-storage variable: Biomarker storage time in days = Re-assay Date (July 28th 2020) – Original Assay Date (sometime in 2017/2018). Across the 96 samples, time in storage ranged from 868 to 998 days. We then estimated the following equation:

$$Diff(B_2 - B_1) = \alpha + \beta(\text{storage time})$$

where $Diff(B_2 - B_1)$ denotes the difference between batches, and storage time is measured in days since July 1st, 2018, the mean assay date of Batch 1.

Second, we drew a random sample of about 8,000 observations from Batch 2 DBS and randomly matched those with observations from Batch 1 DBS that had experienced similar fieldwork conditions and then estimated the same regression equation as above. This produced the coefficients in Table 5-1.

Both methods produce the same result in the sense that the difference between the predicted outcome variables is not statistically different. We use the second method since it incorporates more precisely measured regression coefficients.

We then standardize all Batch 2 observations to Batch 1 values by computing:

$$x_1 = x_0 - \alpha - \beta(\text{storage time}) \text{ if batch} = 2$$

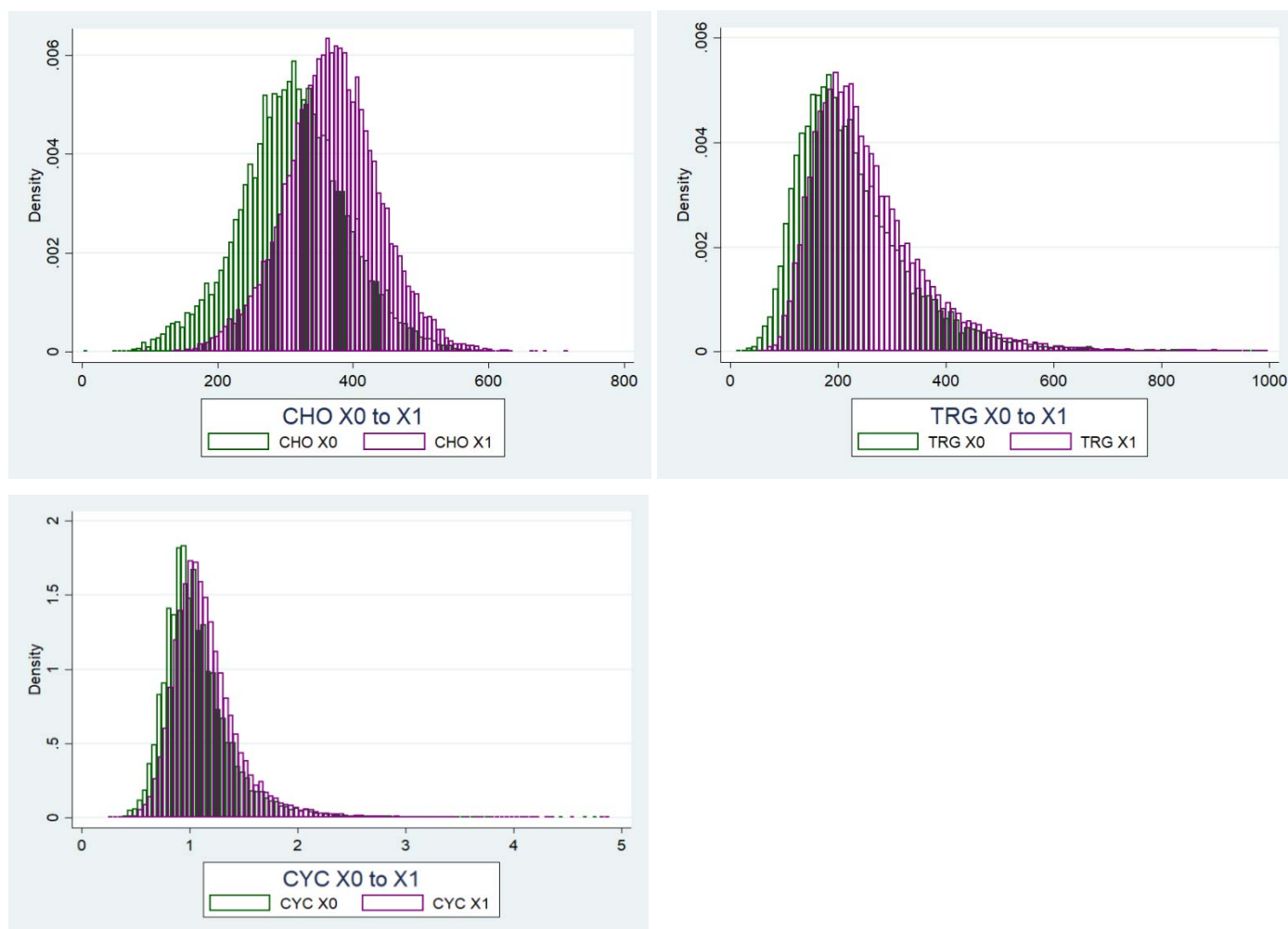
for CHO, TRG, and CYC. All other biomarkers are left unchanged. Results are shown in Figure 5-2. While the R-squares for the three regressions are rather small, storage time significantly reduces the marker values for CHO and TRG. We do not find a significant storage-time effect for CYC.

Table 5-1: Effect of storage time on difference in analyte value between batches

VARIABLES	(1) ydiff cho	(2) ydiff trg	(3) ydiff cyc
Storage time	-0.175*** (0.0436)	-0.173** (0.0845)	0.000291 (0.000232)
Constant	96.45** (44.82)	139.5 (86.93)	-0.414* (0.238)
Observations	6,918	6,845	6,918
R-squared	0.002	0.001	0.000

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Figure 5-2: Effects of correction for storage time



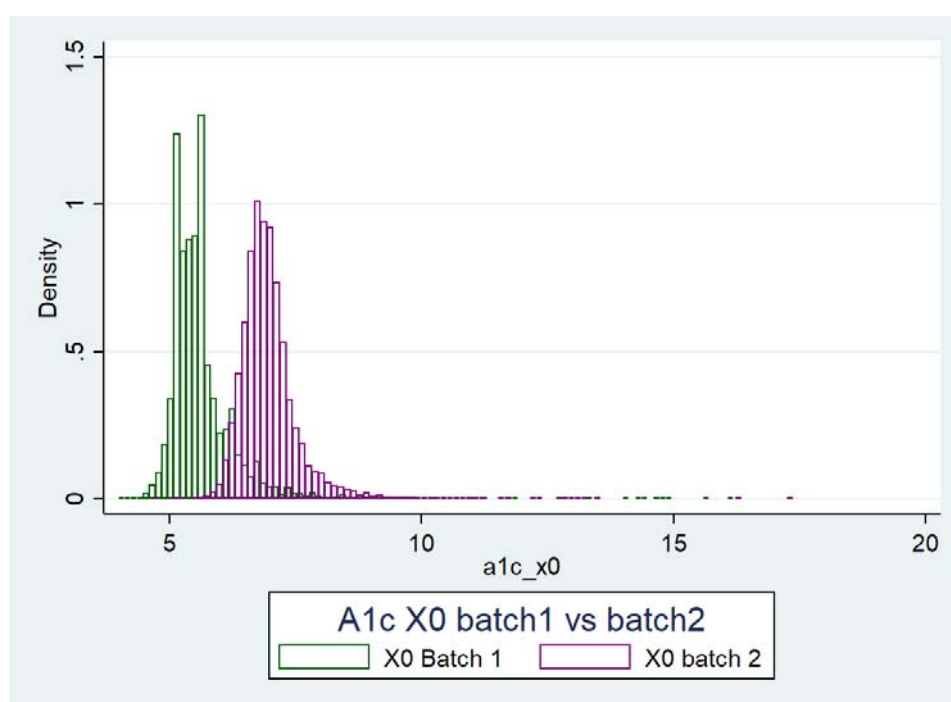
For all three biomarkers, the Batch-2 values are corrected for the degradation during storage, which increases the biomarker values and thereby shifts each distribution to the right. The percentage degradation was strongest for CHO and smallest for CYC.

5.2 Instrument correction for HbA1c

The glycosylated hemoglobin (HbA1c) percentage in SHARE-DBS samples was measured by high performance liquid chromatography (HPLC) systems. HPLC methods are considered one of the standard methods of HbA1c measurement. Batch 1 was analyzed in 2018 with a Variant II ion-exchange HPLC system. Given innovations in technology, in the middle of the study period the lab at the University of Washington received a new ion-exchange HPLC system that specifically analyzed HbA1c. Therefore, whereas Batch 1 HbA1c data was analyzed on the Variant II hemoglobin testing system in 2018, Batch 2 was analyzed on a D-100 hemoglobin testing system in 2020. As depicted in Figure 5-3, the distributions of the biomarker values had very similar standard

deviations (std.dev=0.55%) but the mean biomarker values produced by the new D-100 instrument (mean=6.8%) was shifted significantly to the right relative to the biomarker values produced by the Variant II instrument (mean=5.6%). Since the selection into first and second batch was random, the cause for this difference must be in the instrument, the reagents used, or similar lab-generated differences between the biochemical analyses in 2018 and 2020. Moreover, both the Variant II and the D100 instruments indicated an unexpectedly high number of non-hemoglobin A peaks. These atypical peaks may affect the direct HbA1c value. Despite testing the impact of various field conditions on lab-created DBS samples, we could neither recreate nor explain the occurrence of the non-hemoglobin A peaks.

Figure 5-3: Distribution of HbA1c raw values



In order to resolve these issues, we turned in 2023 to an immunoassay-based HbA1c measurement with a DxC 600 AU analyzer (Beckman Coulter, CA, USA). We expected the immunoassay to be more robust and show less interference due to pre-analytic variation.

To cross-validate the results obtained from the Variant II and D100 instruments, we drew a stratified random subsample of 998 observations from both batches, oversampling very low and very high HbA1c raw values, with about equal sample size in each batch, and re-analyzed those DBS with the DxC 600 instrument. The re-analysis yielded a mean HbA1c blood value of 6.1% with a standard deviation of 0.7%. We then regressed the DxC 600 values on the Variant II and D100 values (see Table 5-2).

Table 5-2: Effects of instruments on HbA1c values (regression coefficients)

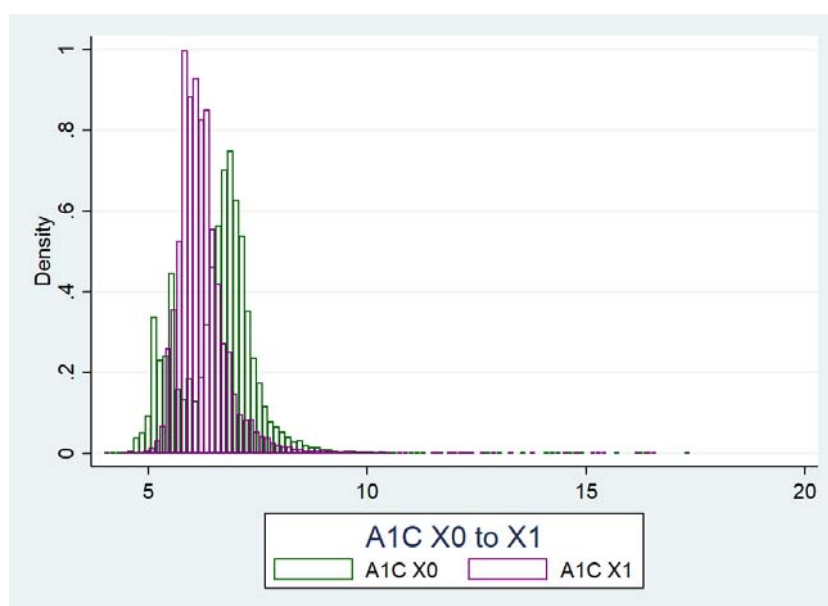
X1=DxC 600	Batch 1: Variant II in 2018			Batch 2: D100 in 2020		
	Coef.	Std.Err.	t	Coef.	Std.Err.	t
X0	1.023768	.0417441	24.52	.920823	.0370295	24.87
Constant	.3691073	.2345225	1.57	-.1627274	.2530389	-0.64
Adj R-squared	0.5461			0.5540		
Number obs	500			498		

Based on these regressions, we normalized all biomarker values to the DxC-values using the following equations

$$\begin{aligned}
 x1 &= 1.023768 * x0 + 0.3691073 \quad \text{for batch=1 (x0 from Variant II in 2018)} \\
 &= 0.920823 * x0 - 0.1627274 \quad \text{for batch=2 (x0 from D100 in 2020)}
 \end{aligned}$$

Results are shown in Figure 5-4:

Figure 5-4: Effect of correction for instrument change



The distribution of the HbA1c raw data (A1c_x0) has two peaks: one for Batch 1 and one for Batch 2. The instrument correction shifts the values of both batches such that the biomarker means of Batches 1 and 2 coincide with the mean generated by the DxC 600 instrument. This creates a single-peaked distribution of the lab-corrected HbA1c_x1.

5.3 Spot Size Corrections

After these preliminary corrections for storage time and analyzer change, we correct for small spot sizes. As described in Groh et al. 2022, a unique innovation in SHARE DBS collection was an algorithm that precisely measured the blood-covered area of a DBS spot, as each of the DBS samples were photographed during the punching process. This study included the installation of monitoring mechanisms that enabled the establishment of a new algorithm that automated the precise measurement of the blood-covered area of each DBS spot. For this approach, we took advantage of the fact that all SHARE DBS samples selected for analyses were photographed during the punching process. This process allowed for ex-post spot-size estimation for a very large number of DBS data. Findings from Groh et al. 2022 show that spot size impacts biomarkers differently and in a non-linear way.

Groh et al. 2022 provide a table with the percentage deviations in biomarker values from the values that would have been obtained if the spot actually collected in the field had a standard size of 1.0 cm. We approximate these deviations by a quadratic polynomial

$$marker_{dev} = \alpha + \beta_{size}(size) + \beta_{size^2}(size^2)$$

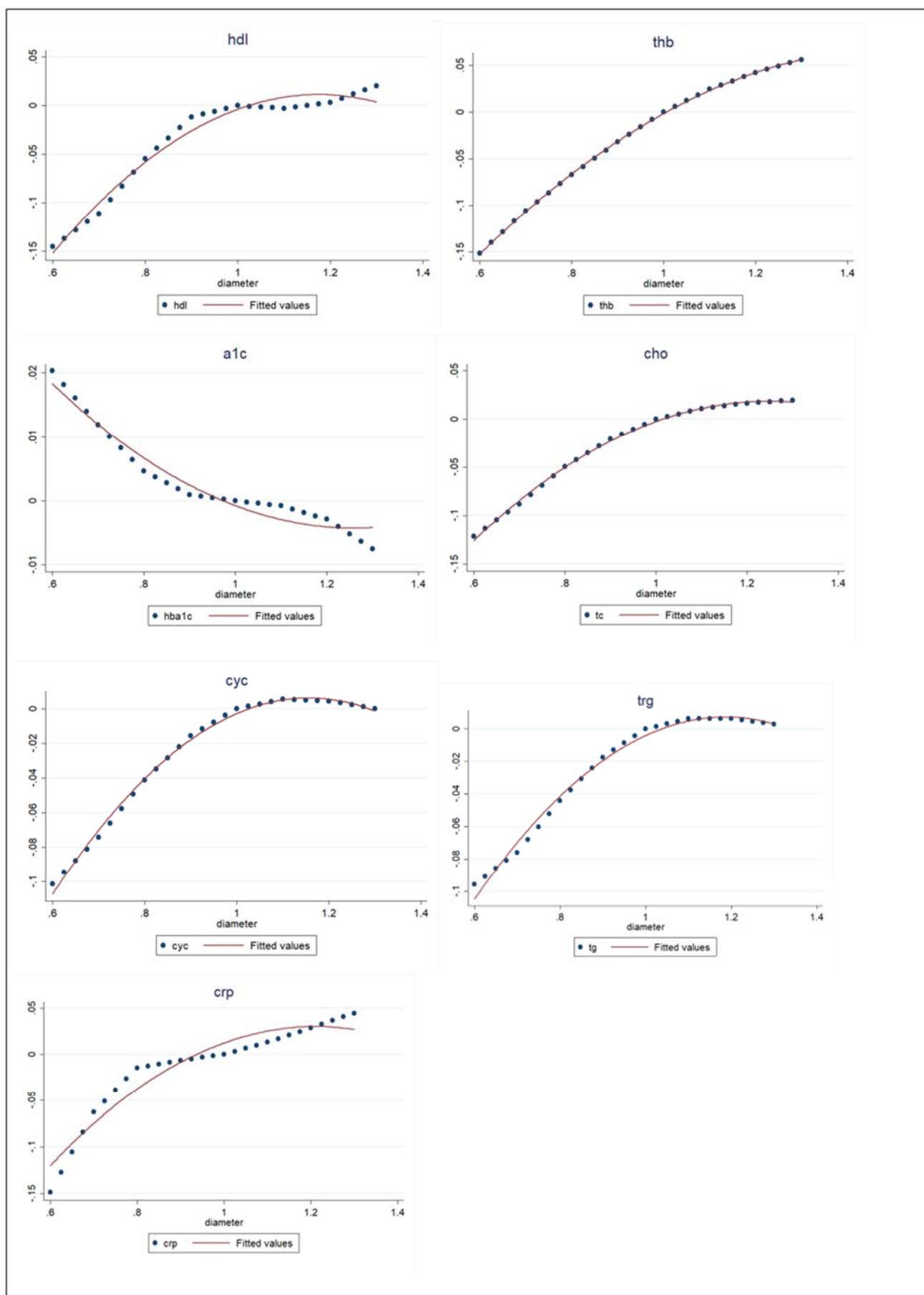
Since measurements of A-markers required two punches, depending on spot size, size is the average diameter between spots punched for HDL, THB and HbA1c. Table 5-3 and Figure 5-5 show the estimated and approximated relationship between spot size and analyte value. Note that THB has a perfect fit, and CHO, TRG and CysC nearly perfect fit.

Table 5-3: Effect of spot size on biomarker values (regression coefficients)

VARIABLES	(1) HDL	(2) THB	(3) A1C	(4) CHO	(5) TRG	(6) CRP	(7) CYC
Size	1.198*** (0.0881)	0.801*** (0.00781)	-0.121*** (0.0159)	0.886*** (0.0159)	0.821*** (0.0277)	0.834*** (0.124)	0.877*** (0.0184)
Size ²	-0.512*** (0.0456)	-0.265*** (0.00404)	0.0469*** (0.00820)	-0.358*** (0.00823)	-0.349*** (0.0143)	-0.334*** (0.0641)	-0.381*** (0.00950)
Constant	-0.689*** (0.0412)	-0.538*** (0.00365)	0.0730*** (0.00741)	-0.530*** (0.00743)	-0.476*** (0.0129)	-0.490*** (0.0579)	-0.499*** (0.00858)
Observations	28	28	28	28	28	28	28
R-squared	0.970	1.000	0.947	0.999	0.994	0.923	0.997

Standard errors in parentheses
 *** p<0.01, ** p<0.05, * p<0.1

Figure 5-5: Effect of spot size on biomarker values (percentages)



Spot-size corrections are then applied as follows:

$$x_2 = x_1 * (1 - marker_{dev})$$

where x_2 denotes the spot-corrected values, x_1 denotes storage- and instrument-corrected values, and $marker_{dev}$ denotes the percentage deviation from the 1cm-standard-size DBS value per Figure 5-5. Since smaller spot sizes lead to lower biomarker values except for HbA1c, the corrected biomarker values (“as if the spot size had been 1 cm”) are larger for all biomarkers except for HbA1c where it is the opposite. Results are shown in Figure 5-6.

Figure 5-6: Effects of correction for small spot sizes

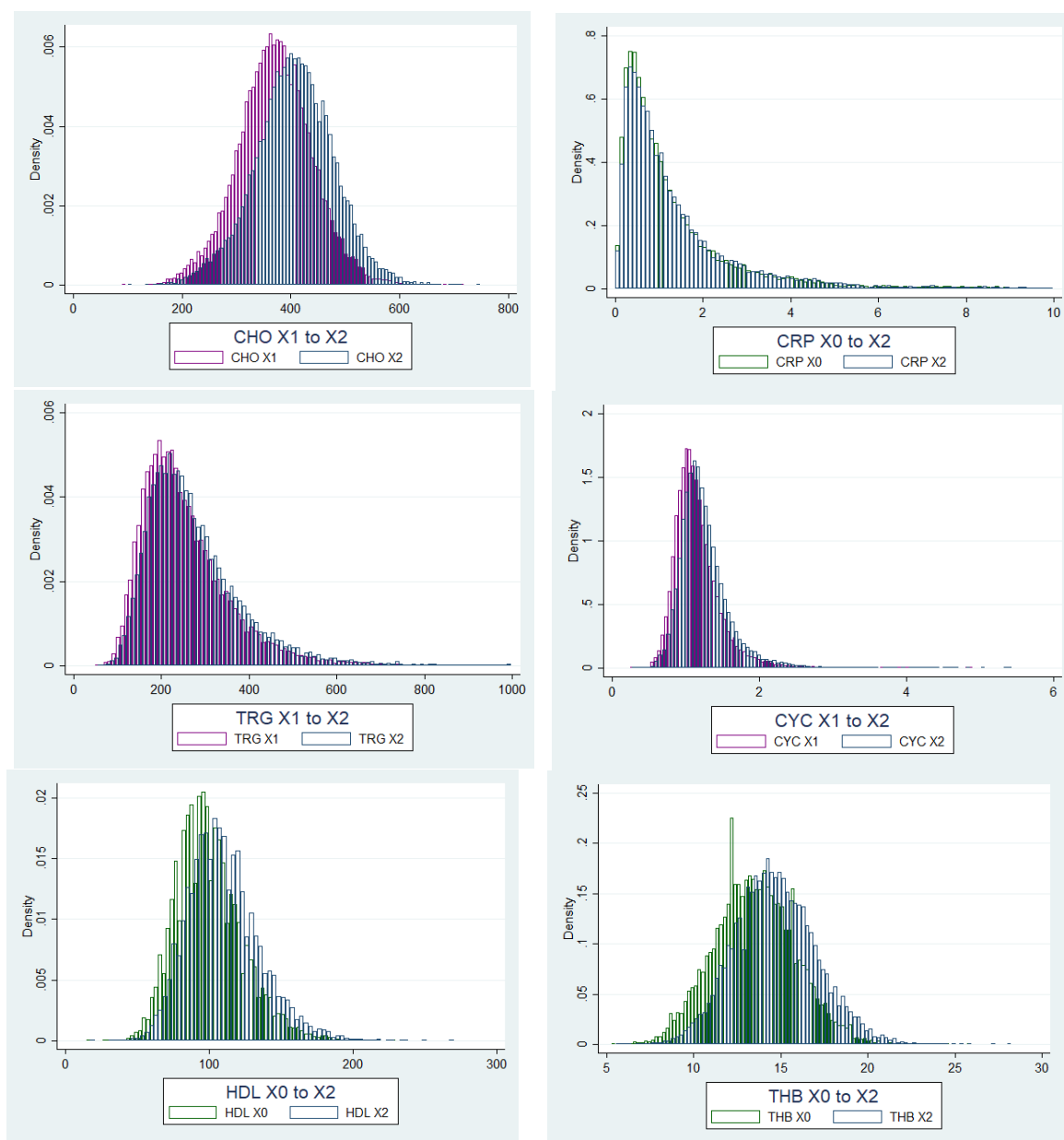
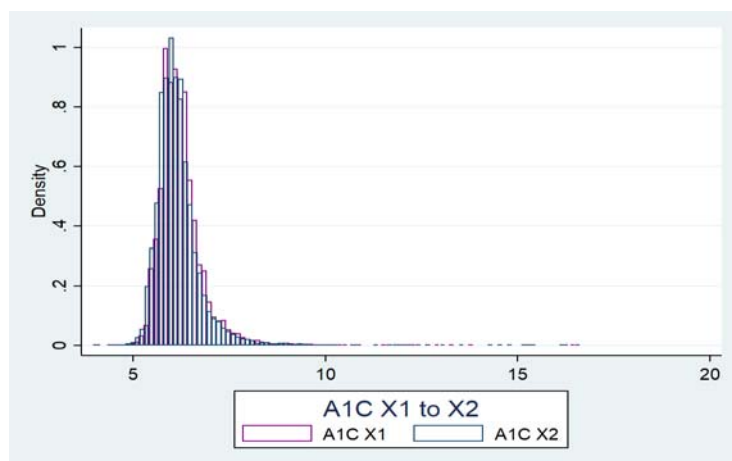


Figure 5-6: Effects of correction for small spot sizes (continued)



The correction for small spot sizes shifts the biomarker distributions to the right, most strongly for CHO, HDL and THB, while the other biomarkers were less sensitive to small spot sizes.

5.4 Adjustment for fieldwork conditions

Being aware of the influence of field conditions and sample quality on the DBS analyte values, Börsch-Supan et al. (2021) performed controlled laboratory experiments at UW by systematically exposing DBS to adverse fieldwork conditions. The key idea was to simulate in the lab the conditions that were prevalent in the actual fieldwork and to then, through multiple regression analysis, determine the influence of those conditions on biomarker measurements. The regressions are used to transform the analyte values measured by the laboratory into the analyte values that would have been obtained under ideal sample quality and field conditions.

Figure 5-7 illustrates the concept of the validation experiments. The steps in the figure correspond to those in Figure 5-1. The left panel shows what occurred in the field: blood was collected from the finger of a respondent and dropped onto DBS card filter paper, often generating blood spot of relatively small sizes. We denote those samples as DBS_C for DBS created from capillary blood. No laboratory value could be obtained at this stage since the DBS were collected in the field and had to be transported to the lab. During transportation, the blood was exposed to prevailing temperature and humidity, and shipped to the lab under occasionally unfavorable conditions. The samples arriving at the lab were designated to be DBS_F (for DBS created under fieldwork conditions) and were analyzed, producing a value of x_0 , the initial value for our adjustments (see Figure 5.1).

In the validation experiments (right panel, Figure 5-7), DBS and venous blood were collected from 20 volunteers and used for three different purposes. First, the DBS were analyzed in the lab without adverse fieldwork conditions, DBS-C yielding the value d_3 . Second, the venous blood samples (VBS) were analyzed under “gold standard” conditions, yielding the value v_4 . Third and most importantly,

venous blood was pipetted onto filter paper to simulate DBS. These venous-blood-based DBS (thus named DBS-V in Figure 5-7) were exposed to the same range of field conditions that were observed during collection of the SHARE Wave 6 DBS, such as temperature, humidity, transportation time, etc. (see Table 5-4), totaling 171 different field conditions.

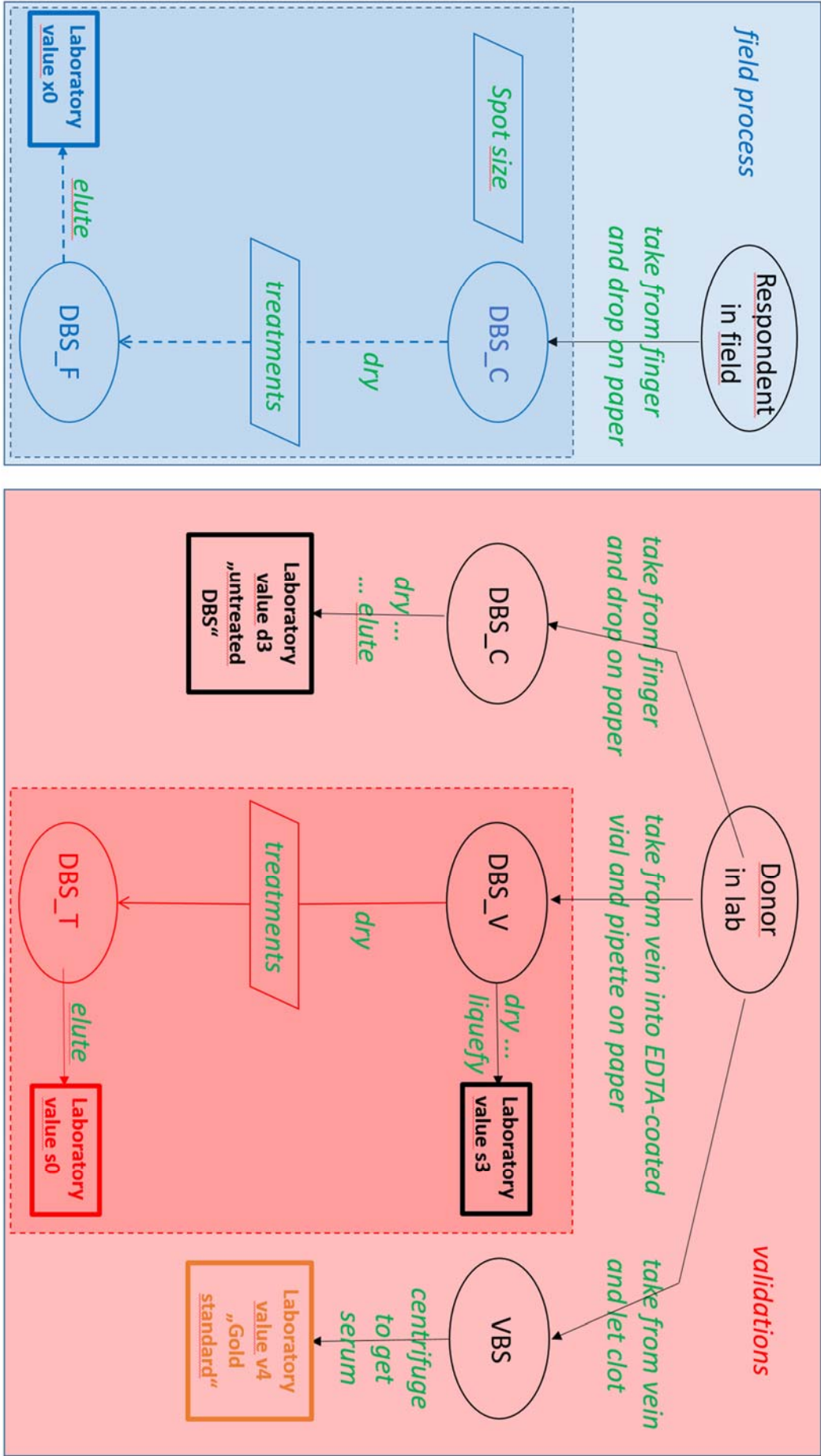
Table 5-4: Treatment levels in the simulated fieldwork conditions

Fieldwork Condition	
Drying Time (minutes)	5, 20 , 35
Shipment Time (Days)	3, 7 , 14
Outside Temperature (C)	5, 20 , 35
Humidity Protection	0 ,1 for open bag; 0 ,1 for desiccant missing

Note: Bold figures are used as standard values

These treated DBS (hence DBS_T) yielded laboratory values s_0 . As controls, some venous-blood-based DBS were analyzed without exposure to adverse fieldwork conditions (DBS-V), yielding the value s_3 . The use of venous blood rather than capillary blood from a finger prick was necessary to create sufficient numbers of DBS for exposure to the 171 different field conditions. Apart from being based on venous blood, the treatment procedures in the validation experiment (dark red in Figure 5-7) mirrored the field conditions (dark blue in Figure 5-7). Hence, s_0 in the validation experiments corresponds to x_0 in the field. We do not have field values for all biomarker values obtained from the laboratory experiments (i.e., s_3 , d_3 , v_4). The ultimate aim of the validations is to compute a value x_4 that corresponds to the “gold standard” v_4 . This requires the correction for the simulated fieldwork conditions (s_0 to s_3 , corresponding to the steps x_0 to x_3), which is discussed in this subsection. The final step is the blood conversion from d_3 to v_4 , corresponding to the step x_3 to x_4 , which is discussed in the next section.

Figure 5-7: Field process and validation experiments



The validation experiment produced 3420 outcomes (171 different treatments for each of the 20 donors). A second sample of 30 donors was also included to diversify the sample, namely by adding more diabetics. However, due to constraints, this sample of 30 was not subjected to the full treatment procedure, resulting in only 210 outcomes. We estimated one statistical relationship between these outcomes and the fieldwork conditions for each biomarker by using the following regression equation

$$s1 = \beta_1 + \beta_2 \cdot s0 + \beta_3 \cdot \text{dry} + \beta_4 \cdot \text{ship} + \beta_5 \cdot \text{bag} + \beta_6 \cdot \text{desiccant} + \beta_7 \cdot \text{temp} + \beta_8 \cdot \text{size} + \beta_9 \cdot \text{size}^2 + \beta_{10} \cdot \text{dry}^2 + \beta_{11} \cdot \text{ship}^2 + \beta_{12} \cdot \text{temp}^2 + \beta_{13} \cdot \text{bag} \cdot \text{desiccant} + \beta_{14} \cdot \text{bag} \cdot \text{dry} + \beta_{15} \cdot \text{dry} \cdot \text{size}$$

where $s0$ is the biomarker value of the treated DBS_T, $s1$ is the biomarker value of the untreated DBS_V, and the coefficients vary by biomarker. Results are shown in Table 5-5.

Table 5-5: Estimation results for simulated fieldwork conditions in the validation sample

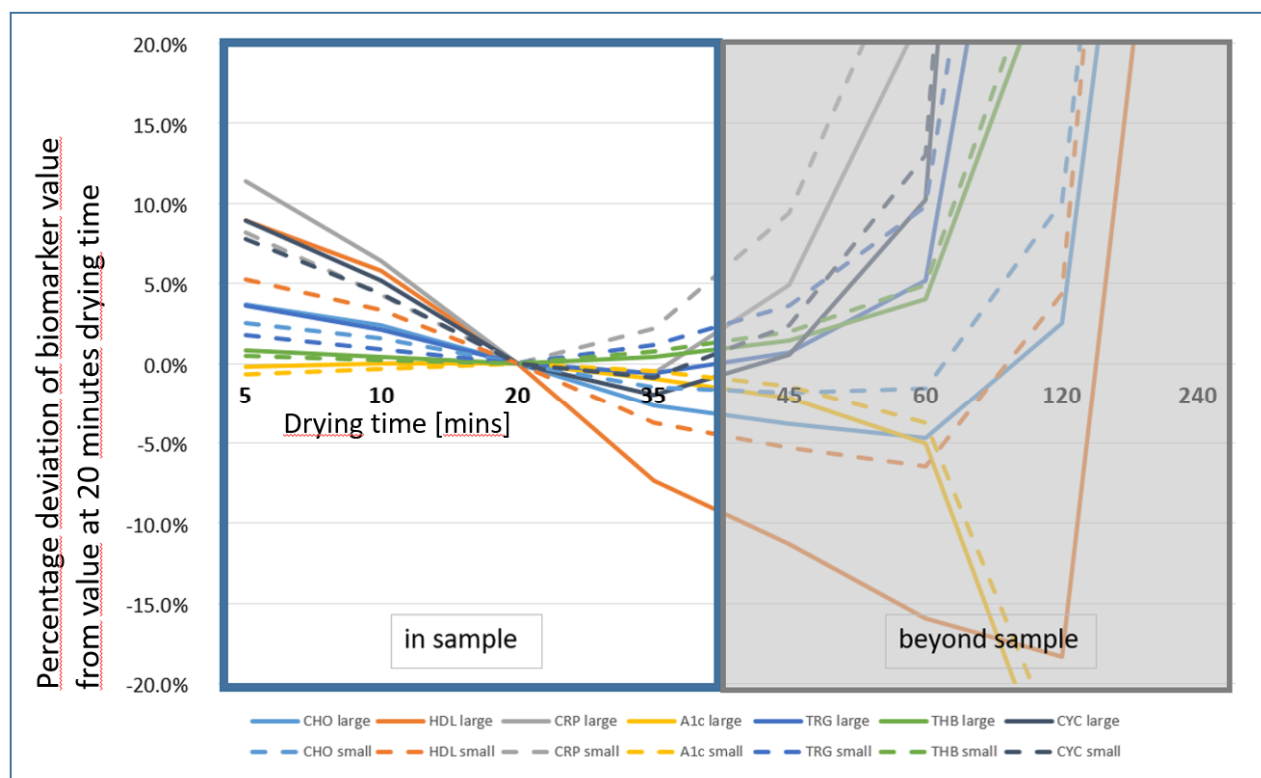
VARIABLES	(1) CHO dbs v	(2) HDL dbs v	(3) CRP dbs v	(4) A1C dbs v	(5) TRG dbs v	(6) THB dbs v	(7) CYC dbs v
Dry	-0.251 (0.248)	-0.0733 (0.171)	-0.0290*** (0.00577)	0.00929*** (0.00244)	-0.226 (0.313)	-0.00986 (0.0113)	-0.00692*** (0.00106)
Dry ²	0.00520 (0.00479)	0.00233 (0.00330)	0.000793*** (0.000112)	-0.000152*** (4.50e-05)	0.0139** (0.00607)	0.000411* (0.000218)	0.000134*** (2.06e-05)
Shipment	2.123*** (0.687)	1.733*** (0.475)	-0.0619*** (0.0160)	0.0203*** (0.00648)	-0.309 (0.870)	-0.0196 (0.0313)	-0.00328 (0.00293)
Shipment ²	-0.126*** (0.0388)	-0.0825*** (0.0268)	0.00399*** (0.000905)	-0.00149*** (0.000365)	0.0214 (0.0491)	0.00205 (0.00177)	0.000356** (0.000166)
Open Bag	-1.537 (1.986)	-0.422 (1.372)	0.105** (0.0464)	-0.0449** (0.0191)	-2.433 (2.514)	0.165* (0.0907)	0.0113 (0.00847)
Desiccant incl.	11.12*** (3.442)	5.488** (2.361)	0.0282 (0.0802)	-0.291*** (0.0281)	-35.62*** (4.385)	0.0141 (0.156)	0.0438*** (0.0146)
Spot Size	-124.2*** (17.04)	-71.54*** (11.81)	-2.893*** (0.393)	1.236*** (0.183)	-62.04*** (21.41)	-8.914*** (0.779)	-0.188*** (0.0723)
Spot Size ²	47.01*** (8.819)	39.48*** (6.146)	1.430*** (0.204)	-0.496*** (0.105)	20.15* (11.12)	3.425*** (0.401)	0.0861** (0.0376)
Temperature	0.533*** (0.196)	0.570*** (0.136)	-0.0113** (0.00457)	0.0146*** (0.00183)	-0.255 (0.248)	0.00774 (0.00896)	0.00771*** (0.000841)
Temperature ²	-0.0103** (0.00479)	-0.00866*** (0.00331)	0.000438*** (0.000112)	-0.000415*** (4.49e-05)	-0.00587 (0.00606)	0.000189 (0.000218)	-2.51e-05 (2.04e-05)
Open Bag * Desiccant	-4.594 (5.104)	3.010 (3.519)	-0.0722 (0.119)	-0.0679 (0.0413)	-13.41** (6.471)	0.336 (0.232)	-0.0583*** (0.0217)
Open Bag * Dry	-0.0251 (0.0849)	-0.0109 (0.0586)	-0.000684 (0.00198)	-0.000447 (0.000814)	0.0431 (0.107)	-0.00193 (0.00388)	-0.000172 (0.000361)
Dry * Spot Size	-0.421*** (0.150)	-0.372*** (0.104)	-0.0160*** (0.00350)	-0.00469** (0.00187)	-0.620*** (0.190)	-0.00854 (0.00685)	-0.00159** (0.000639)
CHO dbs t	0.565*** (0.00944)						
HDL dbs t		0.617*** (0.0145)					
CRP dbs t			1.250*** (0.00439)				
A1C dbs t				0.993*** (0.0121)			
TRG dbs t					1.131*** (0.00946)		
THB dbs t						0.640*** (0.0150)	
CYC dbs t							0.960*** (0.0162)
Constant	285.6*** (8.564)	101.9*** (5.801)	2.095*** (0.195)	-0.939*** (0.109)	57.12*** (10.58)	12.31*** (0.402)	0.432*** (0.0368)
Observations	3,569	3,402	3,569	2,602	3,562	3,402	3,569
R-squared	0.502	0.348	0.958	0.728	0.803	0.356	0.502

Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

We find significant treatment effects for almost all treatments that simulate the fieldwork effects in the lab. Some effects are non-linear, and some treatments significantly interact with other treatments. Small spot size significantly decreases all biomarker values except HbA1c where it increases the value. This replicates the results from the previous subsection.

Drying time turns out to be more complex (Figure 5-8). For short drying times (less than 20 minutes), the linear term dominates the quadratic term and we find a significant negative effect on all biomarker values except HbA1c where drying time has a positive effect. This means that lengthening very short drying times increases all biomarker values except HbA1c, where it slightly decreases the value. However, for drying times exceeding 20 minutes, the quadratic term dominates the linear term and we find the opposite effects. If extrapolated beyond the 5-40 minutes drying time range of the SHARE DBS, the positive effect is strongly exaggerated (grey area in Figure 5-8). Moreover, there is a significant interaction with size. For small spot sizes (6 mm, broken lines), the effects of drying time are attenuated relative to standard spot sizes (10 mm, solid lines).

Figure 5-8: Non-linear and interaction effects for drying time



Exposure to high temperatures does not significantly affect TRG or THB values but significantly increases values of CHO, HDL, HbA1c, and CYC, and decreases CRP values. Leaving the plastic storage bag open has significant but small effects on CRP and HbA1c values, while failing to include the desiccant in the bag decreases CHO, HDL and CYC values, and increases HbA1c and TRG values. Shipment time has large effects on CHO and HDL, with longer shipment durations increasing

biomarker values. The quadratic terms for temperature and shipment time indicate that the effects are less than linear for high temperatures and long shipment times.

We then applied the resulting equation to the fieldwork data to compute biomarker values corrected for the fieldwork conditions, i.e., the biomarker values that would have been obtained if the fieldwork conditions had been as close as realistically possible to what they could have been in the laboratory. These conditions are bolded in Table 5-4.

$$x3 = \beta_1 + \beta_8 \cdot (\text{average size}) + \beta_9 \cdot (\text{average size})^2 + \beta_2 \cdot x2 - (\beta_3 \cdot (\text{dry} - 20) + \beta_4 \cdot (\text{ship} - 7) + \beta_5 \cdot \text{bag} + \beta_6 \cdot \text{desiccant} + \beta_7 \cdot (\text{temp} - 20) + \beta_{10} \cdot (\text{dry}^2 - 20^2) + \beta_{11} \cdot (\text{ship}^2 - 7^2) + \beta_{13} \cdot (\text{temp}^2 - 20^2) + \beta_{14} \cdot \text{bag} \cdot \text{desiccant} + \beta_{15} \cdot (\text{bag} \cdot \text{dry} - 20) + \beta_{16} \cdot \text{dry} \cdot \text{size})$$

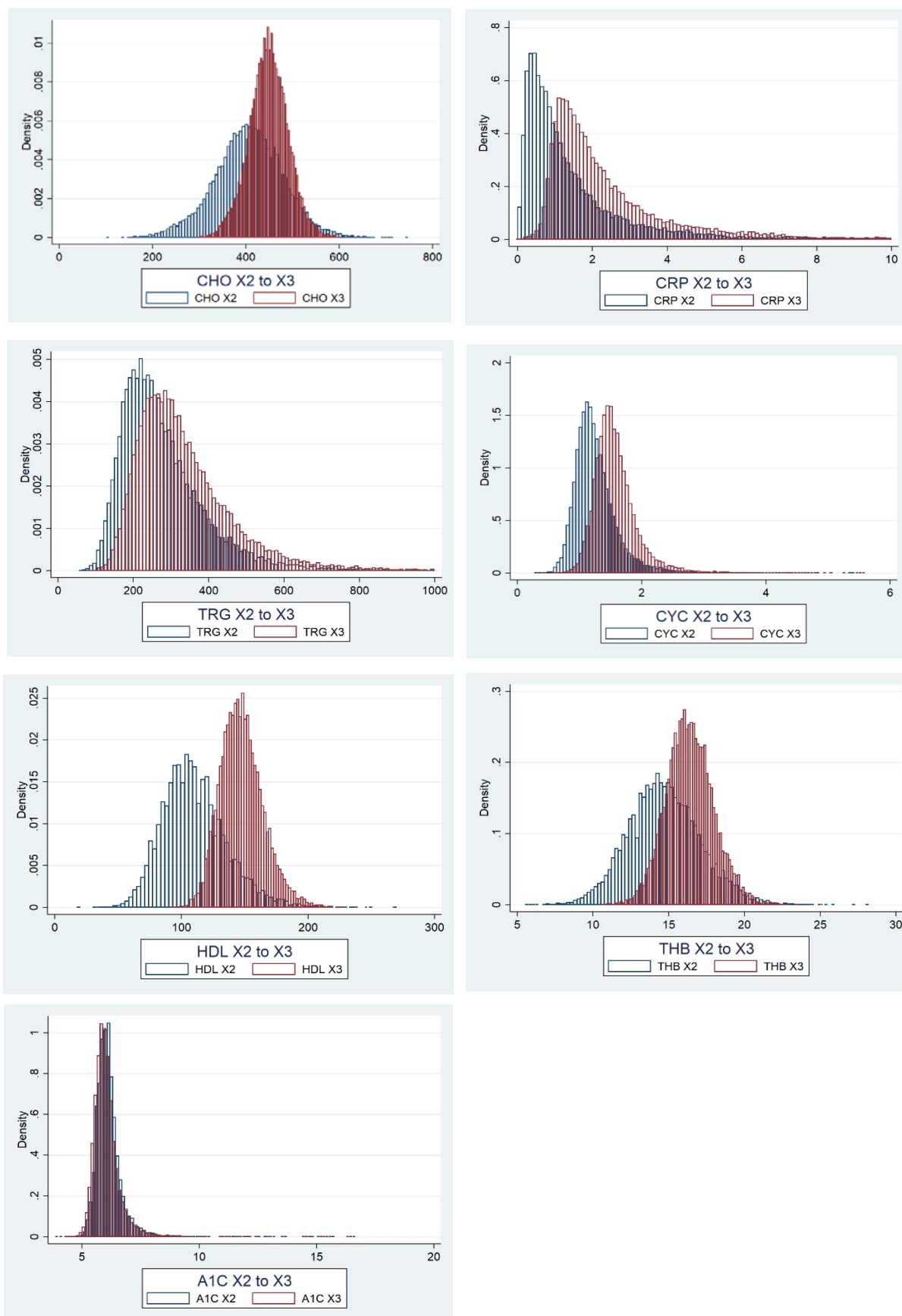
where x2 is the size-corrected fieldwork biomarker value and x3 the resulting fieldwork-corrected biomarker value.

To test the obtained equations, we applied them to a dataset from a small field experiment conducted in Poland during the SHARE Wave 6 pretest in which nurses collected DBS samples and venous blood samples from 37 respondents. The experiment is described in Weiss, Börsch-Supan et al., Chapter 38, in: Börsch-Supan, A et al. (Eds.), 2019. We used the obtained equations to convert the DBS marker values into venous blood values and compared them to the actual venous blood values; no statistically significant differences were found. This test is encouraging; however, its statistical power is low due to the small sample size.

As discussed in Section 5-3, innovations in spot-size calculation (Groh et al. 2022) led to an individual spot-size correction applied to each biomarker, which yielded value x2 (above). Hence, including the spot-size terms in the fieldwork correction equation for x2 values would be redundant so the fieldwork-correction equation generating x3 values omits the spot-size terms. To ensure that the expected values remain identical on both sides of the regression equations, we adjusted the constant to include the average spot size of the validation experiments (.969 cm²) as a component of the constant.

Figure 5-9 shows the resulting distributions of x3. For all biomarkers except HbA1c, adverse fieldwork conditions lead to lower biomarker values when compared to the biomarker values obtained from samples created under ideal conditions in a laboratory. Correcting for these conditions therefore leads to a substantial shift of the biomarker distribution to the right, i.e., higher biomarker values. As in the spot size correction, the effect of the fieldwork corrections on HbA1c is small and in the opposite direction.

Figure 5-9: Effects of correction for fieldwork effects



5.5 Adjusting for blood-specimen type and converting to standard equivalents

While the validation experiments described in the previous section try to mimic the fieldwork treatments as close as possible, they had to be conducted on measurement of EDTA-anticoagulated venous blood pipetted onto DBS filter paper rather than on DBS created from capillary blood because only venous blood could be obtained in an amount sufficient for all 171 different conditions to be simulated.

The final adjustment addresses this difference in blood specimens. In Figure 5-1, this is represented by the conversion from x3 to x4. It is based on a regression that estimates the relationship between DBS_C (measured by its laboratory value d3, see Figure 5-7) and VBS serum values (measured by its laboratory value v4, Figure 5-4; VBS THB values are measured as whole blood equivalents). No adjustments for blood-specimen type are necessary for HbA1c since the HbA1c values are obtained as ratio measurements.

$$v4 = \gamma_1 + \gamma_2 \cdot d3$$

Table 5-6 shows the estimation results for this last conversion step, based on the validation sample:

Table 5-6: Estimation results for blood-specimen type conversion in the validation sample

VARIABLES	(1) cho serum	(2) hdl serum	(3) crp serum	(4) trg serum	(5) thb blood	(6) cyc serum
cho_dbs_c	0.597*** (0.00722)					
hdl_dbs_c		0.539*** (0.00510)				
crp_dbs_c			1.000*** (0.00178)			
trg_dbs_c				0.809*** (0.00665)		
thb_dbs_c					0.814*** (0.00851)	
cyc_dbs_c						0.381*** (0.00828)
Constant	-44.46*** (2.713)	-11.19*** (0.625)	0.187*** (0.00786)	-52.33*** (1.579)	1.429*** (0.130)	0.387*** (0.00812)
Observations	3,690	3,690	3,690	3,683	3,690	3,720
R-squared	0.649	0.752	0.989	0.801	0.713	0.362

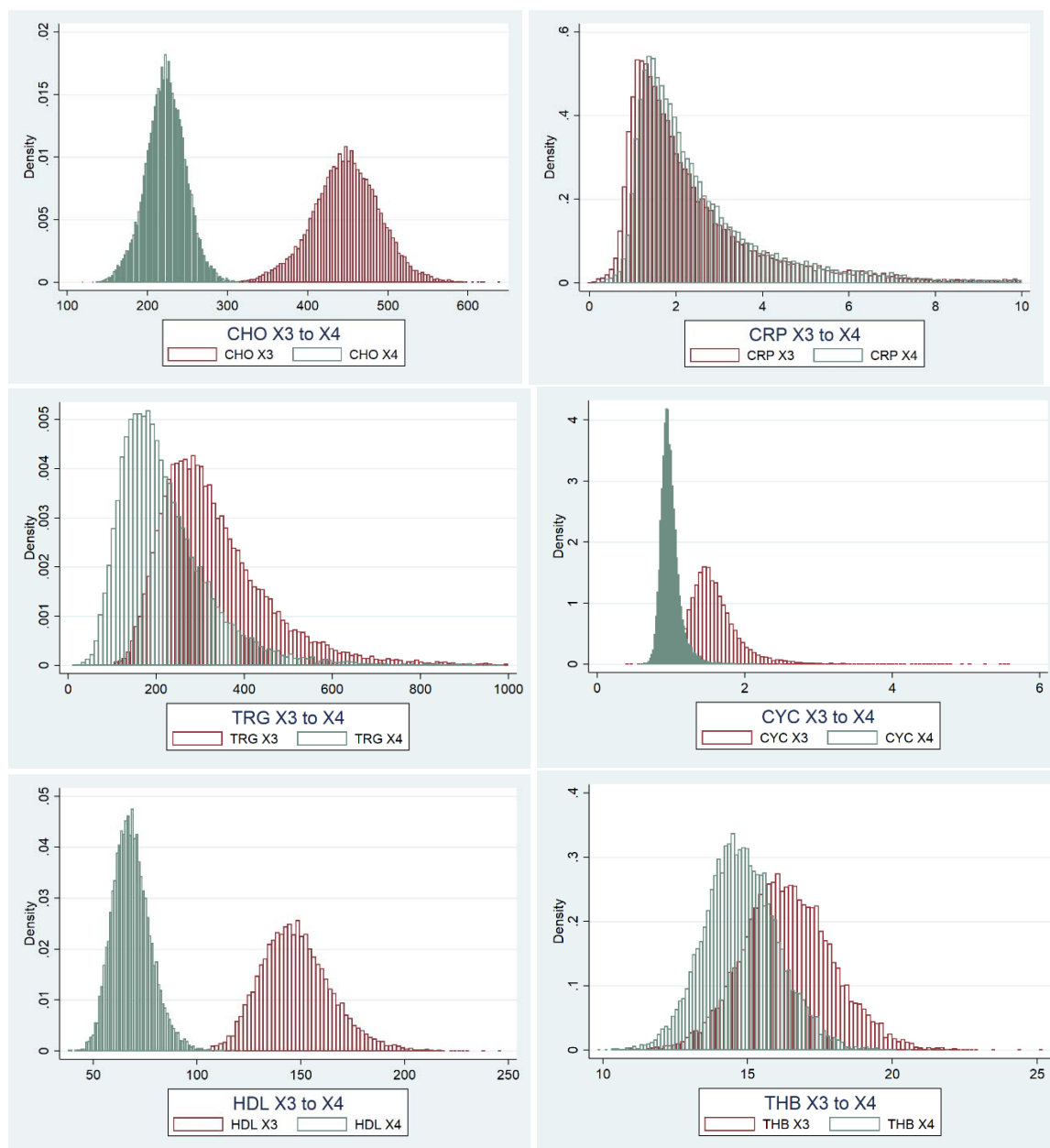
Standard errors in parentheses
 *** p<0.01, ** p<0.05, * p<0.1

The regressions yield statistically precise results with high R-squares for all biomarkers except a moderate R-square for CYC. The equations are used to compute the final correction step 4 in Figure 5.1:

$$x4 = \gamma_1 + \gamma_2 \cdot x3$$

for all respondents and biomarkers in the SHARE sample. The x4 values are the result from the sequence of conversions, starting with the raw data x0. The resulting distributions of the marker values are shown in Figure 5-10 and Table 5-7.

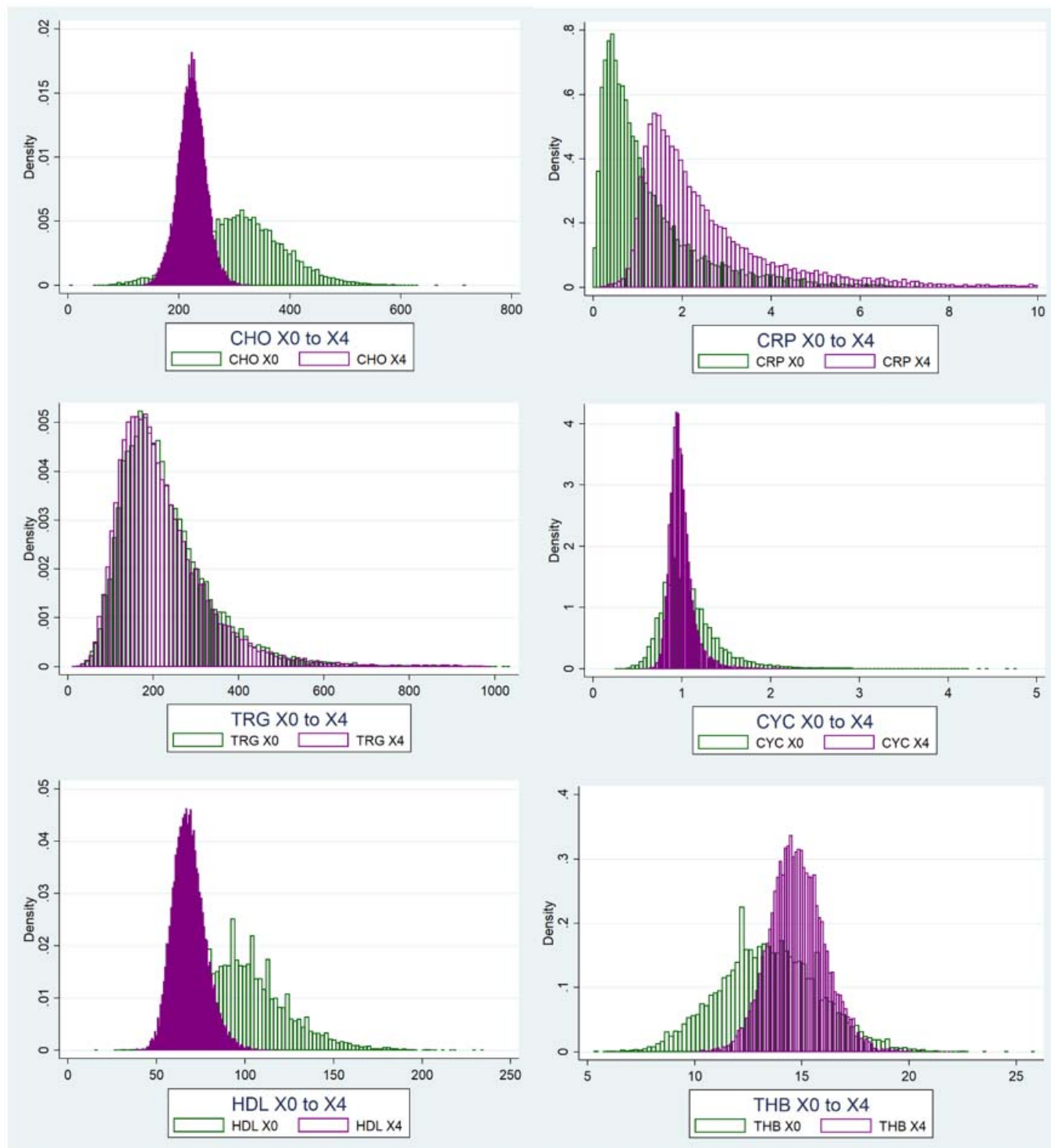
Figure 5-10: Effect of correction for blood-specimen-type



The blood-specimen type corrections have large effects on most biomarkers. The effects of the final corrections are particularly pronounced for the lipids CHO and HDL, shifting the distribution to the left, while they are small for CRP.

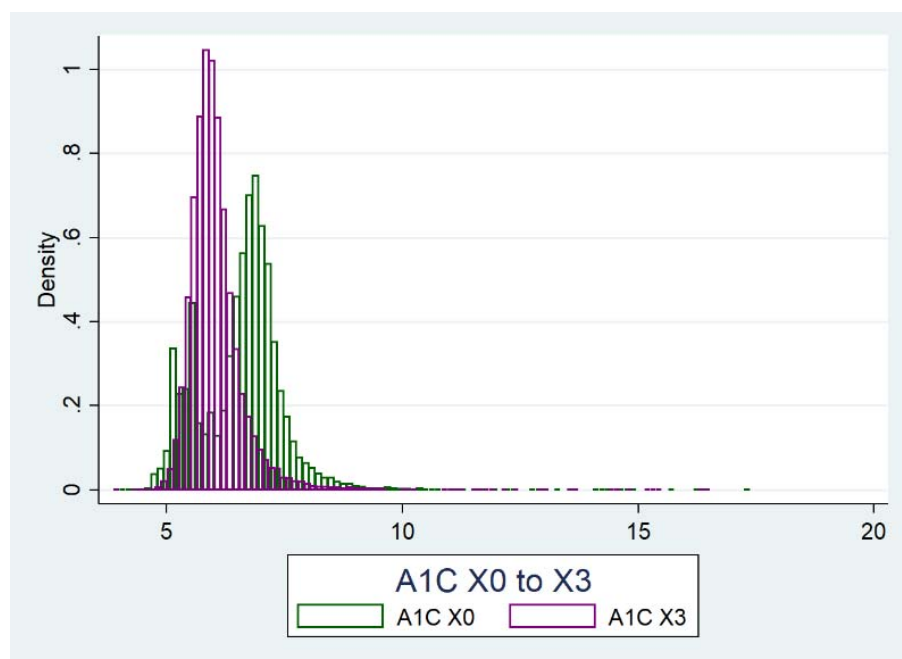
Finally, Figures 5-11 and 5-12 summarize all conversion steps. In general, correcting for the storage, spot size, fieldwork and blood-specimen type effects narrows the distribution since these influences are adjusted for. This is particularly pronounced for CHO, CYC and HDL, much less so for the other markers. The distribution of TRG remains almost unchanged. For CYC, the mean value remains almost unchanged while the width of the distribution is substantially narrower. The means of CHO and HDL are shifted to smaller biomarker values, the reverse holds for CRP and THB.

Figure 5-11: Effect of all conversion steps from raw data (x0) to final data (x4)



For HbA1c, the instrument correction dominates the result. A1c_x0 exhibits the different peaks from the two batches, while A1c_x3 has a single-peaked distribution generated by the DxC600 instrument. The other conversion steps have negligible effects on the HbA1c biomarker value, and there is no need for the final step of the blood-specimen type conversion.

Figure 5-12: Effect of the conversion steps for HbA1c



The main descriptive statistics (mean, median, standard deviation, and range) of the distributions of the x4 values are displayed in Table 5-7.

Table 5-7: Final conversion results x4 of steps 1 through 4

Variable	Unit	Obs	Mean	Median	Std. Dev.	Min	Max
a1c_x4	Percent	19,788	6.03	5.95	0.60	3.86	16.52
cho_x4	mg/dL	23,061	223.01	223.1	24.85	118.30	337.72
crp_x4	mg/L	22,728	3.51	2.08	6.55	0.14	139.40
cyc_x4	mg/L	23,061	0.99	0.97	0.14	0.53	2.52
hdl_x4	mg/dL	14,902	68.51	67.85	9.22	37.90	121.88
thb_x4	g/dL	14,902	14.78	14.73	1.28	9.81	21.95
trg_x4	mg/dL	22,989	222.98	196.65	127.6	9.79	3385.60

Our sample includes a small number of observations with very large values for CRP and TRG. These are also the only biomarkers with means and medians that substantially differ. There are 14 observations with a CRP value above 100, and 21 observations with a TRG value above 1500. These values may indicate the presence of rare diseases or health conditions but may also be artifacts. In addition, there are three observations with a negative CRP value which are certainly artifacts. CRP values above 20 mg/L may be evidence of acute inflammation and TRG values above

500 mg/dL of dyslipidemia, reasonable possibilities considering the gender distribution and advanced age of the SHARE respondents. CRP values above 10 mg/L but below 20 mg/L may indicate an elevated risk of CVD. TRG values above 500 mg/dL are considered high and above 1000 mg/dL very high, potentially indicating a lipid metabolism disorder (see <https://www.nhlbi.nih.gov/health/high-blood-triglycerides>).

6 Description of Variables

6.1 Descriptive statistics by socio-demographics

Table 6-1 shows the averages of each of the seven biomarkers by sex, age and country, while Table 6-2 presents regressions results (regressing the marker values on socio-economic characteristics and country dummies) in order to ascertain which of these differences are statistically significant. The study sample excludes 17 observations with a CRP value outside the [0,100] interval and 21 observations with a TRG value above 1500 as pointed out at the end of the previous section.

Table 6-1: Biomarker averages by sex, age and country

	thb [g/dL]	hdl [mg/dL]	cho [mg/dL]	crp [mg/L]	cyc [mg/L]	trg [mg/dL]	a1c [%]
All	14.69	66.80	222.2	3.42	0.980	227.4	6.06
Male	15.09	65.59	219.5	3.39	0.977	230.6	6.07
Female	14.36	67.79	224.6	3.44	0.982	224.7	6.06
Age 50-64	14.81	67.35	226.5	2.96	0.926	231.8	6.00
Age 65-74	14.77	66.96	221.5	3.48	0.970	231.3	6.07
Age 75+	14.40	65.77	217.5	3.94	1.069	215.2	6.14
Educ low	14.51	64.74	218.4	3.56	0.995	222.0	6.13
Educ medium	14.83	67.87	225.0	3.61	0.972	235.1	6.06
Educ high	14.77	68.52	224.8	2.82	0.964	224.3	5.96
BMI<24	14.47	69.23	225.4	2.73	0.961	202.3	5.94
BMI medium	14.76	66.31	222.5	3.36	0.973	231.2	6.06
BMI>30	14.81	64.96	218.0	4.31	1.015	247.2	6.21
Germany	14.76	67.49	224.3	3.55	0.988	232.1	6.07
Sweden	15.04	73.25	233.3	3.09	1.025	229.3	6.03
Spain	14.22	61.60	214.7	3.77	0.940	212.1	6.18
Italy	14.55	63.96	217.8	3.31	0.991	220.8	6.09
France	14.98	69.86	227.6	3.28	0.975	240.1	5.97
Denmark	15.00	71.88	226.9	3.68	0.995	207.0	5.89
Greece	13.91	61.58	218.7	2.91	0.960	220.1	6.32

Switzerland	15.00	72.20	231.4	3.03	1.012	241.2	5.98
Belgium	15.00	69.60	222.5	3.31	0.992	223.5	5.97
Israel	14.16	60.92	203.1	3.35	0.939	211.7	6.13
Slovenia	14.97	67.03	222.3	3.33	0.985	237.9	6.15
Estonia	14.63	67.78	223.2	3.47	0.986	206.7	6.02

Table 6-2 shows, that the differences between men and women are statistically significant for all biomarkers except CRP and TRG. Men have higher THB, CYC and HbA1c, women higher HDL and CHO biomarker levels.

Age effects are statistically significant for all biomarkers except CRP and CHO. They are depicted in Figure 6-1. HDL, THB, CHO (not significant), and TRG generally decline at older ages, while CRP (not significant), CysC and HbA1c generally increase. However, most biomarkers show a U-shaped pattern. At very old ages, pattern reversal such as in HbA1c is most likely due to selective mortality.

Table 6-2: Regression analysis of biomarkers

VARIABLES	(1) thb	(2) hdl	(3) cho	(4) crp	(5) cyc	(6) trg	(7) alc
female	-0.550*** (0.0203)	2.654*** (0.142)	6.240*** (0.318)	-0.0625 (0.0792)	-0.00811*** (0.00170)	1.008 (1.532)	-0.0199** (0.00860)
age	0.0921*** (0.0123)	0.348*** (0.0862)	0.00323 (0.194)	-0.0256 (0.0483)	-0.0110*** (0.00103)	3.786*** (0.936)	0.0331*** (0.00526)
age2	-0.000759*** (8.90e-05)	-0.00289*** (0.000623)	-0.00201 (0.00140)	0.000470 (0.000348)	0.000120*** (7.46e-06)	-0.0305*** (0.00675)	-0.000200*** (3.80e-05)
13.country	0.275*** (0.0430)	5.691*** (0.301)	8.515*** (0.660)	-0.547*** (0.165)	0.0348*** (0.00352)	-4.657 (3.179)	-0.0528*** (0.0176)
15.country	-0.533*** (0.0576)	-5.274*** (0.403)	-9.071*** (0.796)	0.0269 (0.199)	-0.0311*** (0.00425)	-17.26*** (3.839)	0.143*** (0.0222)
16.country	-0.205*** (0.0508)	-3.385*** (0.355)	-7.091*** (0.738)	-0.230 (0.184)	0.000229 (0.00394)	-10.09*** (3.559)	0.0162 (0.0201)
17.country	0.171* (0.0905)	2.300*** (0.633)	2.159* (1.281)	-0.392 (0.320)	-0.00542 (0.00684)	4.623 (6.183)	-0.0670* (0.0357)
18.country	0.197*** (0.0400)	4.306*** (0.280)	2.659*** (0.643)	0.0341 (0.160)	0.0125*** (0.00343)	-25.89*** (3.094)	-0.168*** (0.0168)
19.country	-0.827*** (0.0656)	-5.678*** (0.459)	-4.355*** (1.044)	-0.776*** (0.259)	-0.0417*** (0.00557)	-10.61** (5.022)	0.212*** (0.0294)
20.country	0.186*** (0.0454)	4.532*** (0.318)	7.905*** (0.710)	-0.568*** (0.177)	0.0274*** (0.00379)	12.47*** (3.419)	-0.0793*** (0.0190)
23.country	0.185*** (0.0405)	2.070*** (0.283)	-2.360*** (0.622)	-0.148 (0.155)	0.0121*** (0.00332)	-8.316*** (2.995)	-0.0878*** (0.0166)
25.country	-0.622*** (0.0669)	-6.213*** (0.468)	-20.01*** (0.941)	-0.409* (0.236)	-0.0543*** (0.00502)	-19.51*** (4.551)	0.0352 (0.0310)
34.country	0.0785* (0.0468)	-0.820** (0.327)	-2.646*** (0.693)	-0.172 (0.173)	0.00726** (0.00370)	3.066 (3.336)	0.0901*** (0.0191)
35.country	-0.0728* (0.0384)	-0.243 (0.268)	-1.293** (0.611)	-0.290* (0.152)	-0.0107*** (0.00326)	-23.37*** (2.941)	-0.0543*** (0.0161)
Constant	12.40*** (0.422)	55.91*** (2.953)	229.4*** (6.644)	3.243* (1.656)	1.173*** (0.0355)	116.6*** (32.08)	4.778*** (0.180)
Observations	14,902	14,902	23,061	22,714	23,061	22,968	19,788
R-squared	0.104	0.156	0.090	0.005	0.177	0.013	0.033

Standard errors in parentheses; *** p<0.01, ** p<0.05, * p<0.1

Country differences only slightly differ when age and sex differences across countries are controlled for, see Table 6-3.

Table 6-3: Biomarker values by country controlling for socio-demographics

	thb	hdl	cho	crp	cyc	trg	alc
Germany	14.76 ref	67.49 ref	224.3 ref	3.55 ref	0.99 ref	232.1 ref	6.07 ref
Sweden	15.04 ***	73.18 ***	232.8 ***	3.01 ***	1.02 ***	227.5 ns	6.01 ***
Spain	14.23 ***	62.21 ***	215.2 ***	3.58 ns	0.96 ***	214.8 ***	6.21 ***
Italy	14.56 ***	64.10 ***	217.2 ***	3.32 ns	0.99 ns	222.0 ***	6.08 ns
France	14.93 *	69.79 ***	226.4 *	3.16 ns	0.98 ns	236.7 ns	6.00 ns
Denmark	14.96 ***	71.79 ***	226.9 ***	3.59 ns	1.00 ***	206.2 ***	5.90 ***
Greece	13.93 ***	61.81 ***	219.9 ***	2.78 ***	0.95 ***	221.5 **	6.28 ***
Switzerland	14.95 ***	72.02 ***	232.2 ***	2.98 ***	1.02 ***	244.6 ***	5.99 ***
Belgium	14.95 ***	69.56 ***	221.9 ***	3.40 ns	1.00 ***	223.8 ***	5.98 ***
Israel	14.14 ***	61.28 ***	204.2 ***	3.14 **	0.93 ***	212.6 ***	6.10 ***
Slovenia	14.84 ns	66.67 ***	221.6 ***	3.38 ns	1.00 *	235.2 ns	6.16 ***
Estonia	14.69 **	67.25 ***	223.0 ***	3.26 **	0.98 ***	208.7 ***	6.01 ***

*** p<0.01, ** p<0.05, * p<0.1, ns=not statistically significant, ref=reference country

Most country effects are statistically significant, less so for CRP. Controlling for socio-demographics, THB is highest in Sweden and lowest in Greece and Israel; similarly, HDL, CysC and CHO; the latter is low in all Mediterranean countries IL, GR, IT, and ES. CRP is highest in Spain, Germany and Denmark, lowest in Greece. TRG is highest in France and Switzerland, and lowest in Denmark. HbA1c is highest in Greece and Israel, and lowest in Denmark with a remarkable North-South gradient.

Finally, the age and sex patterns derived from the regressions are shown in Figure 6-1.

Figure 6-1: Biomarker values by age and sex

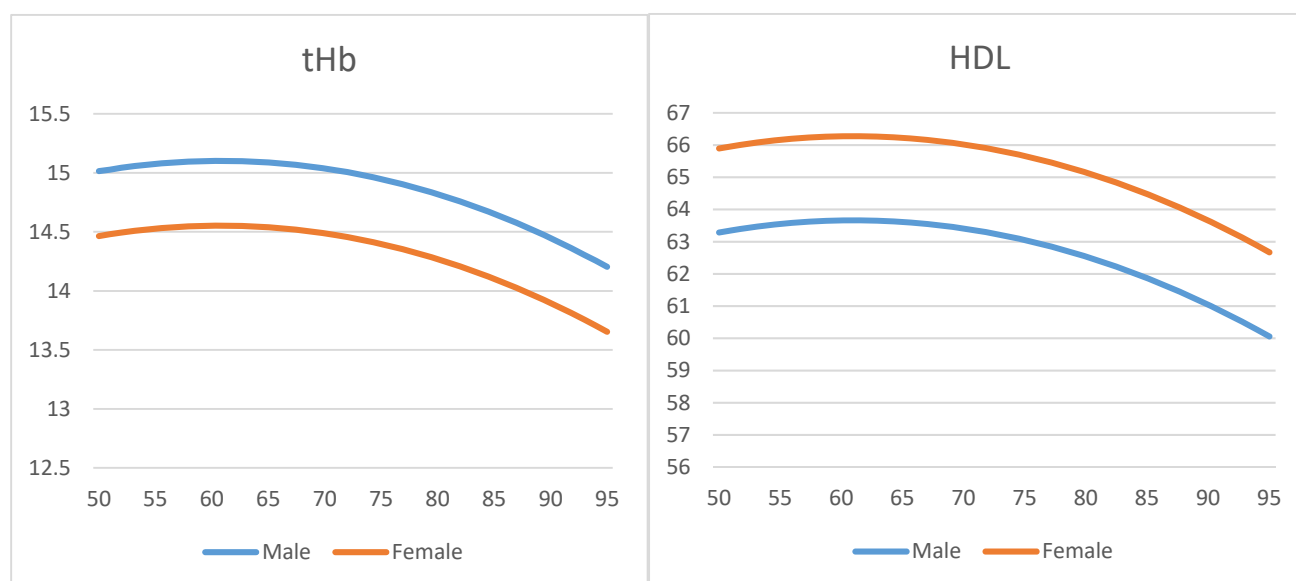
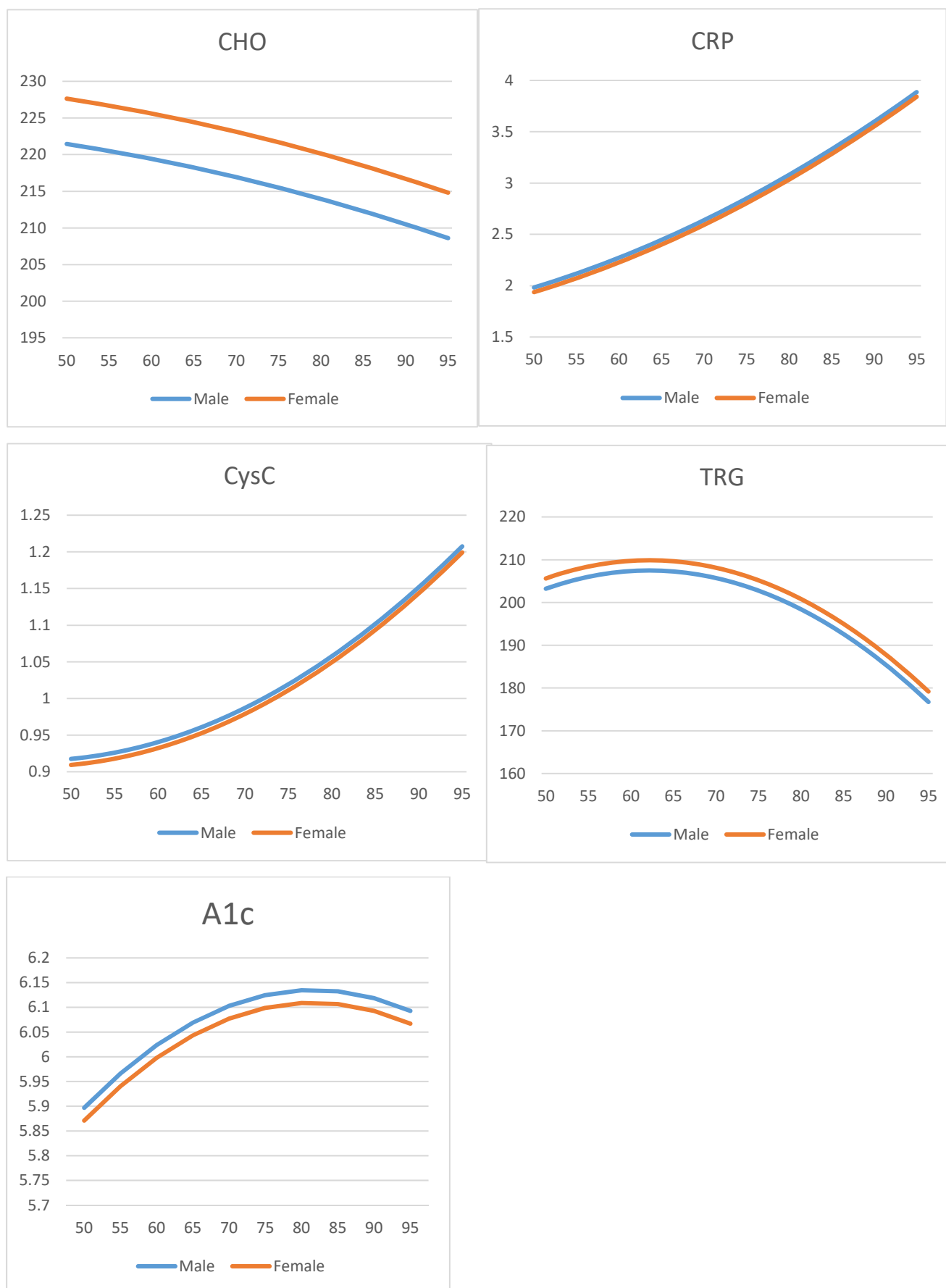


Figure 6-1: Biomarker values by age and sex (continued)



6.2 Variables in data set

The remaining tables in this section are technical and provide the number of available observations by biomarker, variable explanations for the DBS data set and a codebook with descriptive statistics of each variable.

Table 6-4: Number of observations

Marker Observations	Raw (all batches)	Final Corrected Values (all batches)
HDL	14,904	14,902
THB	14,904	14,902
A1c	19,790	19,788
CHO	23,063	23,061
TRG	22,991	22,989
CRP	22,730	22,728
CYC	23,063	23,061

Table 6-5: Variable definitions

Variable	Description
mergeid	Person identifier across modules and waves
country	Country identifier
age_int	Age in 2015
female	Female=1, male=0
batch	Batch 1 analyzed 2018, batch 2 in 2020
t_ship	Shipment time (in days) to Odense, DK
t_dry	Drying time (minutes)
t_temp	Temperature at time of collection, via interviewer
t_open	Bag opened before/during shipment, 0=No, 1=Yes
t_desiccant	Desiccant missing, 0=No, 1=Yes
t_sizeA	Average size of SizeA1 and SizeA2, censored .5 < size <1.5
t_sizeB	SizeB, censored .5 < size <1.5
*_x0 (a1c_x0, cho_x0, crp_x0, cyc_x0, hdl_x0, thb_x0, trig_x0)	Raw DBS_F values collected in the field (see Figures 5-1 and 5-6)
*_x1	Instrument and storage corrected values
*_x2	Spot-size corrected biomarker values
*_x3	Fieldwork-corrected biomarker values
*_x4	Final marker values corrected for fieldwork conditions, spot size, and storage time, then converted to standard equivalent (serum) values. These values should be used for analyses.
astar	Flag for whether HbA1c was analyzed on a small or smeared/overlapping spot (HbA1c-only)

storage_time	time spent in storage (days)
cciw_dbs	Calibrated cross-sectional individual weights corrected for DBS eligibility, consent and analyzability. These weights should be used for analysis.

Table 6-6: Basic descriptive statistics

Variable	Obs	Mean	Std.Dev.	Min	Max
mergeid	24,224				
country	24,224	21.36674	8.113054	12	35
age_int	24,224	67.86959	9.677573	30	101
female	24,224	0.5706324	0.4949961	0	1
batch	24,224	1.666612	0.4714337	1	2
t_ship	24,224	5.478575	3.829548	1	20
t_dry	24,222	20.66374	9.685071	5	40
t_temp	24,224	13.99211	8.22408	0	35
t_open	24,224	0.1060931	0.3079631	0	1
t_desiccant	24,224	0.0180812	0.1332481	0	1
t_sizeA	21,966	0.6946213	0.2917866	0.5	1.5
t_sizeB	23,081	0.6512868	0.2796448	0.5	1.5
cho_x0	23,063	312.4353	78.53884	2	718
crp_x0	22,730	1.967718	4.85489	0	100
trg_x0	22,991	232.4817	128.9938	11	3375
cyc_x0	23,063	1.082851	0.3484396	0.245	4.78
hdl_x0	14,904	100.0961	23.69677	15	235
alc_x0	19,790	6.581493	0.8802127	4	17.38
thb_x0	14,904	13.50003	2.357856	5.3	25.9
cho_x1	23,063	370.7431	68.51305	89.49326	718
cyc_x1	23,063	1.163231	0.3443091	0.245	4.894264
trg_x1	22,991	259.5729	127.7578	49.40471	3375
alc_x1	19,790	6.229294	0.609026	4.128308	16.95415
thb_x2	14,904	14.7105	2.402218	5.526941	28.23277
hdl_x2	14,904	110.2035	25.33835	17.32097	270.2072
cho_x2	23,063	406.5936	73.01137	100.8933	747.385
crp_x2	22,730	2.124303	5.241437	0	110.9296
cyc_x2	23,063	1.262478	0.3719677	0.271834	5.430316
trg_x2	22,991	281.5219	139.3721	54.76728	3741.335
alc_x2	19,790	6.152999	0.5985471	4.055694	16.65594

Variable	Obs	Mean	Std.Dev.	Min	Max
cho_x3	23,061	448.2954	41.65019	272.7499	640.5481
crp_x3	22,728	3.322064	6.552898	-0.0419732	139.2073
trg_x3	22,989	340.2067	157.6598	76.76746	4248.312
cyc_x3	23,061	1.587663	0.3663996	0.3843897	5.599596
hdl_x3	14,902	147.8058	17.11526	90.97936	246.7778
alc_x3	19,788	6.036386	0.6037	3.869202	16.52445
thb_x3	14,902	16.39956	1.573586	10.3011	25.20331
cho_x4	23,061	223.0142	24.85059	118.2749	337.7218
crp_x4	22,728	3.508892	6.553229	0.1446853	139.401
trg_x4	22,989	222.9802	127.5853	9.793603	3385.591
cyc_x4	23,061	0.9914428	0.1394587	0.5334539	2.518461
hdl_x4	14,902	68.51533	9.22905	37.87286	121.8839
thb_x4	14,902	14.78214	1.281285	9.816497	21.95055
alc_x4	19,788	6.036386	0.6037	3.869202	16.52445
astar	21,966	0.3144405	0.4643032	0	1
storage_time	16,147	1028.031	25.92818	982	1068
cciw_dbs	23,635	5221.305	12457.79	42.38616	275961.5

APPENDIX

Description of SHARE dried blood spot (DBS) Assays performed at the University of Washington Department of Laboratory Medicine and Pathology (Seattle, WA; UWLMP)

Methodology

Gold-standard assays were performed on liquid (i.e., serum, plasma, and/or whole blood) specimens to establish the target concentrations of analytes in calibrators, quality control (QC) specimens, and assay validation specimens. Total cholesterol (CHO), triglycerides (TRG), HDL cholesterol (HDL), and C-reactive protein (CRP) concentrations were measured by an AU680 Chemistry Analyzer (Beckman Coulter, Miami, FL, USA). Total hemoglobin (THB) concentration was measured by a Sysmex XE-5000 hematology analyzer (Sysmex, Mundelein, IL). Glycosylated hemoglobin A1c (HbA1c) percentage was measured by a Variant II and/or D100 ion-exchange high-performance liquid chromatography system (HPLC; Bio-Rad Laboratories, Hercules, CA), and, for confirmation of HbA1c values, by an immunoturbidimetric method run on a UniCell DxC 600 Chemistry Analyzer (Beckman Coulter). Cystatin C (CYC) concentration was measured by a sandwich enzyme-linked immunosorbent assay (ELISA; BioVendor, Candler, NC, USA).

The dried blood spot (DBS) assays for CHO, TRG, and HDL were modified versions of the gold-standard enzymatic methods. In brief, a filter paper disc(s) punched from a DBS card was eluted with a buffer solution to elute (re-liquefy) the dried blood. An aliquot of DBS eluate was incubated with an assay-specific enzyme to produce hydrogen peroxide, which then reacted with horseradish peroxidase (HRP) and a fluorogen to produce a fluorescent fluorophore. The fluorescent intensity of the solution, proportional to the concentration of analyte in the specimen, was measured by a spectrophotometer (BioTek, Winooski, VT). The DBS assays for C-reactive protein (CRP) and cystatin C (CYC) were modified versions of well-established sandwich ELISA methods. For those two assays, in brief, an aliquot of DBS eluate was incubated in a reaction plate coated with an antibody specific to the biomarker of interest. The plate was then washed, a detector antibody coupled to HRP was added, excess HRP was washed away, chromogenic substrate was added, and light absorbance proportional to the amount of analyte in the specimen was measured by a spectrophotometer (BioTek). HbA1c was measured in DBS eluates by HPLC and/or by immunoturbidimetry. THB concentration was determined by incubating an aliquot of DBS eluate with 12.5% Triton X-100/0.1% sodium hydroxide (NaOH), and absorbance was then measured by a spectrophotometer (BioTek). Assay-specific details are given below.

Calibrators and Quality Control (QC)

Liquid plasma assay calibrators were constructed from high concentration pooled human plasma (UWLMP) serially diluted with 7% bovine serum albumin in phosphate buffered saline (BSA/PBS;

Sigma Aldrich, St. Louis, MO) to the desired final analyte concentrations. 50µL aliquots of each calibrator were pipetted onto CF12 filter paper (Whatman, Piscataway, NJ), dried for four hours at room temperature (RT), sealed in Ziploc bags with desiccant packs, and stored at -80°C to create dried plasma assay calibrators. Two DBS QC samples (i.e., sample controls) were constructed to have assay-specific clinically-relevant high and low analyte concentrations by mixing pools of human plasma and washed red blood cells in equal volumes (i.e., 50% hematocrit), pipetted in 75µL aliquots onto CF12 filter paper, dried for four hours at RT, sealed in Ziploc bags with desiccant packs, and stored at -80°C. Prior to spotting, the analyte concentrations of the liquid calibrators and QC samples were determined by analysis using the gold-standard methods described above.

Assay Details

To accommodate the limited material available from SHARE DBS, the samples were analyzed using modified versions of UWLMP standard DBS assays. HDL, THB, and HbA1c were measured in shared eluent obtained from two 3.2mm diameter discs punched from a DBS (A markers). CHO, TRG, CRP, and CYC were measured in shared eluent obtained from a single 3.2mm disc punched from a DBS (B markers). For SHARE DBS with very limited material, HbA1c alone was measured in eluent obtained from a single 3.2mm disc punched from a DBS (Am1punch).

A markers

Two 3.2mm discs were eluted by shaking for one hour at RT in 400µL double-distilled water (ddH₂O). A 200µL aliquot of the eluate was transferred to a pierceable microvial followed by addition of 550µL ddH₂O to a final volume of 750µL, and HbA1c then measured by HPLC using the proprietary software provided with the instrument (Bio-Rad). To 200µL of the remaining initial eluate, 200µL ddH₂O was added to a final volume of 400µL. 40µL of that solution was incubated with 75µL Reagent 1 (Beckman Coulter) for 30 minutes at 37°C. 50µL of Reagent 2 (Beckman Coulter), diluted 1:1 with ddH₂O and containing 0.01% 10-acetyl-3,7-dihydroxyphenoxazine (ADHP; Cayman Chemical, Ann Arbor, MI), was then added. After 30 minutes incubation at 37°C, the fluorescence emission of the solution was measured at 590nm using 540nm excitation by a Synergy HT spectrophotometer (BioTek). Gen5 software (BioTek) was used to construct a standard curve by plotting the fluorescence values of the standards against the known HDL concentrations. Using the standard curve, the fluorescence values of the DBS were interpolated to determine HDL concentrations of the samples. THB was measured by incubating 50µL of the above 400µL DBS eluate solution with 100µL 12.5% Triton X-100/0.1% NaOH (Sigma Aldrich) for 15 minutes at RT and reading absorbance at 405nm using a spectrophotometer (BioTek). A standard curve was constructed by plotting the absorbance values of the standards against the known THB concentrations. Using the standard

curve, the absorbance values of the DBS were interpolated to determine THB concentrations of the samples. To measure HbA1c immunoturbidimetrically, a single 3.2 mm punch was eluted in 300 μ L Hemolyzing Reagent (Beckman Coulter, Inc.) by shaking at 1000 rpm for 1 hour at room temperature. An aliquot of 200 μ L was transferred to a 0.3 mL sample cup and loaded onto the DxC 600 (Beckman Coulter) and assayed using the HbA1c- reagent kit (Beckman Coulter A87905) without modification beyond the DBS sample type.

B markers

One 3.2mm diameter disc was eluted by shaking for one hour at RT in 400 μ L PBS/0.1% Triton X-100/0.01% Tween-20. Total cholesterol reagent (Synermed) was prepared following manufacturer directions and 0.01% ADHP was then added. CHO was measured by combining 20 μ L DBS eluate and 100 μ L reagent, incubated for 30 minutes at 37°C, and fluorescence read on a spectrophotometer as given above for HDL. Triglycerides reagent (Beckman Coulter) was prepared by mixing the contents of cartridge A and C and bringing the pH to 7.4. Just prior to assaying, 5% v/v cartridge B and 0.01% ADHP was added to the A/C mixture. TRG was measured by combining 40 μ L sample with 100 μ L reagent, incubating for 30 minutes at 37°C, and fluorescence then read on a spectrophotometer as given above for HDL. ELISAs were used to measure CRP (BioCheck Inc., South San Francisco, CA) and CYC (BioVendor) by assaying 80 μ L and 100 μ L DBS eluate, respectively, with each assay performed following ELISA manufacturer directions.

Am1punch (HbA1c only)

One 3.2mm disc was eluted by shaking for one hour at RT in 400 μ L ddH₂O. 350 μ L ddH₂O was then added for a final volume of 750 μ L, and HbA1c measured by HPLC (Bio-Rad).

Assay Performance

A markers and HbA1c

The HDL within-assay imprecision (CV) was 7.4% and between-assay imprecision was 11.5%. The HDL concentrations of 50 DBS-paired serum and plasma samples assayed on the AU680 correlated with the HDL concentrations of capillary blood DBS and EDTA venous blood DBS ($R^2 = 0.77$ and 0.72 , respectively). The THB within-assay imprecision (CV) was 5.2% and between-assay imprecision was 7.4%. The THB concentrations of 50 DBS-paired blood samples assayed by Sysmex correlated with the THB concentrations of capillary blood DBS and EDTA venous blood DBS ($R^2 = 0.59$ and 0.43 , respectively). The HbA1c within-assay imprecision (CV) was 1.3% and between-assay imprecision was 2.4%. The HbA1c values of 50 DBS-paired blood samples assayed

on the HPLC correlated with the HbA1c values of capillary blood DBS and EDTA venous blood DBS ($R^2 = 0.94$ and 0.96 , respectively). The HbA1c values of 36 DBS-paired blood samples correlated with the HbA1c values of capillary DBS and assayed on the DxC 600 ($R^2 = 0.95$).

B markers

The CHO within-assay imprecision (CV) was 6.2% and between-assay imprecision was 8.9%. The CHO concentrations of 50 DBS-paired serum and plasma samples assayed on the AU680 correlated with the CHO concentrations of capillary blood DBS and EDTA venous blood DBS ($R^2 = 0.73$ and 0.73 , respectively). The TRG within-assay imprecision (CV) was 5.3% and between-assay imprecision was 13.7%. The TRG concentrations of 49 DBS-paired serum and plasma samples assayed by the AU680 correlated with the TRG concentrations of capillary blood DBS and EDTA venous blood DBS ($R^2 = 0.94$ and 0.89 , respectively). The CRP within-assay imprecision (CV) was 4.2% and between-assay imprecision was 11.9%. The CRP concentrations of 50 DBS-paired blood samples assayed on the AU680 correlated with the CRP concentrations of capillary blood DBS and EDTA venous blood DBS ($R^2 = 0.99$ and 0.99 , respectively). The CYC within-assay imprecision (CV) was 3.7% and between-assay imprecision was 7.0%. The CYC concentrations of 30 DBS-paired serum and plasma samples assayed by ELISA correlated with the CYC concentrations of capillary blood DBS and EDTA venous blood DBS ($R^2 = 0.82$ and 0.75 , respectively).



DBS Short Instructions

7. Place the lancet on the side of the pad on R's fourth finger
8. Activate the lancet on R's finger
 - a. If necessary ask R to gently squeeze his/her finger for a **large** drop of blood to form
 - b. Using the sterile gauze, wipe off the first drop of blood

Filling the Filter Card

9. When the next **large** drop of blood forms (be a little patient), place R's finger above the left-most circle on the card and allow it to drop onto the centre
10. Do NOT touch the filter card with the fingers!
11. Continue this procedure with the following circles
12. Put card aside to dry (until the end of the interview, but 15 minutes minimum)

You only have to fill one card!

After Taking the Blood Spots

13. Provide R with a cotton ball and a plaster
14. Place the used gauze pad, wipes, wrappers, lancet and disposable gloves in the large plastic bag and discard
15. Enter the barcode label from the collection card into the CAPI (questions BS_9_ and BS_10_)

After the End of the Interview – Step by Step

16. Put on a new pair of disposable gloves
17. Write the date on the blood collection card
18. Tuck cover down on the blood spot collection card
19. Insert the blood collection card into the small (brown) envelope
 - a. In case you filled both cards, please put barcodes and dates on both cards and put both cards together into the envelope
20. Attach the temperature strip to the small (brown) envelope
21. Place the small (brown) envelope with the blood spot card in the small plastic bag
22. Remove the desiccant from its plastic cover and place it in the small plastic bag (together with the small, brown envelope)
23. Close the small plastic bag and place it in the large padded envelope
24. Close and seal the large, padded envelope
25. Put an address sticker and a stamp onto the large, padded envelope
26. Mail it on the day of the interview or as soon as possible. Avoid high temperatures.
27. Mail the signed copy of the "DBS Collection" consent form to [Address of CT]

Instructions for Blood Sample Collection

Materials needed

- 2 Blood spots collecting cards
- Disposable protective cloth
- 2 pairs of disposable rubber gloves
- Disposable disinfecting wipes
- 2 semi-automatic lancets
- Cotton balls
- Gauze pads
- Plasters
- Desiccant
- Small plastic bag for blood spot collecting card
- Small envelope for blood spot collecting card
- Large, padded envelope
- Barcode stickers
- Address stickers
- Temperature strip
- 2 copies of the "Dried Blood Spots Collection" consent form
- Large plastic bag

During the Interview – Step by Step

Preparation

Please be careful, **do not to touch the circles** of the filter card at any point of the procedure!

1. Ask if R is willing to do the DBS, if yes
 - a. Ask R to rub hands
 - b. Remove all parts from the Blood Spot Collection kit
 - c. Place them on the disposable protective cloth
2. Place barcodes on
 - a. blood collection card
 - b. the small (brown) envelope
 - c. both "DBS Collection" consent forms
3. Put on a pair of rubber gloves

Taking the Blood Spots

4. Wipe of R's fourth finger (the ring finger) on R's non-dominant hand with the disinfecting wipe
5. Wait till the finger is dry
6. Remove the blue cap of one of the lancets

1

Biomarker Checklist

"DBS Collection" Consent Form:

- ☐ Barcode on consent form
- ☐ Consent form signed by respondent
- ☐ Copy of consent form handed to respondent

Blood Collection cards:

- ☐ Barcodes on the filter card
- ☐ Barcodes on the small envelope for the filter card

After blood collection:

- ☐ Blood allowed to dry until the end of the interview
- ☐ Collection card put into small envelope, sealed
- ☐ Temperature strip affixed on small envelope
- ☐ Small envelope placed in plastic bag
- ☐ Desiccant placed in plastic bag
- ☐ Plastic bag placed in padded envelope, sealed

Material for Lab/Country Team (CT):

- ☐ Sent padded envelope with collection card(s) to lab
- ☐ Sent signed "DBS Collection" consent form to the SHARE Country Team

Who participates?

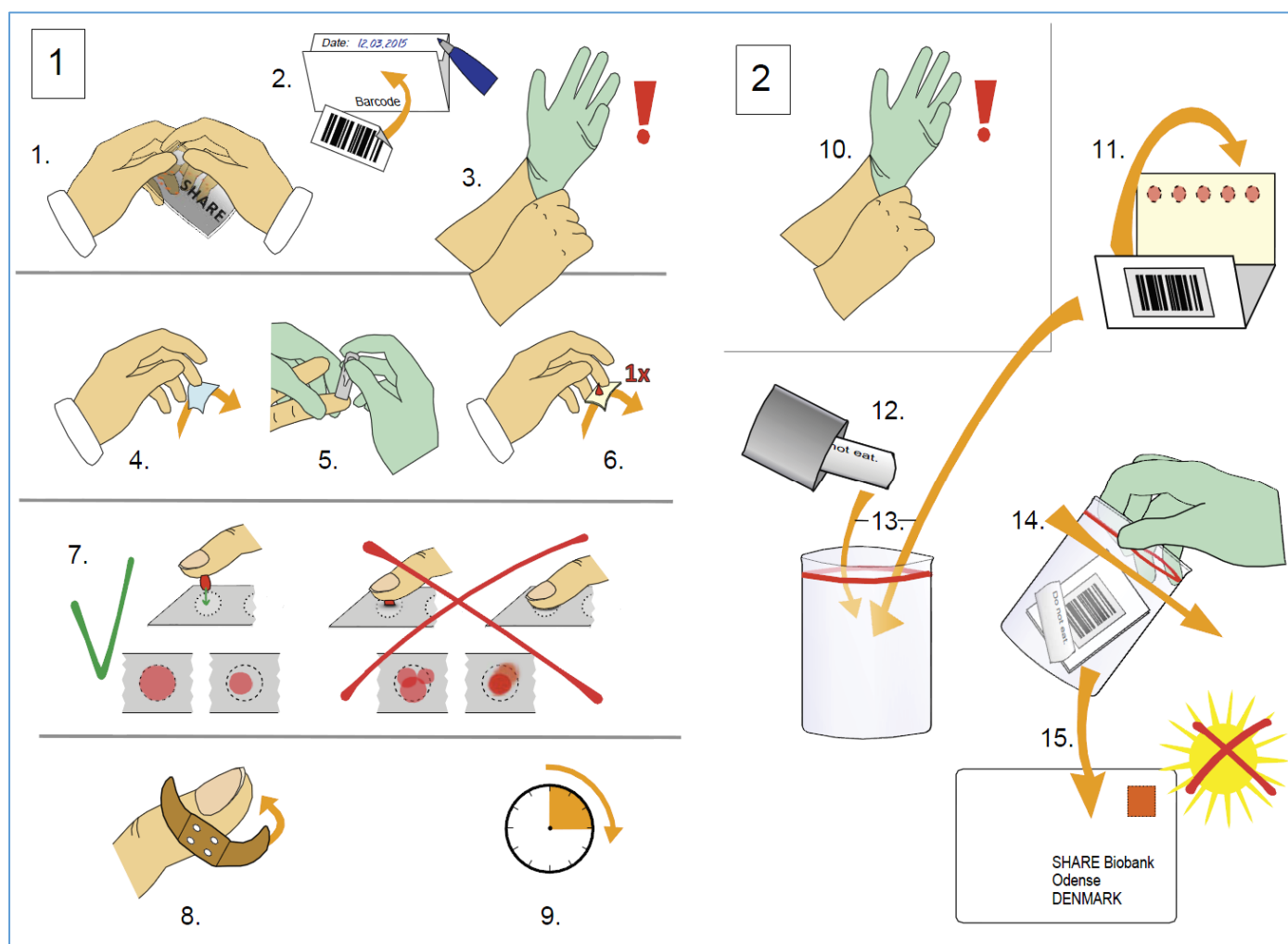
We will ask panel respondents as well as refresher respondents, who do an interview for the first time, to participate in the dried blood spot collection.

The participation in this part of the interview is **voluntary**. Should a respondent refuse to have blood spots taken, please make notes in the CAPI and continue with the rest of the interview.

We will NOT collect blood in case

- a) the interview is a proxy-interview
- b) the respondent is not able to agree to having blood spots collected
- c) the respondent refuses to have their blood spots taken

SHARE-W6 DBS Instruction_illustrated Flyer



Consent Form

The Survey of Health, Ageing and Retirement in Europe



Dried Blood Spots Collection – Consent Form

I hereby consent to take part in the Dried Blood Spots Collection in SHARE and to provide a small blood sample via a finger prick for the analyses as explained in the "Collection of Dried Blood Spots" information leaflet. Furthermore, I agree that the blood drops collected on a filter card will be sent to the SHARE Biobank at the University of Southern Denmark in Odense and to the analysing laboratory.

- a) I understand that my finger will be pricked and a few drops of my blood will be collected. I confirm that I have read and understood the information from the leaflet. The interviewer has explained the procedure and all my questions have been answered to my satisfaction.
- b) I understand that there is no reward for completing this measure, and I also can choose not to complete this part of the study. Although I will not receive direct benefit from my participation, others may ultimately benefit from the knowledge obtained in this part of the study.
- c) I understand that my participation is completely voluntary. I will complete the collection of dried blood spots only if I agree to do so. I can stop my participation at any time without specifying my reasons.
- d) I was offered to ask questions about the procedure at any time and can expect truthful answers. I can ask the interviewer, or contact **[put in name, email address, and telephone number of CTL]**.
- e) I agree that my blood sample will be owned by the SHARE Consortium and, as well as the data retrieved from it, will be stored in the SHARE Biobank at the University of Southern Denmark for scientific research within the context of SHARE. I agree that the storage will extend beyond the ending or termination of the Dried Blood Spots Collection in SHARE.
- f) I understand that the results of the blood analyses will be kept strictly confidential, just in the same way as the rest of the interview data. The data are processed and released in an anonymised form in accordance with all relevant data protection laws.
- g) I also understand that if – at any time – I want to revoke my participation and to withdraw consent for storing the blood sample, I can do so. I can contact **[put name of CTL in here]** (contact details see information leaflet) in writing and request that my blood sample will be destroyed.
- h) I want to exclude the following analyses from being done (*in capital letters*):
.....
.....
- i) I received a copy of this form and of the "Collection of Dried Blood Spots" information leaflet.

Date of birth of participant

Full name of participant (*please use capital letters*)

Date, place

Signature of participant

INTERVIEWER

PUT BARCODE HERE

I confirm that the collection was conducted correctly and all relevant information was given.

Name of interviewer (*in capital letters*)

Signature of interviewer

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